# Rent Inflation and Monetary Policy

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**Abstract**

Rental prices enter the inflation indices that central banks target, yet rents behave unlike other consumer prices after a monetary tightening. We build a Heterogeneous-Agent New Keynesian model in which households rent, own, or become landlords, with endogenous house prices and rents, and sticky leases. A tightening keeps would-be buyers renting, and rents rise until enough owners supply the extra rental housing. About two-thirds of the effect of higher interest rates on the rental market is supply-driven: higher financial returns and mortgage costs make supplying rental housing less attractive. We study how policy should treat rent inflation. A Taylor rule that targets a rent-inclusive price index deepens the consumption contraction. Judged when the shock hits, the welfare-optimal coefficient on rent inflation is negative under both general demand and supply shocks: central banks should react to rent increases with accommodative policy. Redistribution toward renters and mortgagor-owners drives this answer: a Ramsey planner eases even without a shock, and once it cannot surprise households, it mostly stabilizes.

**JEL codes:** D31, E21, E31, E52, E58, E61, G51, R21, R31.

**Keywords:** Rent inflation, optimal monetary policy, Ramsey policy, redistribution, heterogeneous-agent New Keynesian models, housing tenure choice, sticky rental leases, inflation targeting, rental market.

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# 1 Introduction

Shelter costs are a large part of the price indices that central banks target. In the euro area, the Harmonised Index of Consumer Prices (HICP) targeted by the European Central Bank (ECB) includes the actual rents paid by tenants, about 6 percent of the basket.[1] In the United States, the weight of rents is larger because they are used to impute owner-occupied housing costs: they account for about 15 percent of headline personal consumption expenditures and 31 percent of headline CPI (Bolhuis et al., 2022). The post-pandemic episode showed how much these weights matter. Market rents surged in 2021 and 2022; the shelter components of the indices were still climbing when central banks tightened, and they kept rising long afterward, as existing leases caught up with market rents.

This paper studies how monetary policy should respond to rent inflation. Rents generate trade-offs for monetary policy if they do not fall when policy tightens, unlike other consumer prices. In Spain, we estimate that a monetary tightening by the ECB lowers prices excluding rents within a year and raises real rents. We argue that the tenure choice and portfolios of different participants in rental markets are central to understanding this response. When a tightening lifts the return on financial assets, landlords require a higher rental yield. Higher rates also raise landlords' mortgage costs, a housing-specific form of the cost channel (Barth and Ramey, 2001), and keep would-be buyers renting. We study these forces in a Heterogeneous-Agent New Keynesian (HANK) model in which households choose to rent, to own their home, or to become landlords, and can buy housing with a mortgage. Rental leases are sticky: only new contracts reprice at the market rent.

The model delivers three results. First, the supply side dominates: about two-thirds of the effect of higher interest rates on the rental market comes from higher returns on financial assets and higher mortgage costs for landlords. Second, a rule that targets a rent-inclusive index, as central banks effectively do, keeps reacting to rent pressure of its own making: the consumption contraction becomes deeper. Third, redistribution shapes how monetary policy should treat rent inflation: judged when the shock hits, the welfare-optimal response of a standard Taylor rule to rent inflation is neither to tighten against it nor to look through it, but to ease, under both general demand and supply shocks. Accommodation lowers the cost of supplying rental housing and relieves poorer renters and mortgage borrowers, at the expense of unlevered owners and unlevered landlords.

In the model, households facing uninsurable earnings risk decide each period whether to rent, own their home, or own an additional housing unit that they supply to the rental market as landlords. House prices, market rents, residential construction, and the real interest rate are equilibrium objects. Because only new contracts reprice, the model distinguishes the market rent on new leases from the average rent households actually pay, which is the object measured in

[1] Following its 2021 strategy review, the ECB recommended a roadmap for incorporating owner-occupied housing costs into the index, on the grounds that doing so would better represent the inflation rate relevant for households (European Central Bank, 2021). The ECB's preferred approach is based on net acquisitions, in contrast to the rental-equivalence imputation used in the United States (Lane, 2022). The ECB's 2025 strategy assessment reiterates the Governing Council's support for including owner-occupied housing costs in the HICP (European Central Bank, 2025).

consumer price indices. On the nominal side, unions set wages subject to Rotemberg adjustment costs, yielding a New Keynesian wage Phillips curve, and the central bank sets the nominal rate with a Taylor rule. Our policy experiments vary whether, and how strongly, rent inflation enters the policy rule, either through the inflation index the rule targets or as a separate term.

We calibrate the model to Spain at quarterly frequency and validate it against two pieces of untargeted evidence: tenure transitions from the Spanish survey of household finances, and the responses of prices and rents to ECB monetary policy shocks, whose signs the model reproduces. In Spain, although many new mortgages are now fixed-rate, most outstanding mortgages are still adjustable-rate (Banco de España, 2025), as in our model.

**The rent channel of monetary policy.** We first characterize how a monetary tightening reaches the rental market. A contractionary monetary policy shock lowers consumption, output, and nondurable inflation, and at the same time raises market rents. The market rent on new leases rises, while the average rent, the component that enters the price index, inherits the increase only as leases turn over: it peaks at 4 basis points above its steady-state level eight quarters after a 25-basis-point annualized innovation to the monetary policy rule, long after the shock itself has faded. We show that about two-thirds of the effect on the rental market driven by higher interest rates comes from the supply side: the rise in returns on alternative financial assets reduces the willingness of current landlords to invest in real estate, and the increase in mortgage costs discourages owners who would borrow to buy a rental unit from becoming landlords if rents do not increase. The demand side, would-be buyers who keep renting because mortgages cost more and savings pay more, accounts for the remaining third. The model generates an impact house-price decline of 11 basis points, consistent with the empirical evidence that monetary tightenings lower house prices (Corsetti et al., 2022). This price decline offsets more than half of the upward pressure on rents, and the rental market clears as higher rents draw wealthy owners into becoming landlords. On impact, the largest consumption losses fall on mortgagor-owners, while renters, whose rents rise as leases reset, recover more slowly than unlevered owners and landlords.

**Rent in the inflation target.** We next put measured rent inflation inside the policy rule. We give rent inflation its share of household spending in the inflation index the rule targets, and hold the economy and the monetary shock fixed, so the comparison with the rent-exclusive rule isolates the policy feedback from measured rent inflation to the path of interest rates. Because the rent component reaches the policy index only as leases turn over, the rule keeps responding to the second-round rent pressure of its own tightening long after the shock has faded; most of the additional contraction therefore accumulates after impact, while rents are still rising. Over the first two years, house prices fall, rents rise, and consumption contracts by more, with no additional nondurable disinflation. How owner-occupied housing enters the index matters more than the weight on rents: rental equivalence more than triples the feedback relative to the model's rent share, and the net-acquisitions approach shifts the additional tightening to the housing recovery.

**The optimal response to rent inflation.** We let the central bank choose the coefficient on rent inflation in an additive Taylor rule, which adds a term $\phi_r \pi_t^{\text{rent}}$ to a conventional rule that responds to nondurable inflation. We pick $\phi_r$ to maximize utilitarian welfare under two general demand shocks, to household patience and to government spending, and two supply shocks, to rental supply and to cost push. The optimal coefficient is negative under all four: $-0.49$ under the patience and government-spending shocks; $-0.18$ under the rental supply shock; and $-0.48$ under the cost-push shock. A negative coefficient means that the rule eases when rent inflation rises, relative to a rule that ignores it. It lowers rents through the transmission mechanism: easier policy reduces the cost of supplying rental housing and leans against the shock that raised rents, whether or not it is housing-specific. It also relieves mortgage borrowers and lowers the government's interest bill.

The poorest renters, who hold the least financial wealth and have the highest marginal utility of consumption, benefit most from the lower rents and higher transfers, while landlords lose rental income and indebted homeowners gain from lower mortgage costs. Tenure is a key determinant of who gains, but net asset positions also matter: rich renters and unlevered homeowners lose from the lower returns on their savings.

The Dávila and Schaab (2025) decomposition of these gains into efficiency and redistribution shows that the redistributive motive is what pushes the monetary authority to respond negatively to rent inflation under the patience and rental supply shocks, and does most of the work under the government-spending shock, while under the cost-push shock both motives favor a negative response.

Under three of the four shocks, a Taylor rule that targets only nondurable inflation but is less hawkish overall cannot achieve such gains. Under the patience and rental supply shocks, no such rule comes close to the gain from responding to rent inflation; under the government-spending shock, a less hawkish rule helps but falls short of it. The negative response also survives interest-rate smoothing, an output-gap term in the rule, a transfer proportional to labor productivity, and shorter leases. At a practical level, easing in response to rent inflation means that the part of measured inflation that comes from rents deserves less weight than other price increases: when rents drive a substantial share of inflation, central banks should tighten by less than they otherwise would.

Finally, we solve for Ramsey policy. A planner who chooses policy when the shock hits eases, and would ease even if no shock had hit: households have already decided how much to borrow and whether to rent or own, and easing then redistributes toward the poorest renters, who value resources most. The optimized rule captures about a fifth of this gain after a patience shock. Following Dávila and Schaab (2023, DS hereafter), we separate this redistributive motive from the pure stabilization motive by solving for the policy of a planner that cannot surprise households in this way. That planner mostly stabilizes, as DS find in a one-asset HANK economy. Judged from its perspective, the rule with the best rent coefficient leans slightly against persistent rent inflation. Optimal policy thus has two answers, as in DS. We emphasize the first because its redistributive

motive has no counterpart in a representative-agent economy. Easing in response to rent inflation is the answer of a central bank that chooses its response when rents start to rise, and it rests largely on redistribution.

**Related literature.** Our paper relates to three literatures. First, we build on general-equilibrium models of housing and rental markets with endogenous tenure choice (Sommer et al., 2013; Kaplan et al., 2020; Castellanos et al., 2026). Greenwald and Guren (2025) show that the slope of the tenure-supply curve, which reflects landlords' willingness to sell rental properties to owner-occupiers, governs how credit shocks move the price–rent ratio and the homeownership rate; in our model, this slope is the endogenous landlord margin. These models are real, whereas New Keynesian models with housing, starting with Iacoviello (2005), emphasize housing collateral and mortgage costs, whose role in monetary transmission depends on mortgage-market structure and is stronger where mortgages are adjustable-rate (Calza et al., 2013; Garriga et al., 2017). Empirically, the response of euro-area rents to a monetary tightening is small and imprecisely estimated (Corsetti et al., 2022), new-lease rents fall in Germany (Groiss and Syrichas, 2026), advertised rents rise in the United Kingdom (Lazarowicz and Richard, 2023), and rents rise in the United States (Dias and Duarte, 2019; Abramson et al., 2025). We add evidence for Spain: after a monetary tightening, prices excluding rents fall and real rents rise. Tighter mortgage supply also raises rents (Gete and Reher, 2018). The consumption response to monetary policy differs among renters, mortgagors, and outright owners (Cloyne et al., 2020) and across household balance sheets in the euro area (Slacalek et al., 2020). Rents are sticky (Genesove, 2003; Baker and Wroblewski, 2026), and recorded shelter inflation lags market rents because lease turnover is slow (Bolhuis et al., 2022; Ball and Koh, 2025). Castellanos et al. (2026) show that an exogenous rise in the real rate raises rents and lowers homeownership. We embed this channel in a HANK economy with sticky leases, so that the real rate, inflation, and rents respond jointly to policy, and we show why rents rise when policy tightens: mainly because supplying rental housing becomes less attractive.

Second, we relate to work on which inflation central banks should stabilize. Since Aoki (2001) and Woodford (2003), the prescription has been to stabilize inflation in the sectors where prices are stickiest, with weights that also reflect each sector's cyclical sensitivity and shocks (Mankiw and Reis, 2003).[2] Rents are sticky, so a mechanical application of this logic would give rent inflation a large weight. The logic does not carry over to leases in our economy: a lease keeps its real rent until it turns over, so inflation creates no relative-price distortion in rents for policy to correct. A recent literature asks directly how monetary policy should treat housing inflation. Bianchi et al. (2024) argue that housing services are rationed through search and inelastic short-run supply, so that tightening against a housing-demand shock inefficiently depresses the goods sector, and optimal policy stabilizes nonhousing inflation while disregarding housing inflation; Chodorow-Reich and Mehrotra (2026) apply this logic to U.S. shelter inflation and argue for placing less weight on

[2]See also Benigno (2004) for currency areas, Eusepi et al. (2011) for a welfare-based price index, and Bodenstein et al. (2008) for core versus headline inflation.

shelter than its expenditure share. Dias and Duarte (2022) show in a two-agent model with tenure choice that a rent-inclusive index creates a policy feedback loop, in which the rule responds to rent movements that monetary policy itself induced. In an estimated New Keynesian model of the United States with a rental stock owned by entrepreneurs, De Llanos (2026) finds that a tightening shifts constrained households toward renting and that, among the rules he considers, the shelter-inclusive rule has the largest welfare cost. Albuquerque et al. (2026) study a HANK economy with individual landlords who, being credit-constrained and slow to update their expectations, largely fail to pass higher interest rates into rents, and find that a central bank minimizing an inflation loss may do better by looking through a shock to rental supply. In our economy, landlords' financing and portfolio costs pass into rents, so a rule that responds to rents raises the very costs that make them supply less, and we find that the welfare-maximizing response to measured rent inflation is not to exclude or merely down-weight it but to ease.

Third, we build on the Heterogeneous-Agent New Keynesian literature (Kaplan et al., 2018). On the normative side, optimal monetary policy in these economies trades off price stability against insurance and inequality (Bhandari et al., 2021; Acharya et al., 2023), and a utilitarian planner faces an incentive to redistribute toward indebted, high-marginal-utility households (Dávila and Schaab, 2023). Wealth-poor working households prefer a rule that stabilizes unemployment, while the wealth-rich prefer inflation targeting (Gornemann et al., 2021). Characterizing optimal rules in HANK, McKay and Wolf (2026) find that distributional concerns barely change optimal monetary policy, whose distributional effects are small. We add a discrete tenure choice and an endogenous landlord margin, so that a tightening transmits through tenure decisions and the rents that clear the rental market, and we characterize an optimal simple rule (Schmitt-Grohé and Uribe, 2007) and compare it with Ramsey policy, with and without the planner's incentive to surprise households. In our economy, monetary policy moves a relative price between two groups of households, the rent that renters pay to landlords, and the redistribution it brings about shapes the optimal response to rent inflation.

We begin in Section 2 by developing the model. Two features of the housing block matter throughout: the endogenous tenure margin, which makes rents rise when policy tightens, and sticky leases, which separate the market rent on new contracts from the average rent that price indices measure. Section 3 calibrates the economy to Spain and validates it against untargeted evidence. Section 4 isolates the rent channel of a monetary tightening, decomposes the rent response into channels, and quantifies its consumption incidence across households. Section 5 puts rent inflation in the policy index and quantifies the cost of the resulting policy feedback. Section 6 characterizes the optimal coefficient on rent inflation and who gains from it. Section 7 solves for Ramsey policy and evaluates the optimized rules when the central bank cannot surprise households. Section 8 concludes.

# 2 A HANK Model with Housing Tenure Choice

We study a discrete-time Heterogeneous-Agent New Keynesian economy with endogenous housing tenure choice, sticky rental leases, and housing supply. These margins determine how monetary shocks affect market rents, average rents, house prices, and households across tenure states. Time is indexed by $t \in \{0, 1, 2, \ldots\}$. The economy is populated by heterogeneous households, labor unions that introduce nominal rigidities through sticky wages, a representative consumption-good producer, a construction sector, a government that issues debt, and a central bank. The final good's price acts as the numeraire.

## 2.1 Households

We consider a continuum of infinitely lived, ex ante identical households that become heterogeneous ex post along four dimensions: financial wealth, housing tenure status, idiosyncratic labor productivity, and a lease state. Each period, households consume a nondurable good, choose a financial asset position, and make a discrete housing tenure decision. The housing tenure choice determines whether a household rents, owns a single housing unit, or owns two housing units, in which case the household becomes a landlord and rents out one unit to the rental market, similarly to the framework in Castellanos et al. (2026).

**Idiosyncratic risk.** Households face two sources of idiosyncratic uncertainty. First, labor productivity follows a persistent stochastic process. Second, the discrete housing choice is subject to type-I extreme-value taste shocks, which smooth the tenure decision. There is no aggregate uncertainty. Aggregate shocks are modeled as one-time unanticipated MIT shocks: they are unexpected on impact, and the economy follows a deterministic transition thereafter.

**Preferences.** Households derive utility from nondurable consumption and housing services, and they dislike supplying labor. They also bear a cost of wage adjustment that unions pass on to them (Section 2.3). They value these flows according to the per-period utility function

$$\mathcal{U}_t(c, h, n) = u(c, h) - v(n) - \frac{\theta_w}{2}(\pi_t^w)^2, \qquad u(c, h) = \frac{(c\, f(h))^{1-\gamma}}{1-\gamma}, \quad v(n) = \psi_n \frac{n^{1+\eta}}{1+\eta}, \tag{1}$$

where $\gamma$ is the coefficient of relative risk aversion, $\eta$ is the inverse Frisch elasticity, $\psi_n$ scales the disutility of labor, the wage-adjustment cost $\frac{\theta_w}{2}(\pi_t^w)^2$, in which $\pi_t^w$ is nominal wage inflation and $\theta_w \geq 0$ governs wage rigidity, is the same for every household and affects none of its decisions, and $f(h)$ captures the housing services enjoyed by homeowners relative to renters,

$$f(h) = \mathbb{1}_{h=0} + \mathbb{1}_{h>0} \left(\theta\, \tilde{h}_2 / \tilde{h}_1\right)^{\alpha_h}. \tag{2}$$

Here $\tilde{h}_1 < \tilde{h}_2$ denote the two housing-quality (equivalently, size) tiers, with renters occupying the low tier and homeowners (owner-occupiers and landlords) the high tier, while $\theta > 1$ raises the service flow from ownership relative to renting and $\alpha_h < 1$ captures diminishing returns to housing quality. Housing consumption varies only through tenure: there is no intensive margin, so a renter cannot absorb higher rents by renting a smaller dwelling, and rents must move by more to clear the rental market. Labor supply $n$ is set by labor unions, as described in Section 2.3.

The marginal utility of consumption, $u_c(c,h) = f(h)^{1-\gamma}c^{-\gamma}$, depends on the household's tenure state through the housing-services shifter $f(h)$. Because $\gamma > 1$, the term $f(h)^{1-\gamma}$ is decreasing in $f(h)$: at a given level of consumption, a homeowner, who enjoys more housing services ($f(h) > 1$), values an additional unit of consumption less than a renter does. Consumption and housing services are therefore substitutes: a household is willing to accept lower nondurable consumption in exchange for the housing services of ownership.

**Budget constraint.** Households allocate their savings across housing $h$ and financial assets $a$, which can be negative, reflecting the net debtor position of households with mortgages. Households consume the nondurable good, and renters pay rent. Thus, their budget constraint reads:

$$c + a' + \psi_t(h,h') \leq \left(1 + r_t + \kappa\, \mathbb{1}_{a<0}\right)a + (1-\tau_y)\, w_t y n + T_t(y) + \widehat{p}^r_t(s)\,(h-1), \tag{3}$$

where a prime denotes a next-period variable, $r_t$ is the real interest rate on balances carried from $t-1$ into $t$ (Section 2.6), $\kappa$ is a mortgage spread that makes the borrowing rate exceed the rate on savings by a constant margin, $w_t$ is the real wage, $\tau_y$ is a flat tax on labor income, $T_t(y)$ is a government transfer (Section 2.7), and $\widehat{p}^r_t(s)$ is the rent paid or received under a lease in state $s$ (see Section 2.2). The term $\widehat{p}^r_t(s)(h-1)$ captures the rental market: renters ($h = 0$) pay rent, owner-occupiers ($h = 1$) have no rental cash flow, and landlords ($h = 2$) receive rent. The function $\psi_t(h,h')$ summarizes all housing costs, as detailed below.

**Collateral constraints.** Borrowing is subject to collateral and income constraints. A household that upgrades its housing position ($h' > h$) faces loan-to-value (LTV) and loan-to-income (LTI) limits. Let $p^h_t$ denote the real price per unit of housing quality. Following Castellanos et al. (2026), houses of different qualities can be freely converted into one another, so a housing unit of quality $\tilde{h}_i$ is worth $p^h_t\,\tilde{h}_i$. Let $q_t(h,h')$ denote the quantity of housing quality acquired when moving from housing state $h$ to $h'$,

$$q_t(h,h') = \begin{cases} \tilde{h}_2, & \text{if } (h,h') = (0,1), \\ \tilde{h}_1 + \tilde{h}_2, & \text{if } (h,h') = (0,2), \\ \tilde{h}_1, & \text{if } (h,h') = (1,2), \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Thus, $p_t^h q_t(h, h')$ is the value of housing acquired in the transaction. An existing mortgage borrower who retains the same housing state must satisfy a minimum amortization requirement. Formally,

$$a' \geq \begin{cases} \max\left\{ -\lambda_{\text{LTV}} p_t^h q_t(h, h'),\ -\lambda_{\text{LTI}} w_t y n \right\}, & \text{if } h' > h, \\ (1-m)\, a, & \text{if } h' = h > 0 \text{ and } a < 0, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Households can borrow only when upgrading their housing position. The LTV constraint prevents them from borrowing more than a fraction $\lambda_{\text{LTV}}$ of the value of the housing units purchased, while the LTI constraint caps borrowing at $\lambda_{\text{LTI}}$ times current labor income. Existing borrowers who retain the same housing state may roll over their mortgage but must reduce its balance by at least a fraction $m$ each quarter. Their minimum payment, $(r_t + \kappa + m)\,|a|$, therefore covers interest and amortization and moves one for one with the mortgage rate, as the installment of an adjustable-rate mortgage does when its rate resets. Households who become renters or move to a lower housing state must repay their debt in full. We abstract from refinancing and mortgage equity extraction.[3]

**Housing costs.** The housing cost function $\psi_t(h, h')$ captures depreciation and the costs of buying and selling housing units. An owner-occupier holds the single unit $\tilde{h}_2$, a landlord holds both $\tilde{h}_1$ and $\tilde{h}_2$, and a renter holds none. Let $p_t(h)$ denote the market value of the housing position held in state $h$, $p_t(h) = p_t^h\, \tilde{h}_2\, \mathbb{1}_{h=1} + p_t^h\, (\tilde{h}_1 + \tilde{h}_2)\, \mathbb{1}_{h=2}$, and recall that $q_t(h, h')$ is the quantity purchased and $q_t(h', h)$ the quantity sold when moving from $h$ to $h'$. Housing costs are

$$\psi_t(h, h') = p_t(h') - (1 - \delta_h)\, p_t(h) + \tau_h\, p_t^h \left[ q_t(h, h') + q_t(h', h) \right], \tag{6}$$

where $\delta_h$ is the depreciation rate and $\tau_h$ is a transaction cost, which the buyer pays on the units purchased and the seller on the units sold. The first two terms are the change in the value of the household's housing position net of depreciation; the last is the transaction cost, which accrues to the government. For example, an owner-occupier who sells and becomes a renter ($h = 1$, $h' = 0$) receives $(1 - \delta_h - \tau_h)\, p_t^h\, \tilde{h}_2$, and a renter who buys ($h = 0$, $h' = 1$) pays $(1 + \tau_h)\, p_t^h\, \tilde{h}_2$.

**Recursive problem.** We denote the individual state by $(a, h, y, s)$. At the beginning of each period, the household observes a vector of choice-specific taste shocks $\varepsilon = \{\varepsilon(h')\}_{h'}$ and solves

$$V_t^{\varepsilon}(a, h, y, s) = \max_{c, a', h'} \mathcal{U}_t(c, h, n) + \sigma_\varepsilon \varepsilon(h') + \beta_t\, \mathbb{E}_t \left[ V_{t+1}(a', h', y', s') \,\middle|\, y, h, s, h' \right]. \tag{7}$$

[3] Mortgages in the model reset automatically with the real rate, so refinancing is not needed for rate changes to pass through to debt service. Using the Spanish Survey of Household Finances (*Encuesta Financiera de las Familias*, EFF), 2014–2022 waves, we find that only 3.3 percent of owner-occupiers hold debt secured on real estate for purposes other than buying property, compared with 37.9 percent with a loan for the main residence. International Monetary Fund (2008) likewise classifies mortgage equity withdrawal in Spain as limited.

The shocks $\varepsilon(h')$ are drawn independently across households, housing choices, and dates from a type-I extreme-value distribution with unit scale and zero mean, and $\sigma_\varepsilon > 0$ scales the taste shocks. The continuation value $V_t(a,h,y,s) = \mathbb{E}_\varepsilon[V_t^\varepsilon(a,h,y,s)]$ integrates over these shocks, yielding logit transition probabilities across housing states. Appendix A.1 provides the corresponding formulas. The discount factor $\beta_t$ may vary over time.

**Distribution.** We denote the distribution over the idiosyncratic state space $(a,h,y,s)$ by $g_t(a,h,y,s)$ and the stationary distribution by $g_{ss}(a,h,y,s)$. The share of households with housing state $h$ and lease state $s$ is $\mu_t^{h,s} = \iint g_t(a,h,y,s)\,da\,dy$, and the share with housing state $h$ is $\mu_t^h = \sum_s \mu_t^{h,s}$.

## 2.2 Sticky Rental Leases

In practice, the rents paid by households do not adjust instantaneously to market conditions, both because existing contracts take time to be updated and because tenancy law limits when their rent may change. We capture these frictions with leases that retain their signing rent in real terms for a fixed number of quarters, unless they are renegotiated or either party changes tenure first. Their nominal rent is therefore fully indexed: it is updated every quarter by current nondurable inflation.

**Leases as contracts.** A lease is a contract between one renter and one landlord. New leases are signed at the market rent: households entering a rental tenure, and households whose previous lease has ended, are matched at random in the market for new leases. Each quarter, a lease is renegotiated with probability $1-\varphi$; otherwise it continues into the next quarter as long as both parties remain in their rental tenure. A renegotiation resets the lease to the market rent. When either party changes tenure, the contract ends, and a partner who stays in its rental tenure signs a new lease at the market rent. Lease survival thus depends on both parties' endogenous tenure decisions. Households take aggregate tenure-retention rates as given; in equilibrium, these rates coincide with those generated by their choices. Appendix A.1 gives these rates and the associated continuation values.

We track the lease with a state $s \in \{0,1,\ldots,J\}$: $s < J$ denotes a lease signed $s$ quarters ago, and $s = J$ pools leases signed $J$ or more quarters ago, which allows us to keep the state space finite. A continuing lease ages to $\min\{s+1, J\}$, and a new lease starts at $s = 0$. Owner-occupiers hold no lease; their state is normalized to $s = 0$ and has no effect on payoffs. The rent paid by the renter and received by the landlord is

$$\widehat{p}_t^r(s) = \begin{cases} p_{t-s}^r, & s < J, \\ \widetilde{p}_t^r, & s = J, \end{cases} \tag{8}$$

where $p_t^r$ is the market rent, determined by rental-market clearing, and $\widetilde{p}_t^r$ is the pooled rent, the average of the rents at which the leases in the pool were signed, which is the same for renters and landlords. Appendix A.2 gives its updating rule and the evolution of lease masses. The two sides

hold the same contracts, so renters' aggregate payments equal landlords' aggregate receipts at every date.

**Average rent and rent inflation.** The average rent is computed over the distribution of renters across lease vintages:

$$\overline{p}_t^r = \sum_{s=0}^{J-1} \frac{\mu_t^{0,s}}{\mu_t^0} p_{t-s}^r + \frac{\mu_t^{0,J}}{\mu_t^0} \widetilde{p}_t^r. \tag{9}$$

Rent inflation in the consumer price index is computed from the average rent $\overline{p}_t^r$, rather than the market rent $p_t^r$. This is the model counterpart of an index measured from outstanding leases and the relevant object for a central bank targeting an HICP-based index (Section 5). Lease turnover delays the transmission of market rents into this index.

Because $\overline{p}_t^r$ is expressed in units of the consumption good, the nominal average rent is $\overline{P}_t^r = P_t^c \overline{p}_t^r$, where $P_t^c$ is the nominal price of the consumption good. With $\pi_t^c = P_t^c / P_{t-1}^c - 1$ denoting nondurable inflation, nominal rent inflation satisfies

$$1 + \pi_t^{\text{rent}} = (1 + \pi_t^c) \frac{\overline{p}_t^r}{\overline{p}_{t-1}^r}. \tag{10}$$

## 2.3 Labor Market

As is standard in New Keynesian models with sticky wages, both without household heterogeneity (Erceg et al., 2000) and with it (Auclert et al., 2024), a continuum of labor unions sets nominal wages and determines the hours households work.[4] Hours are rationed equally across households,

$$n_t = N_t, \tag{11}$$

where $N_t$ is aggregate labor. Unions adjust nominal wages subject to a quadratic Rotemberg (Rotemberg, 1982) cost, which they pass through to their members as a utility loss. The flow payoff that the union internalizes is therefore the household utility (1), this cost included, with $\pi_t^w = (W_t - W_{t-1})/W_{t-1}$.

Each union sets its wage to maximize the aggregate lifetime utility of the households it represents. Optimal nominal wage setting yields a nonlinear New Keynesian wage Phillips curve (see Appendix A.3 for the derivation),

$$\pi_t^w(1 + \pi_t^w) = \frac{\epsilon_t}{\theta_w} N_t \left( \nu'(N_t) - (1 + \tau_L)(1 - \tau_y) \frac{\epsilon_t - 1}{\epsilon_t} w_t \int y_{it}\, u_c(c_{it}, h_{it})\, di \right)$$
$$+ \beta_t \pi_{t+1}^w (1 + \pi_{t+1}^w). \tag{12}$$

[4] Placing the nominal rigidity in wages avoids the countercyclical profits that sticky-price heterogeneous-agent models generate and would have to assign to households (Broer et al., 2020; Auclert et al., 2025): with perfect competition and flexible prices, profits are zero at all dates. Symmetric rationing serves a separate purpose, shutting down the endogenous labor-supply margin so that labor income moves proportionally across households.

Here $c_{it}$ and $h_{it}$ denote the consumption and housing of household $i$, and $\int(\cdot)\,di$ averages over the continuum of households. Equivalently, one can integrate with respect to the distribution $g_t(a, h, y, s)$. The parameter $\epsilon_t$ is the elasticity of substitution across labor varieties and $\tau_L$ is an employment subsidy. With a subsidy that offsets the union's market power in the steady state, namely $(1+\tau_L)(\epsilon-1)/\epsilon = 1$ at the steady-state elasticity $\epsilon$, this equation shows that unions aim to equate the marginal disutility of labor to the marginal utility of the after-tax income earned in an additional hour of work.[5] The elasticity $\epsilon_t$ may vary over time.

Because goods prices are flexible, the model has no dispersion in relative prices across producers: inflation is costly through the wage-adjustment term in (1), and sticky leases matter for welfare through the rents households pay and the tenure choices they make.

## 2.4 Consumption-Good Producer

A representative firm produces the final nondurable consumption good using only labor.[6] The technology is linear:

$$Y_t^c = A^c N_t, \tag{13}$$

where $A^c$ is total factor productivity. Under perfect competition and flexible prices, profits from production are zero and the real wage equals the marginal product of labor,

$$w_t = A^c. \tag{14}$$

Nominal wage inflation $\pi_t^w$ and nondurable inflation $\pi_t^c$ are linked through the labor market. Since the real wage equals productivity, $w_t = A^c$, and the nominal wage is $W_t = w_t P_t^c$, nominal wage inflation satisfies $\pi_t^w = \pi_t^c$.

## 2.5 Construction Sector

The construction sector produces new housing units $Y_t^h$ using land $L_t$ and housing structures $S_t$ (priced in units of the consumption good) according to a Cobb–Douglas technology,

$$Y_t^h = A^h L_t^{\alpha_L} S_t^{1-\alpha_L}, \tag{15}$$

where $A^h$ is total factor productivity in construction and $\alpha_L$ is the land share. Under perfect competition, the construction firm solves

$$\max_{S_t, L_t} \; p_t^h A^h L_t^{\alpha_L} S_t^{1-\alpha_L} - p_t^L L_t - S_t, \tag{16}$$

[5] We assume, as in Erceg et al. (2000), that employment in the steady state is subsidized to eliminate the monopolistic distortion associated with the wage-setting power of unions.

[6] We adopt the simplest production structure, as in Dávila and Schaab (2023). An alternative is to assume firms produce differentiated varieties under monopolistic competition, as in Auclert et al. (2024). This generates positive profits that must be redistributed, adding a layer of complexity without affecting the main transmission channels we study.

where $p_t^L$ is the price of land in consumption units. Land permits are in fixed supply, $L_t = \overline{L}$, and $p_t^L$ adjusts to clear their market. The government issues the permits, and their revenue $p_t^L \overline{L}$ enters its budget (Section 2.7). Because houses of different qualities can be freely converted into one another, we do not need prices that clear the market for each individual housing type; it suffices to clear the market for the aggregate housing stock. Aggregate housing demand implied by the tenure distribution is $H_t = \mu_t^1 \tilde{h}_2 + \mu_t^2 (\tilde{h}_1 + \tilde{h}_2)$. The housing market clears when this demand equals the available aggregate stock. Since new construction augments the existing stock, which depreciates at rate $\delta_h$, market clearing requires

$$H_{t+1} = (1 - \delta_h) H_t + Y_t^h. \tag{17}$$

Together with rental-market clearing, $\mu_t^0 = \mu_t^2$, and $\mu_t^0 + \mu_t^1 + \mu_t^2 = 1$, the definition of $H_t$ ties the tenure distribution to the housing stock,

$$\mu_t^0 = \mu_t^2 = \frac{\tilde{h}_2 - H_t}{\tilde{h}_2 - \tilde{h}_1}. \tag{18}$$

When an owner-occupier becomes a renter and another becomes a landlord, the housing they demand falls from $2\tilde{h}_2$ to $\tilde{h}_1 + \tilde{h}_2$, so a higher renter share requires a smaller housing stock, and hence construction below depreciation. Because the stock adjusts slowly, the number of leases can change only gradually: in the short run, house prices and rents adjust so that households hold the existing stock in this tenure mix, and shifts in rental demand or in landlords' willingness to supply rental housing move rents rather than the number of rental units.

## 2.6 Financial Market

We consider a closed economy in which the wealth of savers ($a_t > 0$) finances the mortgages of borrowers ($a_t < 0$) and the public debt $B_t$. Aggregate net financial wealth is $A_t \equiv \sum_{h,s} \int\int a\, g_t(a, h, y, s)\, da\, dy$. In each period $t$, the real interest rate $r_t$ adjusts so that the financial market clears,

$$A_t = B_t. \tag{19}$$

We adopt the following timing convention: the household's net position $a_t$ is a one-period riskless claim rolled over each period. We fix the real payoff on the balance carried into $t$ at the rate agreed when that balance was chosen, so monetary shocks realized at $t$ affect the return on debt issued from $t$ to $t+1$ but do not revalue the balance brought into $t$. Throughout, $r_t$ denotes this real rate between $t-1$ and $t$.[7]

Real and nominal returns are linked by the Fisher equation. Along the perfect-foresight path after the shock, this claim is equivalent to a nominal bond with gross nominal rate $1 + i_t$, and the

[7] Because the payoff is fixed in real terms, surprise inflation does not redistribute wealth between nominal creditors and debtors, a channel emphasized by Doepke and Schneider (2006) and Auclert (2019); monetary policy redistributes in the model through the paths of real rates, rents, and house prices.

real gross return in consumption units is $1 + r_{t+1} = (1 + i_t)\, P_t^c / P_{t+1}^c$. The nominal interest rate $i_t$, the real interest rate $r_{t+1}$, and nondurable inflation $\pi_{t+1}^c$ satisfy

$$1 + i_t = (1 + r_{t+1})(1 + \pi_{t+1}^c). \tag{20}$$

Mortgages are adjustable-rate: debt carried from $t$ to $t+1$ bears the rate $r_{t+1} + \kappa$, where $\kappa$ is the constant spread in (3).[8] Through the Fisher equation (20), the nominal policy rate $i_t$ determines the real rollover rate $r_{t+1}$ and hence mortgage financing costs and the minimum payment required by (5).

## 2.7 Government and Central Bank

**Government.** The government issues one-period bonds, finances purchases that may vary over time, pays an employment subsidy, and taxes labor income at the flat rate $\tau_y$. It also collects the transaction costs on housing purchases and sales, the mortgage spread, and the revenue from land permits. Deficits are financed with debt, and the budget balance is passed to households through a lump-sum transfer that responds gradually to the level of debt. The government pays an employment subsidy $\tau_L(1 - \tau_y) w_t\, y\, n_t$ to each household with labor productivity $y$. To finance this subsidy, it raises a tax proportional to labor productivity, equal to $\tau_L(1 - \tau_y) w_t\, y\, N_t$. In equilibrium this scheme does not redistribute resources across households, because hours are rationed symmetrically and $n_t = N_t$.[9]

Let $B_t$ denote the stock of public debt carried into period $t$, $G_t$ government purchases, $\overline{T}_t$ the lump-sum transfer paid to every household, and $M_t(h, h')$ the mass of households moving from housing state $h$ to $h'$. Transaction-cost revenue $\mathcal{T}_t^h$ and aggregate mortgage debt $A_t^-$ are

$$\mathcal{T}_t^h \equiv \tau_h\, p_t^h \sum_{h,h'} M_t(h,h') \left[ q_t(h,h') + q_t(h',h) \right], \quad A_t^- \equiv \sum_{h,s} \iint a\, \mathbb{1}_{a<0}\, g_t(a,h,y,s)\, da\, dy < 0.$$

The government also collects the labor tax, $\tau_y w_t N_t$, and the spread on mortgage debt, $-\kappa A_t^- > 0$. Its budget constraint is

$$B_{t+1} = (1 + r_t)\, B_t + G_t + \overline{T}_t - \tau_y w_t N_t - \mathcal{T}_t^h + \kappa A_t^- - p_t^L \overline{L}, \tag{21}$$

and the transfer follows the fiscal rule

$$\overline{T}_t = \overline{T}_{ss} - \Gamma_B \left( B_t - \overline{B} \right), \tag{22}$$

[8] Spanish variable-rate mortgages are indexed to Euribor plus a contractual margin. We represent this structure with a constant spread over the real rate and a quarterly reset, abstracting from the annual or semiannual reviews usual in Spanish contracts.

[9] Making both the subsidy and its financing proportional to labor productivity prevents this fiscal arrangement from redistributing income across households. This allows the employment subsidy to offset the steady-state wage markup without creating an additional fiscal transmission channel.

where $\overline{B}$ is steady-state debt and $\Gamma_B$ governs how fast debt returns to it, as in the fiscal rules of Leeper (1991). In the stationary equilibrium, debt equals $\overline{B}$, purchases equal $G_{ss}$, and the transfer $\overline{T}_{ss}$ balances the budget. Along the transition, a change in the interest bill or in tax revenue first changes the stock of debt, and households receive it through the transfer over the following years rather than in the quarter in which it occurs.[10] The transfer schedule $T_t(y)$ combines the employment-subsidy scheme with the lump-sum transfer $\overline{T}_t$.

**Central bank.** Because the model contains only two priced components of household consumption, nondurable consumption and rental housing services, we define the model counterpart of HICP inflation, $\pi_t^{\text{hicp}}$, as a weighted average of nondurable and rent inflation. House purchases are asset transactions and do not enter the consumption price index. The central bank sets the nominal interest rate using a Taylor rule that targets this inflation measure,

$$i_t = r_{ss} + \phi_\pi \pi_t^{\text{hicp}} + \varepsilon_t^m, \qquad \pi_t^{\text{hicp}} = (1-\omega_r)\,\pi_t^c + \omega_r\,\pi_t^{\text{rent}}, \tag{23}$$

where $r_{ss}$ is the steady-state real interest rate, $\phi_\pi$ is the inflation response coefficient, $\varepsilon_t^m$ is a monetary policy shock, $\omega_r$ is the weight on rents in the index, and $\pi_t^{\text{rent}}$ is the nominal inflation rate associated with the *average rent* $\overline{p}_t^r$ defined in (9). Interest and inflation rates in (23) are quarterly, and steady-state inflation is zero.[11]

## 2.8 Equilibrium

We now define the competitive equilibrium for this economy. Aggregate consumption is $C_t \equiv \int c_{it}\, di$. The equilibrium requires that household and firm choices be individually optimal, that all markets clear, and that the distribution of households evolve consistently with individual decisions and aggregate prices.

**Definition 1** (Competitive Equilibrium)**.** *Given a sequence of shocks, a competitive equilibrium consists of household value and policy functions* $\{V_t, V_t^\varepsilon, c_t, a_{t+1}, h_{t+1}\}$*, prices* $\{w_t, r_t, p_t^h, p_t^r, \widetilde{p}_t^r, \overline{p}_t^r, p_t^L, i_t, \pi_t^c, \pi_t^{\text{hicp}}, \pi_t^w, \pi_t^{\text{rent}}\}$*, aggregate quantities* $\{N_t, C_t, Y_t^c, Y_t^h, H_t, A_t, L_t, S_t, B_t\}$*, lease-survival objects* $\{\xi_{h,t}^s, \lambda_t\}$*, a transfer schedule* $T_t$*, and a distribution* $g_t$ *such that:*

*(i)* ***Household optimality.*** *Given prices and retention rates, household policy functions solve the Bellman equation (7), with continuation value (A.4), subject to the budget constraint (3) and the borrowing constraints (5).*

*(ii)* ***Consumption-good firm optimality.*** *The representative firm maximizes profits, yielding* $w_t = A^c$*.*

[10] Kaplan (2025) emphasizes that fiscal assumptions are not innocuous in HANK models: the way monetary-policy-induced fiscal effects are financed can shape aggregate and distributional transmission.

[11] In Spain, the ECB sets the interest rate in response to euro-area inflation. Spanish inflation and inflation in the rest of the euro area share one cycle, with a correlation of 0.82 since 1999, so that, seen from Spain, the ECB's rule acts as a rule on Spanish inflation with a coefficient above one. Appendix D.10 derives this mapping.

(iii) ***Construction firm optimality.*** *The construction firm maximizes profits as in* (16).

(iv) ***Labor union optimality.*** *Labor unions ration hours, so* $n_t = N_t$*, and nominal wage inflation* $\pi_t^w$ *satisfies the New Keynesian wage Phillips curve* (12).

(v) ***Market clearing.*** *Four markets clear:*[12]

(a) Housing market: $H_{t+1} = (1 - \delta_h)H_t + Y_t^h$.

(b) Rental market: *the total share of renters equals the total share of landlords,* $\mu_t^0 = \mu_t^2$*; by* (A.5)*, the two sides then coincide within each lease state.*

(c) Financial market: $A_t = B_t$.

(d) Land market: $L_t = \overline{L}$.

(vi) ***Government budget.*** *Public debt evolves according to* (21)*, and the lump-sum transfer follows the fiscal rule* (22).

(vii) ***Law of motion for the distribution.*** *The distribution* $g_t$ *evolves according to household policy functions, the idiosyncratic income transition probabilities, and the lease-state transition rules of Section 2.2.*

(viii) ***Lease consistency.*** *The retention rates households take as given equal those implied by their choices,* (A.3)*, for every lease state, and the pooled rent satisfies* (A.6).

(ix) ***Monetary policy.*** *The nominal interest rate satisfies the Taylor rule and HICP-index definition* (23)*, and the Fisher equation* (20) *links the nominal and real interest rates.*

## 2.9 Timing and Rent Measurement

The lease structure of Section 2.2 determines when aggregate shocks can move rents and how rents enter measured inflation. We collect these implications here, since the quantitative sections use them repeatedly.

**Predetermination.** We index the initial stationary equilibrium by $t = 0$ and the impact period by $t = 1$. Aggregate shocks are realized after period-1 leases have been signed: leases are signed at the start of the quarter, before the shock is observed. Every incumbent renter and landlord enters the impact period at the steady-state rent: the market rent satisfies $p_1^r = p_{ss}^r$, every outstanding lease was signed at $p_{ss}^r$, and by (A.6) the pooled rent equals $p_{ss}^r$ as well. The average rent is then predetermined, $\overline{p}_1^r = p_{ss}^r$, regardless of the distribution of households across lease states.

[12] Goods-market clearing, $Y_t^c = C_t + S_t + G_t$, follows from Walras's law given the other clearing conditions (Appendix A.4).

**Remark 1** (Impact equivalence)**.** Since $\overline{p}_1^r = \overline{p}_0^r = p_{ss}^r$, the rent-inflation identity (10) implies that measured rent inflation coincides with nondurable inflation on impact,

$$\pi_1^{\text{rent}} = \pi_1^c \qquad \Longrightarrow \qquad \pi_1^{\text{hicp}} = \pi_1^c \quad \text{for any } \omega_r. \tag{24}$$

The rent weight in the Taylor rule (23) is therefore irrelevant for the mapping from inflation to the policy rate on impact. Its effects operate only from $t = 2$ onward and, at $t = 1$, through expectations of the future path.

**Rent-inflation decomposition.** Solving the rent-inflation identity (10) for $\pi_t^{\text{rent}}$ and subtracting nondurable inflation gives

$$\pi_t^{\text{rent}} - \pi_t^c = \left(1 + \pi_t^c\right)\left(\frac{\overline{p}_t^r}{\overline{p}_{t-1}^r} - 1\right) \approx \frac{\overline{p}_t^r}{\overline{p}_{t-1}^r} - 1. \tag{25}$$

Measured rent inflation exceeds nondurable inflation exactly when the real average rent grows. A rule that responds separately to $\pi_t^{\text{rent}}$ and $\pi_t^c$ therefore responds to real average rent growth over and above its response to nondurable inflation. Sections 5 and 6 use this decomposition.

**Turnover lag.** A lease ends when it is renegotiated, at rate $1 - \varphi$ per quarter, or when either party changes tenure. In the stationary equilibrium a fraction of about $\varphi\, \xi_{0,ss}\, \xi_{2,ss}$ of leases therefore survives each quarter, where $\xi_{h,ss}$ is the stationary tenure persistence of side $h$, and the average rent inherits movements in the market rent with a mean lag equal to the mean age of outstanding leases (Appendix A.2). Because lease survival responds to tenure decisions, a shock that leads more households to change tenure also shortens that lag. Market rents can move sharply while measured rent inflation responds only gradually.

# 3 Taking the Model to the Data

We take the model to Spanish data at quarterly frequency. Spain is particularly well suited to our analysis because it is an economy with a high homeownership rate, where the margin between renting and owning is active, and because most outstanding mortgages and some newly originated ones are adjustable-rate, as in our model. We interpret the calibrated economy as a Spain-like economy whose central bank sets policy for it alone.

We divide the calibration between externally assigned parameters and seven internally calibrated parameters. We choose $(A^h, \delta_h, \theta)$ to reproduce the Spanish house-value-to-income ratio,[13] residential investment over GDP,[14] and the renter share; the transaction cost $\tau_h$ and the

[13] The target is the mean value of homeowners' main residences divided by mean annual gross household income (ECB Household Finance and Consumption Survey, HFCS). Its model counterpart is the price of a reference dwelling of unit quality relative to annual mean income; see Appendix D.1.

[14] Gross fixed capital formation in dwellings divided by GDP at market prices (Eurostat national accounts). Its

taste-shock scale $\sigma_\varepsilon$ to reproduce the quarterly flows from renting to owner-occupation and from landlord to owner-occupation; $\beta$ to reproduce a 2 percent annual real interest rate; and the labor tax $\tau_y$ to reproduce the ratio of government consumption to GDP. The steady-state lump-sum transfer balances the government budget. All tenure moments refer to households that rent their main residence, own it, or own it and rent out another dwelling.[15] We then assess the model against moments it does not target, among them the remaining entries of the quarterly tenure-transition matrix, estimated from the panel of the EFF, which govern mobility across tenure states. Section 4.2 compares the model's responses of prices and rents to a monetary tightening with our estimates for Spain.

## 3.1 Externally Calibrated Parameters

Table 1 reports the parameters assigned outside the internal calibration.

**Housing and tenure.** We set the housing qualities to $(\tilde{h}_1, \tilde{h}_2) = (0.905, 1.095)$, whose ratio, 1.21, is the ratio of owner-occupied to buy-to-let house prices in Castellanos et al. (2026), and take the housing preference parameter $\alpha_h = 0.5$ from the same paper. We set $J = 20$ to reflect the five-year minimum term of residential leases for individual landlords under current Spanish tenancy law, and approximate the annual rent updates of Spanish leases by full indexation to current inflation (Appendix D.5). We discipline $\varphi$ using rental-turnover evidence. Our calculations using the EFF suggest that approximately 14 percent of market-rent tenants moved into their dwelling during the previous year.[16] With $\varphi = 0.99$, approximately 17 percent of outstanding model leases were signed within the previous four quarters. In the model, a lease that is not ended by a change of tenure ends when the tenant moves to another rental dwelling or the two parties sign a new contract, and in either case the new lease is signed at the market rent. Since every lease pays the same rent in the stationary equilibrium, neither $\varphi$ nor $J$ affects stationary prices or allocations; they change only the age distribution of leases, so both leave the rest of the calibration unchanged.

**Mortgage and financial parameters.** We set the loan-to-value limit to $\lambda_{\text{LTV}} = 0.8$, consistent with the Spanish evidence in Bover et al. (2019). The loan-to-income limit stands in for the affordability checks that lenders apply to new mortgages. Quantitative housing models usually state such checks as a limit on the ratio of payments to income (Kaplan et al., 2020; Greenwald and Guren, 2025); we use the loan-to-income form, which does not tighten mechanically when the interest rate rises. We set it to $\lambda_{\text{LTI}} = 24$ at quarterly frequency, six times annual income, the level above which

model counterpart is $S_{ss}/Y^c_{ss}$, the value of new housing net of land over consumption-good output, which equals model GDP; as in the national accounts, land is excluded from both.

[15] Authors' calculations using the EFF, 2014–2022 waves. Households that neither own their main residence, receive rental income, nor pay rent account for 7.2 percent of all households, and households that receive rental income without owning their main residence for 1.7 percent. The model has no counterpart for either group. Among the remaining households, 17.0 percent pay rent.

[16] Authors' calculations using the EFF, 2014–2022 waves; Appendix D.5 documents the turnover evidence and Spanish lease provisions.

| Parameter | Description | Value | Source / Note |
|---|---|---|---|
| *Housing and tenure* | | | |
| $(\tilde{h}_1, \tilde{h}_2)$ | Housing qualities | $(0.905,\ 1.095)$ | Price ratio, Castellanos et al. (2026) |
| $\alpha_h$ | Housing preference parameter | 0.5 | Castellanos et al. (2026) |
| $\varphi$ | Quarterly non-renegotiation probability | 0.99 | Rental turnover (EFF) |
| $J$ | Quarters a lease pays its own signing rent | 20 | Five-year minimum lease term |
| *Mortgage and financial markets* | | | |
| $\lambda_{\text{LTV}}$ | Loan-to-value limit | 0.8 | 20% down payment; Bover et al. (2019) |
| $\lambda_{\text{LTI}}$ | Loan-to-income limit | 24 | Max. six times annual income |
| $\kappa$ | Mortgage spread | 0.00345 | 1.38 pp annual spread, as in the data |
| $m$ | Minimum amortization | 0.0190 | 90% repaid over 30 years |
| *Preferences and idiosyncratic income* | | | |
| $\gamma$ | CRRA risk aversion | 2 | Standard |
| $\eta$ | Inverse Frisch elasticity | 2 | Standard |
| $\rho_y$ | Quarterly log-productivity persistence | 0.9498 | Flodén and Lindé (2001) |
| $\sigma_y$ | Std. of quarterly log-prod. innovation | 0.104 | Arellano et al. (2022) |
| *Production and market structure* | | | |
| $\alpha_L$ | Land share in housing production | 0.33 | Castellanos et al. (2026) |
| $\epsilon$ | Elasticity across labor varieties | 10 | Standard |
| $\tau_L$ | Employment subsidy | $1/9$ | Offsets the steady-state wage markup |
| $\theta_w$ | Wage Rotemberg coefficient | 778 | Wage Phillips-curve slope of 0.02 |
| *Government and monetary policy* | | | |
| $\phi_\pi$ | Inflation coefficient | 1.5 | Standard |
| $\overline{B}$ | Government debt | 4 | Debt equal to 100% of annual GDP |
| $\Gamma_B$ | Fiscal-rule coefficient | 0.051 | Burriel et al. (2010) |

**Table 1.** Externally Calibrated Parameters

International Monetary Fund (2026) classifies Spanish mortgages as high-LTI loans, so the limit binds only for the most leveraged buyers. We set the mortgage spread to $\kappa = 0.00345$, the sample average of the annualized spread, 1.38 percentage points.[17] We set the minimum amortization requirement to $m = 0.0190$, so that a borrower who retains the same housing state repays 90 percent of the initial balance over 30 years, $(1-m)^{120} = 0.1$.

**Preferences and earnings process.** We set risk aversion to $\gamma = 2$ and the inverse Frisch elasticity to $\eta = 2$. Idiosyncratic labor productivity follows

$$z_{t+1} = \rho_y z_t + \sigma_y \zeta_{t+1}, \qquad z_t \equiv \log y_t, \qquad \zeta_{t+1} \sim \mathcal{N}(0,1).$$

We set annual persistence to 0.8139, the estimate for Sweden in Flodén and Lindé (2001), which gives $\rho_y = 0.9498$ at quarterly frequency. We choose $\sigma_y = 0.104$ using the approximately 0.52 spread between the 90th and 10th percentiles of men's residualized annual log-earnings changes in

[17] Authors' calculations using the ECB Data Portal. The spread is the average, from May 2001 to April 2026, of the difference between the interest rate on new Spanish variable-rate mortgages with an initial rate fixation of up to one year and the one-year Euribor.

2005–2006 reported by Arellano et al. (2022). Appendix D.7 documents the frequency conversion and compares our persistence choice with the wage autocorrelations reported for Spain.

**Production and wage rigidity.** On the production side, we adopt a land share of $\alpha_L = 0.33$ from Castellanos et al. (2026). We set the elasticity across labor varieties to $\epsilon = 10$ and the employment subsidy to $\tau_L = 1/9$, which offsets the steady-state wage markup. We set the wage Rotemberg coefficient so that the slope of the wage Phillips curve with respect to the log labor wedge, $\epsilon\psi_n/\theta_w$, equals 0.02, which gives $\theta_w = 778$.[18] The slope, rather than $\theta_w$ itself, is the object that governs wage dynamics.

**Government, monetary policy, and normalizations.** We set government debt to $\overline{B} = 4$, corresponding to 100 percent of annual GDP.[19] Steady-state government purchases equal the revenue from land permits, transaction costs, and the labor tax, $G_{ss} = p^L_{ss}\overline{L} + \mathcal{T}^h_{ss} + \tau_y w_{ss} N_{ss}$, and we calibrate the labor tax so that $G_{ss}$ equals 19.0 percent of GDP.[20] We set the fiscal-rule coefficient to $\Gamma_B = 0.051$, the posterior mean that Burriel et al. (2010) estimate for Spain for the response of lump-sum transfers to public debt. Half of any gap between debt and its steady-state level then closes in about four years. The steady-state transfer that balances the government budget is $-1.8$ percent of output: households pay a lump-sum tax, because the interest bill on public debt exceeds the revenue from the mortgage spread. We follow Auclert et al. (2025) in setting the inflation coefficient in the monetary policy rule to $\phi_\pi = 1.5$. We normalize consumption-sector productivity, land supply, and stationary mean labor productivity so that $A^c = 1$, $\overline{L} = 1$, and $\mathbb{E}[y] = 1$. We choose the scale of labor disutility $\psi_n$ so that aggregate employment also equals one in the stationary equilibrium. Mean labor income therefore equals one, and we use it as the unit of income throughout.

## 3.2 Internally Calibrated Parameters and Model Fit

Table 2 reports the seven internally calibrated parameters, chosen jointly to hit the targets listed above.

The internally calibrated parameters pin down distinct margins of the stationary equilibrium. The housing parameters $(A^h, \delta_h, \theta)$ jointly govern housing supply, the user cost of housing, and tenure choices. The discount factor $\beta$ governs desired saving and is chosen to deliver a quarterly real interest rate of 0.5 percent, corresponding to 2 percent annualized. The transaction cost $\tau_h$ and the taste-shock scale $\sigma_\varepsilon$ govern mobility across tenure states. The transaction cost dampens moves that require buying or selling a dwelling, while the taste shocks generate moves in every direction, regardless of prices. The flow from renting to owner-occupation responds mainly to

[18] The wage-setting parameters estimated by Galí et al. (2012), at the posterior mode of their specification including unemployment, imply a quarterly slope of approximately 0.022 on the log wage-markup gap.

[19] Spain's general government gross debt stood at 100.7 percent of GDP at the end of 2025, according to Eurostat.

[20] General-government final consumption expenditure over GDP at current prices (Eurostat, `nama_10_gdp`), averaged over 2014–2022 excluding 2020 and 2021, the period of the EFF tenure targets.

| Parameter | Description | Value |
|---|---|---|
| $A^h$ | Housing-sector productivity | 0.0294 |
| $\delta_h$ | Housing depreciation rate | 0.0046 |
| $\theta$ | Homeownership preference shifter | 1.197 |
| $\beta$ | Discount factor | 0.9881 |
| $\sigma_\varepsilon$ | Taste-shock scale | 0.122 |
| $\tau_h$ | Transaction cost | 0.025 |
| $\tau_y$ | Labor tax | 0.150 |

**Table 2.** Internally Calibrated Parameters

the transaction cost, and the flow from landlord to owner-occupation mainly to the taste-shock scale. The calibrated transaction cost, 2.5 percent of the value of the units traded, is an average over tenure changes, many of which involve no market transaction.[21] The labor tax $\tau_y$ finances government consumption.

Table 3 compares the resulting targeted and untargeted moments with their Spanish counterparts; Appendix D documents the data. The calibrated model reproduces the targeted moments in Table 3 by construction.

**Untargeted moments in Table 3.** The level of rents is not targeted directly. The rental burden, the mean across renters of annual rent divided by the household's gross income, is 30.4 percent in the model, compared with 35 percent in Spain.[22] The gross rental yield, the annual rent of a rental dwelling over its price, is 5.1 percent, compared with the 5.6 percent that Khametshin et al. (2024) estimate for Spain from rents declared to the tax authority and registered sale prices, averaged over 2014–2022.

Turning to household balance sheets, the model generates household net worth equal to 6.58 times annual income, compared with 7.33 in Spain.[23] We measure the model's ratio of net worth to annual income as $(A_{ss} + p^h_{ss} H_{ss}) / (4\mathbb{E}[y])$, where $A_{ss}$ denotes aggregate net financial wealth and $H_{ss}$ aggregate housing in dwelling-size units.

The model matches the overall share of property-owning households, although its composition differs from the data. Owner-occupiers account for 66.0 percent of households and landlords for 17.0 percent, compared with 72.9 and 10.0 percent in Spain. The landlord share equals the targeted renter share mechanically because each landlord supplies one rental unit and each renter occupies one unit. In the data, by contrast, a landlord household may supply several rental properties. The discrepancy concerns the number of landlord households; the size of the rental market itself is matched through the targeted renter share. The model also reproduces the share of

[21] In the EFF panel, 88 percent of owner-occupiers who become landlords between two waves buy no dwelling in between: they rent out a dwelling they already owned or had inherited. Similarly, 53 percent of landlords who become owner-occupiers keep the dwelling they used to rent out. Authors' calculations using the EFF, 2014–2022 waves.

[22] Authors' calculations using the EFF, 2014–2022 waves, for households that pay rent; see Appendix D.1.

[23] Household net worth over income is financial assets plus housing wealth minus liabilities, divided by annual gross disposable household income (ECB Distributional Wealth Accounts and Quarterly Sector Accounts).

| Moment | Model | Spain |
|---|---|---|
| *Targeted moments* | | |
| House value / annual income | 5.25 | 5.25 |
| Residential investment / GDP | 6.9% | 6.9% |
| Renter share | 17.0% | 17.0% |
| Government consumption / GDP | 19.0% | 19.0% |
| Renter $\rightarrow$ owner-occupier, quarterly | 1.17% | 1.17% |
| Landlord $\rightarrow$ owner-occupier, quarterly | 2.55% | 2.55% |
| *Untargeted moments* | | |
| Rental burden (renters paying rent) | 30.4% | 35.0% |
| Gross rental yield (annual) | 5.1% | 5.6% |
| Household net worth / annual income | 6.58 | 7.33 |
| Owner-occupier share | 66.0% | 72.9% |
| Landlord share | 17.0% | 10.0% |
| Landlords with net debt | 32.8% | 34.0% |
| Owner-occupier $\rightarrow$ landlord, quarterly | 0.66% | 0.61% |
| Average quarterly tenure persistence | 0.984 | 0.985 |
| Other tenure transitions | Figure 1 | |

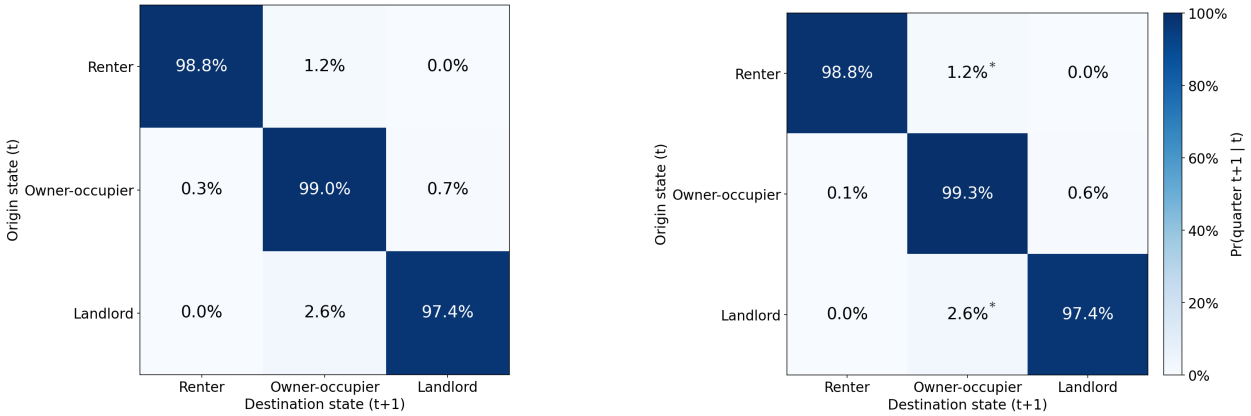
**Table 3.** Targeted and Untargeted Model Moments

**Note.** The 2 percent annual real interest rate is imposed as a calibration restriction. Spanish entries are sample averages. Tenure shares and transition probabilities are computed from the EFF for households that rent their main residence, own it, or own it and rent out another dwelling. In the model, the house value refers to a dwelling of unit quality. The rental burden is the mean across renters of rent relative to the household's own income, and the rental yield is the annual rent of a rental dwelling over its price. Landlords with net debt are landlords whose financial assets fall short of their debts. Sources and construction are documented in Appendix D.

landlords in debt, which governs how mortgage costs pass through to rental supply: 32.8 percent of landlords are net debtors in the model, against 34.0 percent in the EFF.[24] Owner-occupiers carry less debt than in the data, 9.9 against 40.7 percent net debtors, mainly because two-thirds of new owner-occupiers in the model are landlords who sell their rental unit and repay their mortgage (Appendix D.6). Of the owner-occupiers who are net debtors and keep their home, 47 percent pay exactly the minimum installment, which moves one for one with the mortgage rate.

**Wealth distribution.** The upper tail of the model's wealth distribution is less concentrated than in the data, with 18.1 against 53.2 percent of net worth held by the top decile, because households share a common discount factor; heterogeneity in patience would bring it closer (Auclert et al., 2025). The model does reproduce who is hand-to-mouth in Spain, where such households are predominantly wealthy homeowners rather than poor (Kaplan et al., 2014): owner-occupiers and landlords with housing wealth but no positive net financial wealth are 16.0 percent of households, against 15.2 percent in Spain, although the model has fewer poor hand-to-mouth households, since renters cannot borrow. Landlords are the wealthiest tenure group in both, but the model

[24] Authors' calculations using the EFF, 2014–2022 waves, for landlords who own their main residence. A household is a net debtor when its financial assets fall short of its debts, the counterpart of $a < 0$ in the model.




**(a)** Model **(b)** Data

**Figure 1.** Quarterly Tenure-Transition Matrices

**Note.** Panel (b) excludes rent-free households and households that receive rental income without owning their main residence. Rows denote a household's current tenure state, columns denote its tenure state in the following quarter, and entries report the corresponding conditional transition probabilities. Asterisks mark the two transition probabilities used as calibration targets. Panel (a) reports the model matrix, computed from stationary joint tenure flows. Panel (b) reports the data matrix, constructed from households observed in consecutive EFF waves between 2014 and 2022. Each wave-pair matrix is converted to quarterly frequency, and the resulting matrices are pooled using origin-state household counts. Both panels use the same color scale, shown in Panel (b).

compresses the distance between them and their tenants, so it understates the redistributive value of rent relief (Appendix D.3).

**Tenure-transition fit.** Figure 1 compares the quarterly tenure-transition matrix in the model with the one we estimate from the EFF panel (Appendix D.4). Two off-diagonal entries are calibration targets, and the rest test mobility across tenure states: the model reproduces every entry within sampling error except the flow from owner-occupation to renting, which it overstates, and hence the owner-occupier diagonal, which it understates. Two diagonal cells matter most for rent dynamics: the renter and landlord persistence probabilities, 0.9883 and 0.9745 in the model against 0.9880 and 0.9744 in the data, set the retention rates $\xi_{0,ss}$ and $\xi_{2,ss}$ that govern lease survival in (A.4), and hence the mean age of outstanding leases, 22 quarters, which is the mean lag of the measured rent index.

# 4 Monetary Tightening and the Rent Channel

We begin with the benchmark monetary experiment to highlight the transmission mechanism of monetary policy to rent inflation in our model. Throughout this section, the central bank responds only to nondurable inflation, so that $\omega_r = 0$ in the Taylor rule (23). In this environment, a monetary tightening raises the real rate and mortgage costs. Some households delay their move

into ownership and keep renting, and supplying rental housing becomes less attractive, so market rents rise even as aggregate consumption and output fall; the second force is the larger one. We first describe the aggregate response to the monetary shock, then study the housing and rental market mechanism, decompose the rent response into its channels, and finally quantify the consumption incidence of the tightening across households.[25]

## 4.1 A Contractionary Monetary Policy Shock

The experiment starts from the calibrated stationary equilibrium and applies a 25-basis-point annualized innovation to $\varepsilon_t^m$ in the Taylor rule (23). The shock follows an AR(1) process with persistence $\rho_m = 0.7$, which implies a half-life of about two quarters. Figure 2 plots the policy shock, nondurable inflation, the real interest rate, and aggregate real activity.

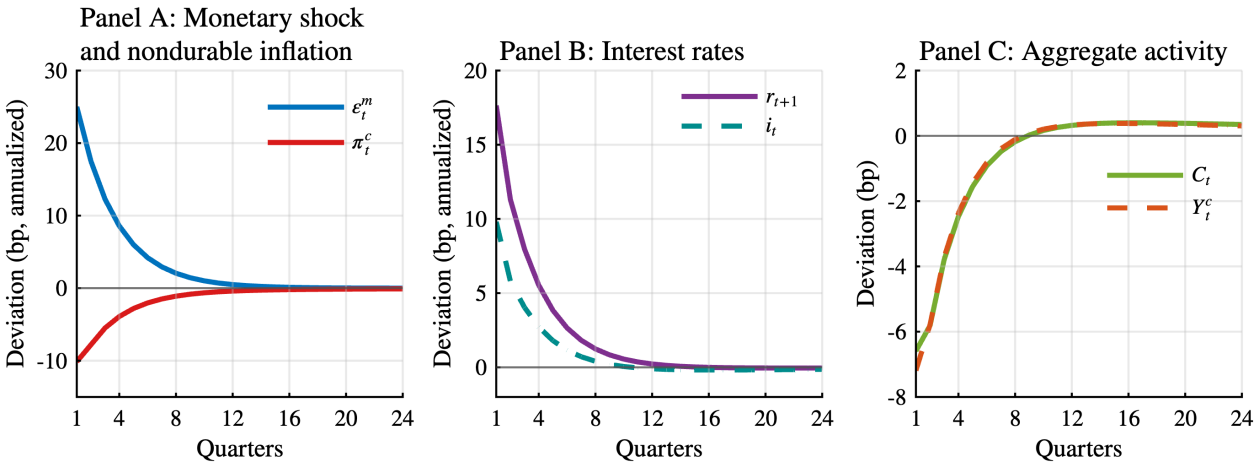



**Figure 2.** Aggregate Response to a Contractionary Monetary Shock

**Note.** All panels report deviations in basis points. Panels A and B annualize the monetary policy innovation, nondurable inflation, and interest rates. Panel C reports relative deviations of aggregate consumption and nondurable output from steady state.

The shock contracts real activity and lowers nondurable inflation. On impact, aggregate consumption falls by 6.6 basis points and nondurable output by 7.2 basis points, while nondurable inflation declines by about 10 annualized basis points. The real interest rate rises by 17.6 annualized basis points on impact, so it absorbs 71 percent of the monetary innovation, comparable to the roughly 60 percent in Kaplan et al. (2018). The policy rate itself rises by only 10 basis points, since the rule responds to the fall in inflation; the real rate rises by more because expected inflation falls.

## 4.2 The Housing and Rental Market Response

The housing-market response is concentrated in rents. Panel A in Figure 3 shows that the rent on new leases, the market rent $p_t^r$, rises as soon as leases are signed after the shock: it is 31 basis points

[25] Castellanos et al. (2026) establish these two forces in a life-cycle model with an exogenous real interest rate.

above steady state in the second quarter and declines gradually afterward, without falling below its steady state. Because a new lease fixes its real rent for up to five years, the market rent is the price of a multi-year contract: it rises by less, and stays high for longer, than it would if leases reset every quarter. The real average rent responds gradually as leases reset (Section 2.9). Because only a fraction of leases reset each quarter, the increase reaches tenants with a delay: the average rent peaks at 4 basis points above steady state in the eighth quarter, when most of the innovation has already faded, and in the twelfth quarter it is still at 93 percent of its peak.[26]

Panel B shows that the house price falls by 11 basis points on impact. With land and construction-sector productivity fixed, the housing-supply block maps this price decline into a fall of 22 basis points in new housing production.[27] The tenure distribution is predetermined on impact, so the house-price decline is forward-looking: it reflects the anticipated reallocation shown in Panel C. Because rental units are smaller than owner-occupied homes, the anticipated shift out of owner-occupation lowers desired holdings of the housing stock. With the stock itself nearly fixed in the short run, this shift appears almost entirely in the price: aggregate housing falls by less than 0.3 basis points over the transition, compared with a decline of 11 basis points in the house price. Section 4.3 shows that the drop in house prices mitigates the increase in rents.

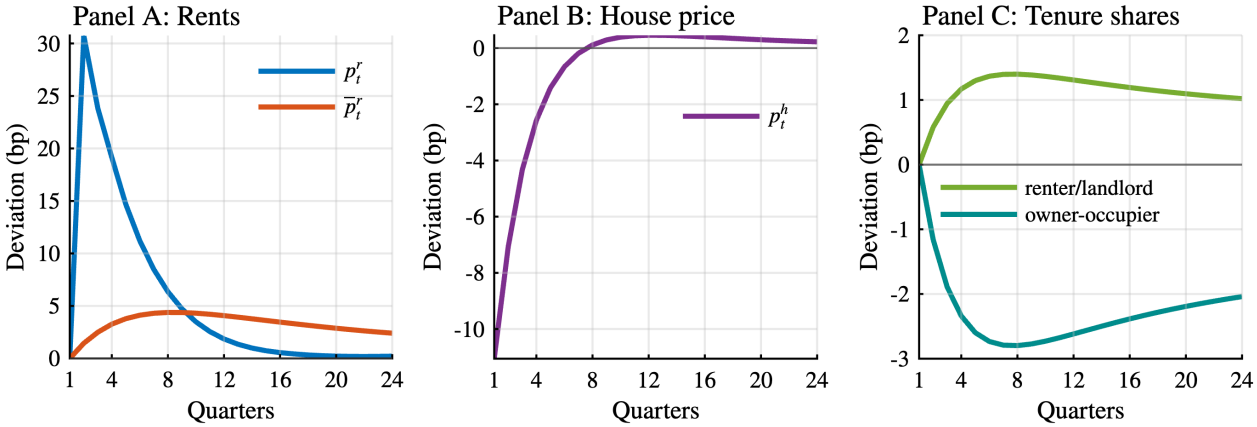



**Figure 3.** Rental and Housing Market Response to a Contractionary Monetary Shock

**Note.** All panels report deviations in basis points. Panel A reports relative deviations of the real market rent, at which new leases are signed, and the real average rent from steady state. Panel B reports the corresponding relative deviation of the real house price. Panel C reports absolute changes in tenure shares.

[26]The length of the period during which a lease keeps its signing rent governs the shape of this response. Figure F.1 in Appendix F.2 repeats the experiment for $J$ between one quarter and five years. With $J = 1$, a new lease pays the market rent for a single quarter, so the market rent must move by more to clear the rental market: it rises more than five times as much in the second quarter and falls below its steady state after a year and a half, and the average rent peaks in the fifth quarter at 15 basis points and is back to 43 percent of its peak in the twelfth quarter.

[27]Combining the construction technology (15) with the construction firm's problem (16) and fixed land supply gives

$$Y_t^h = (A^h)^{1/\alpha_L}\left((1-\alpha_L)p_t^h\right)^{(1-\alpha_L)/\alpha_L}\overline{L}.$$

Hence, conditional on construction-sector productivity, the elasticity of new housing production with respect to the house price is $(1-\alpha_L)/\alpha_L$.

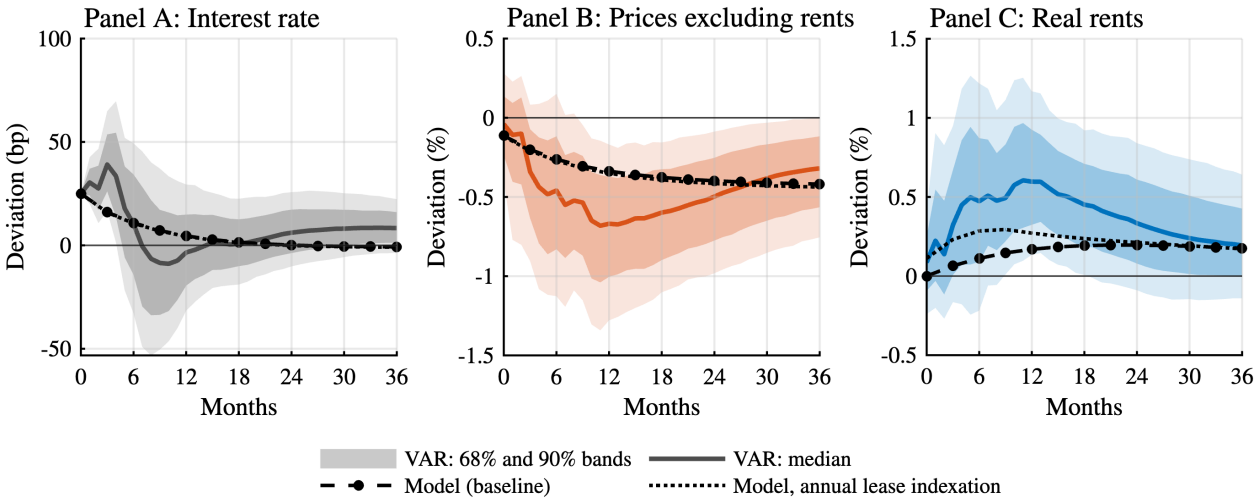



**Figure 4.** Monetary Tightening in Spain: Model and Evidence

**Note.** Responses to a monetary tightening that raises the one-year rate by 25 basis points on impact. VAR: Spain, monthly, 2005–2019; median of 2,000 bootstrap draws with 68 and 90 percent bands (Appendix E). Panel A reports the euro-area one-year rate and, for the model, the annualized deviation of the one-year nominal yield, $\sum_{k=0}^{3}(i_{t+k} - r_{ss})$. Panel B reports the HICP excluding rents and the model's price of the consumption good. Panel C reports real rents, the HICP index of actual rentals relative to the HICP excluding rents, and the model's real average rent $\overline{p}_t^r$. Model responses are scaled so that the one-year yield rises by 25 basis points in the impact quarter. The dotted line updates the rents of continuing leases once a year by the previous year's inflation (Appendix F.10).

Panel C shows the reallocation that generates the rent response. Because mortgages reprice every quarter, the higher real rate raises the cost of servicing existing debt and financing new purchases in the budget constraint (3). Lower labor income also tightens the income limit in (5). Households close to the ownership margin therefore become less likely to enter owner-occupation, raising the renter share and lowering the owner-occupier share. Rental-market clearing requires an equal increase in landlords. Because constrained households cannot take these positions, the market rent rises until households with sufficient balance-sheet capacity are willing to become landlords. The model also reproduces two qualitative features of the U.S. estimates in Dias and Duarte (2022), lower homeownership and a higher rent-to-price ratio after a monetary tightening, and its first-year rental-supply elasticity is close to the low end of available estimates (Appendices F.1 and F.4).

To validate the model results against empirical evidence, we estimate a monthly vector autoregression for Spain over 2005–2019, in which the monetary shock is identified with high-frequency changes in the one-year swap rate around ECB announcements (Altavilla et al., 2019) and normalized to raise the one-year rate by 25 basis points on impact (Appendix E). We use the rental component of the HICP as our measure of rents. Figure 4 compares the estimated responses with those of the model, scaled so that the model's one-year nominal yield also rises by 25 basis points on impact.

Although none of these responses is targeted in the calibration, the model reproduces their main features. In the data, consumer prices excluding rents fall by about 0.7 percent within a

year of the tightening, while real rents rise by about 0.6 percent and remain above their pre-shock level for about two years. In the model, prices fall by about half as much, within the estimated 90 percent bands, and the real average rent rises gradually as leases reset, with the sign of the estimated response. If anything, the model understates the rise in real rents after a monetary tightening. This is partly because, in the model, continuing leases keep the real rent at which they were signed, so the nominal rent falls with the price level, while in the data rents tend to be nominally stickier. When the rents of continuing leases are instead updated once a year by the previous year's inflation, as is common in Spanish contracts, nominal rents stay flat for the first three quarters and the real average rent peaks within the first year, close to the timing of the estimated peak and at about half its size. We keep full indexation in the baseline because it isolates the rent channel: the average rent then moves only through the market rent on new leases. In Appendix F.10 we report further results for the indexation closest to Spanish lease provisions.

### 4.3 Rent Decomposition

In our economy, a tightening affects rents through several channels. On the renter side, it makes it harder for households to get a mortgage, which increases their rental demand. On the landlord side, higher mortgage costs reduce rental supply, and higher returns on financial assets, the alternative investment for potential property owners, may reduce it further. A tightening also reduces labor income through the standard HANK general-equilibrium channels, which may also affect rental demand. Lower house prices push the other way, making housing cheaper to buy.

To separate these channels, we compute excess rental supply in a set of counterfactual scenarios in which we keep all prices constant at their steady-state levels after the monetary shock, with the exception of one, which we vary in turn. For example, in one of the experiments, we change the mortgage rate paid by indebted landlords, while keeping the mortgage rate paid by owner-occupiers, the return on financial savings, labor income, taxes and transfers, the house price, and rents constant. By feeding in the mortgage rate's equilibrium path with all other prices unchanged, we measure the contribution of this channel to rental-market excess supply, i.e., the difference between the amount of housing that landlords are willing to rent out and the amount of rental housing that renters demand. We then do the same with all other prices.[28] Although there are interactions between channels, to first order these single-input responses sum to the equilibrium response. We also report the direct effect of rents. Rents adjust to clear the disequilibrium in the rental market generated by the combined pressure of all the other channels, so that the total effect reported in the decomposition adds up to zero excess supply and the rental market clears. A channel that contracts excess supply therefore contributes to raising rents.

Figure 5, Panel A, shows that reduced rental supply contracts excess supply by about twice as

[28] In equilibrium the three interest-rate inputs comove through the real rate $r_t$ and the mortgage wedge $\kappa$ in the budget constraint (3); the decomposition perturbs them separately to attribute the rent response across the savings, owner-mortgage, and landlord-mortgage margins. Appendix B states the decomposition formally, defines each input, and reports the resulting contributions to excess supply and to the market rent. The equilibrium paths fed into the household problem are those of Figures 2 and 3.

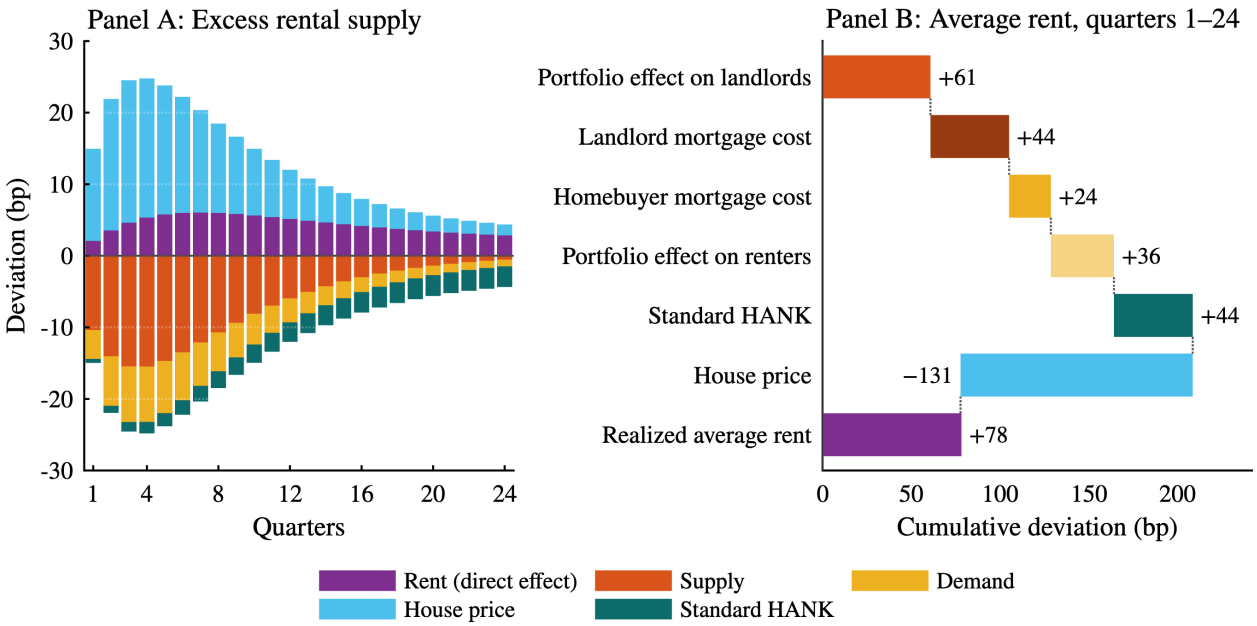



**Figure 5.** Decomposition of the Rent Response

**Note.** Panel A reports each group's contribution to rental-market excess supply, the landlord share minus the renter share, in basis points. Panel B reports each channel's contribution to the relative deviation of the real average rent from steady state, summed over quarters 1–24, in basis points; the portfolio effect on landlords includes that on owner-occupiers. Each channel feeds one equilibrium input path through the household problem, holding the remaining inputs at their steady-state values. Supply groups the mortgage rate on landlord debt and the return on financial savings of landlords and owner-occupiers; demand groups the mortgage rate on owner-occupier debt and the return on savings of renters; standard HANK groups labor income and taxes. Table B.1 reports each channel's contribution to excess supply and to the market rent.

much as increased demand, so supply accounts for about two-thirds of the cumulative contraction generated by the interest-rate channels over the first 24 quarters. The fall in house prices partially offsets these two effects. Housing becomes cheaper to buy for prospective landlords and homebuyers alike, encouraging rental supply and easing access to homeownership. Over the same horizon, this channel offsets more than half of the upward rent pressure generated by the other channels.[29]

The rental market clears through a rise in rents that induces additional supply. On impact, more owners become landlords, more than offsetting a slight increase in exits by incumbent landlords (Appendix F.5). The additional entrants are high-asset households, with net financial assets more than three times those of the average owner-occupier. However, most (86 percent) borrow to buy the rental unit. Expected higher future rents compensate them for the forgone return on financial savings and for the higher cost of that mortgage.

Figure 5, Panel B, digs deeper into the specific channels determining rental supply and

[29]The sign of the rent response depends on the rental yield, which the calibration ties to depreciation through the residential-investment target. If we instead calibrate to lower yields, the house price absorbs more of the tightening. Its impact response is approximately inversely proportional to the yield, as for an asset whose duration is the price–rent ratio, and real rents fall once the yield is below about 3.8 percent, against 5.1 percent in our calibration and 5.6 percent in Spanish data (Appendix F.3).

demand, and translates their contributions into the average rent paid by renters. Two separate channels determine the drop in rental supply. First, at unchanged rents, a higher return on financial savings raises the opportunity cost of holding rental property (portfolio effect on landlords), so incumbent landlords lean toward exit. This channel raises the average rent by 61 basis points. Second, higher mortgage costs make it more expensive to buy and hold a rental unit with debt (landlord mortgage cost), so owners who would need a mortgage to become landlords stay out. Because the channel works through entry rather than through exit by levered incumbents, it would also operate under fixed-rate mortgages to the extent that policy-rate shocks move the rates on new mortgages.[30] This channel is smaller than the former but still relevant: it raises the average rent by 44 basis points.

On the demand side, a higher rate on owner-occupier mortgage debt raises the cost of being a levered owner and lowers the value of becoming one (homebuyer mortgage cost), so would-be buyers stay renters and the pool of renters grows. A higher return on savings also keeps renters renting (portfolio effect on renters), and the standard HANK channel of lower labor income and higher taxes adds to this pressure.[31]

### 4.4 Consumption Incidence across Households

The consumption effects of the tightening depend on both mortgage debt and rental exposure. We measure them by following cohorts defined by tenure and net financial assets at the date of the shock, relative to the same initial cohorts in a no-shock counterfactual.

Mortgagor-owners experience the largest impact contraction as they anticipate higher debt service; levered landlords also cut consumption more than their unlevered counterparts. Renters face an additional housing-cost burden that reduces the resources available for nondurable consumption—the renter cohort's consumption falls more on impact and recovers less strongly than that of unlevered owners and landlords. Sticky leases delay the increase in rental payments, but households already adjust consumption in anticipation of rising rents. We report detailed consumption responses in Appendix F.6.

## 5 Rent in the Inflation Target

So far, we have focused on the case in which the central bank does not react to rent inflation. We now move to the case in which the policy index in rule (23) includes rent inflation with weight $\omega_r$.

[30] Holding all other inputs at their steady-state paths, the higher rate on landlords' debt lowers entry by 1.9 basis points of the population on impact, all of it among owners who would have been net debtors after entry. Over quarters 1–24, lower entry accounts for more than the entire contribution of the channel to rental supply: exits by levered incumbents rise on impact but fall afterward, as fewer levered landlords enter, and on net they slightly raise supply.

[31] The portfolio effect on renters is the demand counterpart of the effect on landlords: a higher return on savings makes remaining a renter, with wealth in the liquid asset, more attractive than buying a house. We group the portfolio effect on owners with supply because both of its margins concern becoming a landlord: the opportunity cost that deters owners from becoming landlords, and the wealth effect of higher interest income, which over time makes becoming a landlord more affordable. The two nearly offset.

Rent inflation enters with its share of household spending on consumption and rent in the steady state, 5.3 percent.[32] The rent component is nominal average rent inflation, $\pi_t^{\text{rent}}$, which reflects the rents paid on leases of every vintage. Because the experiment shocks the rule itself, any term that rises after a tightening amplifies it. The rent term is informative because it rises late and its additional tightening falls on the housing market. The experiment therefore describes how the rent term changes the transmission of a tightening, not whether the rule is desirable; Section 6 studies the optimal response to rent inflation under structural shocks.

Three lessons emerge. First, a rule that responds to rent inflation keeps tightening after the innovation has faded, because measured rents rise only as leases turn over; at HICP weights it adds a fraction of a basis point to the policy rate. Second, the additional contraction operates mainly through lower house prices and higher rents. Responding to rent inflation raises the cost of supplying rental housing and lowers house prices; over the first two years, these housing channels account for the additional fall in consumption. Third, how owner-occupied housing enters the index matters more than the weight on rents: rental equivalence more than triples the feedback relative to the model's rent share, and the net-acquisitions approach shifts the tightening to the housing recovery.

## 5.1 Policy Feedback under a Rent-Inclusive Rule

A rule that includes rent inflation responds to the lagged consequences of its own tightening: the additional tightening follows the rise in rents rather than the monetary innovation, and it persists after the innovation has faded. Table 4 summarizes the responses, and Appendix F.7 plots them and reports Spain's weight.

The feedback operates through the gradual transmission of market rents into rental payments. On impact, the average rent is predetermined, so the policy index coincides with nondurable inflation for any rent weight (Remark 1).[33] Once leases reset at the higher market rent, measured rent inflation exceeds nondurable inflation, so the rule keeps the policy rate above its benchmark path. The higher rate raises landlords' financing costs and the return on their savings, the supply channels of Section 4.3, so rents must rise further to keep rental housing supplied: the rule reinforces the rent pressure of its own tightening.

How long the reinforcement lasts depends on the path of real rents. Summing (25) over time, and using the fact that the average rent is predetermined on impact, the rent term accumulated in

[32] The share exceeds Spain's HICP weight on actual rentals, 3.1 percent over 2014–2022, because the model compresses the dispersion of rental burdens: aggregate rent is 28 percent of aggregate renter income in the model and 20 percent in the EFF, and at the data's aggregate burden the model's share would be 3.9 percent. Appendix D.8 documents the weights and their construction, and Appendix F.7 reports the responses at Spain's weight.

[33] If continuing leases are indexed to past rather than current inflation, as Spanish leases are updated once a year by the previous year's change in consumer prices, measured rent inflation is close to zero on impact while nondurable inflation falls, so the rent term raises the policy rate even on impact, by 0.6 to 0.7 basis points under the model's rent share. The additional tightening stays below one basis point, still lowers house prices, and buys no additional disinflation (Appendix F.10).

| | Benchmark (no rents) | Actual rentals | Owner-occupied housing: Rental equivalence | Owner-occupied housing: Net acquisitions |
|---|---|---|---|---|
| Real interest rate | + | +2.3 | +8.3 | −0.2 |
| Real house price | − | −3.1 | −11.5 | −8.6 |
| Real average rent | + | +3.4 | +12.5 | +14.4 |
| Consumption | − | −2.7 | −9.9 | −9.4 |
| Nondurable inflation | − | 0.0 | +0.3 | −5.6 |

**Table 4.** Rent-Inclusive Targets: Responses Relative to the Benchmark Rule

**Note.** Responses to the monetary policy shock of Section 4.1, cumulated over quarters 1–8. The first column gives the direction of the cumulative deviation from steady state under the benchmark rule, whose index excludes rents ($\omega_r = 0$). Each other entry is the difference between the cumulative deviation under the rule in the column and under the benchmark rule, in percent of the benchmark's absolute cumulative deviation; a negative entry means the variable is lower than under the benchmark rule. Actual rentals enter the index at the model's rent share, 5.3 percent. Rental equivalence weights owner-occupied housing at 13 percent and net acquisitions at 9 percent, with actual rentals at the model's share scaled by $1 - \omega_o$. Appendix F.7 reports Spain's weight and the first five years.

the policy index up to quarter $t$ is

$$\sum_{k=1}^{t} \left( \pi_k^{\text{hicp}} - \pi_k^c \right) \approx \omega_r \ln\left( \frac{\overline{p}_t^r}{p_{ss}^r} \right), \tag{26}$$

the rent weight times the log deviation of the real average rent from its steady state. The rent term adds to the policy index while the real average rent rises and subtracts from it only as the average rent falls back toward its steady state. The average rent rises for eight quarters, as long as new leases are signed above it, and returns slowly because the mean age of outstanding leases is 22 quarters. The real interest rate accordingly stays above its benchmark path from the second to the thirteenth quarter. The additional tightening reaches at most 0.3 basis points on the policy rate after a 25-basis-point innovation. The length of the period at the signing rent decides this timing: when a new lease pays the market rent for a single quarter ($J = 1$), real rents peak and return sooner, and the rule front-loads the tightening instead, gives it back from the third year, and reduces the cumulative fall in consumption over five years (Appendix F.2).

Households anticipate the higher path of rates, so the rent term affects the economy before the rent increase reaches the index. Consumption falls by more on impact under the rent-inclusive rule, although the impact real interest rate is slightly lower.[34]

## 5.2 The Costs of Responding to Rent Inflation

Responding to rent inflation deepens the consumption contraction without delivering additional nondurable disinflation over the first two years, and the additional contraction runs through the housing market. The differences grow with the weight on rents: under the model's rent share they are nearly twice as large as under Spain's HICP weight (Table F.3), and they keep growing almost in proportion to the weight up to $\omega_r = 0.15$ (Appendix F.8). The higher path of real rates lowers house prices further, and lower house prices account for most of the additional fall in consumption; higher rents and lower transfers add to it, while the higher real rates themselves lower consumption only on impact.[35] Nondurable inflation, in contrast, falls by the same amount over two years as under the benchmark rule.[36] Appendix F.9 reports how the additional contraction is distributed across households.

Over five years, the later easing of real interest rates relative to the benchmark is too small to offset the earlier additional tightening. The real interest rate falls below its benchmark path only from the fourteenth quarter, and then by about a hundredth of a basis point, so over five years, under the model's rent share, the cumulative fall in consumption is still 1.4 percent larger than under the benchmark rule, house prices fall by more, and the cumulative rise in the rent tenants pay is 3.1 percent larger. Cumulative nondurable disinflation is larger than under the benchmark rule only until the eighth quarter, while households anticipate the higher rates; by the end of the second year the rent term has bought no additional disinflation, and over five years nondurable inflation falls by 2.7 percent less.

De Llanos (2026) shows, in an estimated model of the United States, that rents rise after a tightening because constrained households shift toward renting and housing is costly to convert to rental use, and that tightening in response to rising rents reinforces that shift. Our economy adds household landlords, whose cost of holding a rental unit rises with the policy rate: the rule's additional tightening raises rents further, and it reaches households mainly through house prices. The feedback scales with the weight of housing in the index, which is several times larger in the U.S. consumer price index, where rents are imputed to owners, than in Spain's HICP.

[34] In a first-order decomposition of the impact consumption difference by household input, the anticipated real rates account for more than a third of the additional fall, and lower house prices, labor income, and transfers, together with higher rents, account for the rest.

[35] In a first-order decomposition of the consumption difference by household input, cumulative consumption over quarters 1–8 under the model's rent share is lower than under the benchmark rule by 0.0058 percent of steady-state consumption. House prices contribute $-0.0057$, rents $-0.0022$, transfers $-0.0016$, labor income $-0.0005$, and real interest rates $+0.0043$.

[36] With interest-rate smoothing of 0.8 and a 25-basis-point annualized innovation, responding to rent inflation still lowers house prices, raises rents, and deepens the consumption contraction relative to the inertial benchmark. These additional effects are smaller than without smoothing. The rule delivers 0.5 percent more nondurable disinflation than the inertial benchmark over two years and 1.3 percent less over five years, under the model's rent share.

### 5.3 Owner-Occupied Housing in the Index

How owner-occupied housing enters the index matters more for the feedback than the weight on actual rentals. The HICP excludes the cost of owner-occupied housing; the ECB's 2021 strategy review concluded that including it, through the net-acquisitions approach, would better represent the inflation rate relevant for households (European Central Bank, 2021; Eiglsperger et al., 2022), and the 2025 assessment reaffirmed the case (European Central Bank, 2025). The alternative, rental equivalence, imputes to owners the rent their home would fetch, as the U.S. CPI does. With owner-occupied housing, the index becomes $(1-\omega_o)\left[(1-\omega_r)\pi_t^c + \omega_r \pi_t^{\text{rent}}\right] + \omega_o \pi_t^o$, where $\pi_t^o$ is nominal average-rent inflation under rental equivalence and nominal house-price inflation under our net-acquisitions approximation.[37] The weight $\omega_o$ is 13 percent under rental equivalence, about the imputed-rent share of Spain's household consumption, and 9 percent under net acquisitions, the euro-area estimate.[38]

Under rental equivalence, the composite raises the weight on rent inflation to 18 percent, and every response in Table 4 other than nondurable inflation is between three and four times as large as under the model's rent share. Under net acquisitions, the index incorporates house prices, which fall on impact and recover while rents are still rising. The rule therefore eases on impact and tightens during the recovery, when it lowers house prices again and reinforces the rise in rents. It is when house prices recover that the inflation term tightens, taking back what it gave on impact; a term on the level of the real house price would instead keep easing until house prices return to their steady state, partly offsetting the tightening (Appendix F.17). Only an index that excludes housing costs avoids the feedback altogether: how owner-occupied housing is measured decides not only how much the rule responds to housing, but also when. Appendix F.7 reports the responses. Under the structural shocks of Section 6, rental equivalence also costs the most welfare among the composites measured in inflation (Tables 5 and F.10).

## 6 The Optimal Response to Rent Inflation

The previous section showed that putting rent inflation directly into the policy index changes the timing and incidence of monetary transmission. We now turn to the normative question: how should a central bank that maximizes welfare treat rent inflation?

To do so, we let the central bank choose the coefficient on nominal rent inflation in an additive Taylor rule,

$$i_t = r_{ss} + \phi_\pi \pi_t^c + \phi_r \pi_t^{\text{rent}}, \tag{27}$$

and report gains relative to $\phi_r = 0$, the rule that responds only to nondurable inflation. The rule

[37] We proxy net-acquisitions inflation by nominal house-price inflation; the official index also includes acquisition and ownership costs and restricts dwelling purchases to those entering the household sector.

[38] The imputed-rent share averages 13.5 percent in Spain over 2014–2022; the 9 percent net-acquisitions weight follows Eiglsperger et al. (2022). All remaining HICP weights are scaled by $1-\omega_o$.

incorporates nominal average rent inflation $\pi_t^{\text{rent}}$, the object observed in a consumer price index.[39]

We evaluate welfare after four shocks, two related to demand and two related to supply, that raise rents. For all four, the welfare-optimal response to rent inflation is negative: the rule should ease when rent inflation rises, relative to a rule that ignores rents. We first introduce the structural shocks used to evaluate policy, then characterize optimal simple rules, and finally decompose their welfare gains across cohorts and channels.

## 6.1 Structural Shocks

We evaluate policy under four shocks that raise rents through different mechanisms. Two are general demand shocks, to household patience and to government spending. They expand activity and raise nondurable inflation, and rents rise because the rule tightens in response. Two are supply shocks. The rental supply shock raises rents at the source, by making it costlier to finance a rental unit, with little nondurable inflation or monetary response. The cost-push shock raises nondurable inflation outside housing; rents again rise through the rule's tightening, but with activity contracting rather than expanding. Together, the four shocks distinguish whether rent pressure originates in the rental market or in the policy response to inflation elsewhere, and whether it comes with a boom or a contraction. Each shock is a one-standard-deviation innovation with the persistence estimated in previous work or on Spanish data, as we detail in Appendix D.9. Appendices F.11, F.12, F.13, and F.14 report the impulse responses.

### 6.1.1 Demand Shocks

A patience shock lowers the discount factor $\beta_t$ by 0.0016 on impact, and a government-spending shock raises purchases $G_t$ by 0.72 percent, financed by debt and, through the fiscal rule (22), by lower transfers over the following years. In both cases, as in a standard demand expansion, output, labor, and nondurable inflation rise (Clarida et al., 1999; Galí, 2015), and the benchmark Taylor rule raises the policy rate, transmitting the expansion to housing decisions through borrowing costs and returns on savings.

Much of the rise in rents comes from the tightening rather than from the expansion itself. Higher mortgage rates keep would-be buyers renting, and higher mortgage rates and returns on savings make supplying rental housing less attractive, through the channels of Section 4.3. Higher labor income works in the opposite direction by relaxing the loan-to-income limit in (5). Under the patience shock, lower patience also reduces the incentive to save for a down payment and lowers the discounted benefits of owning in the future, which adds to the shift toward renting. Under

[39] A rule on the gap of the real average rent from its steady state, $(\overline{p}_t^r - p_{ss}^r)/p_{ss}^r$, also calls for easing under the government-spending, rental supply, and cost-push shocks; under the patience shock, real rents peak after the expansion has passed, and the optimal coefficient is slightly positive. Its gains are smaller than those of the inflation rule after the two demand shocks and, summed over the four shocks, about two-thirds as large. They are larger after the two supply shocks, most of all after the rental supply shock, 0.68 basis points against 0.11, because the rent gap there is created by the shock itself and the level term keeps easing while rents stay high. Appendix F.16 reports the results.

the benchmark rule, the real average rent peaks at about 1.1 percent above steady state after the patience shock and 0.13 percent after the government-spending shock.

### 6.1.2 Supply Shocks

The rental supply shock originates in the rental market. Households in the landlord state who carry mortgage debt pay a spread $\kappa_t^L$ in place of $\kappa$, 22 annualized basis points higher on impact, without any change in owner-occupiers' borrowing costs or in the return on savings; the additional spread revenue accrues to the government (Appendix A.4). At unchanged rents, higher financing costs discourage owners from becoming landlords and encourage indebted landlords to exit. The resulting contraction in rental supply raises rents and shifts households toward owner-occupation. Under the benchmark rule, the real average rent peaks at about 0.39 percent above steady state, while nondurable inflation and the policy rate move little: the shock generates rent pressure with almost no monetary response.

The cost-push shock originates outside the housing market. A transitory fall in the elasticity across labor varieties in the wage Phillips curve (12) raises the desired wage markup by 0.9 percent on impact, and with it wage and nondurable inflation, without directly changing housing costs or borrowing conditions. Under the benchmark rule, nondurable inflation rises by about 0.15 annualized percentage points on impact, the real interest rate rises, and consumption and the house price fall. As after a monetary tightening, higher rates raise rents through the financing and portfolio margins of Section 4.3: the real average rent peaks at about 0.03 percent above steady state. The rent pressure therefore originates, as under the demand shocks, in the rule's own response to inflation, but here inflation rises while activity contracts.

## 6.2 Optimal Simple Rules

We evaluate welfare conditional on a one-time, unanticipated MIT innovation to each structural shock and choose $\phi_r$ to maximize welfare over the resulting transition, in the spirit of the optimal simple rules of Schmitt-Grohé and Uribe (2007),

$$\phi_r^* \in \arg\max_{\phi_r} \mathcal{W}(\phi_r), \qquad \mathcal{W}(\phi_r) = \sum_{t=1}^{\infty} \widetilde{\beta}_t \left[ \int u(c_{it}, h_{it})\, di - v(N_t) - \frac{\theta_w}{2} (\pi_t^w)^2 \right]. \tag{28}$$

Whenever we evaluate welfare, we adopt an equal-weighted utilitarian criterion.[40] The discount factor satisfies $\widetilde{\beta}_1 = 1$ and $\widetilde{\beta}_t = \prod_{\ell=1}^{t-1} \beta_\ell$.[41]

The benchmark is $\phi_r = 0$, under which policy responds only to nondurable inflation. Thus, the exercise asks whether policy should tighten or ease when nominal rent inflation rises, relative

[40] The objective excludes the type-I extreme-value taste shocks $\sigma_\varepsilon \varepsilon(h')$ that smooth the discrete tenure choice (Iskhakov et al., 2017). Their scale is calibrated to tenure flows, not to any preference moment; we treat them as a smoothing device of the solution method, with no welfare content.

[41] The planner discounts with the households' realized discount-factor path, so, under the patience shock, welfare inherits the shocked path—a form of time-dependent discounting in the spirit of Dávila and Schaab (2023).

to that benchmark, holding $\phi_\pi$ fixed. We compute the optimum separately for each of the four shocks[42] and report welfare differences as consumption equivalents, in basis points of permanent consumption. Finally, we also report welfare changes from the benchmark to the rent-inclusive (HICP-style) rule of Section 5.

### 6.2.1 The Optimal Coefficient on Rent Inflation

The welfare-optimal response to rent inflation is negative under all four shocks. The optimal coefficient is $-0.18$ under the rental supply shock and $-0.48$ under the cost-push shock, and a positive coefficient lowers welfare under every shock (Table 5 and Figure C.1). Under the two demand shocks welfare rises as $\phi_r$ falls until the equilibrium ceases to be determinate (Appendix C.2), so the optimal coefficient is the corner, $-0.49$. The welfare gains range from 0.09 to 1.09 basis points of permanent consumption, after a single unanticipated shock. They offset between 7 and 42 percent of the welfare cost of the shock itself.[43]

The rent-inclusive rules of Section 5, which measure housing costs in inflation, do the opposite: they lower welfare and raise the rent burden under all four shocks. The welfare criterion therefore favors a negative response to rent inflation over both excluding rents and including them with their expenditure weight.

How owner-occupied housing enters the index does not change this ranking, but it decides how much welfare the rule gives up. Rental equivalence, which raises the weight on rent inflation to 18 percent, loses about four times as much as actual rentals alone under every shock. Measured in inflation, as in Section 5.3, net acquisitions loses about as much as actual rentals under the demand and rental supply shocks, more under the cost-push shock, and less than rental equivalence under all four (Table F.10). The ECB favors net acquisitions on measurement grounds (European Central Bank, 2021; Eiglsperger et al., 2022); the welfare criterion ranks the two approaches the same way, although, measured in inflation, neither improves on an index without housing costs.

To give net acquisitions its best chance as a welfare-relevant index, we also let the composite respond to the level of house prices, replacing house-price inflation with the gap of the real house price from its steady state.[44] The distinction matters for the same reason as for rents. Summed over time, an inflation term in a relative price adds up to that price's current gap, as in (26), so it only shifts the timing of policy: what the rule gives while the house price falls, it takes back as the price recovers. A level term instead responds for as long as the gap lasts. Under the patience,

[42] We do not optimize $\phi_r$ against the monetary shock itself. Optimizing a rule's response to a shock that originates in the rule conflates responding to rent inflation with reversing the policy innovation.

[43] The welfare cost of a shock is the consumption equivalent that makes the benchmark transition as good as remaining in the stationary equilibrium: 4.5, 4.9, 0.5, and 0.2 basis points for the patience, government-spending, rental supply, and cost-push shocks, of which the optimized rule offsets 25, 7, 20, and 42 percent. The patience shock changes the households' own discount factor, so for this shock we measure the cost and the gain at the steady-state discount factor.

[44] A price index reports a growth rate, which is observable; the steady-state real house price is not. In practice, a central bank could measure the gap against a long trailing moving average of real house prices, which approximates the steady state when shocks are transitory. Using the model's own steady state gives the level rule its most favorable measurement. The same caveat applies to the rule on the real rent level of Appendix F.16.

| | Demand | | Supply | |
|---|---|---|---|---|
| | Patience | Government spending | Rental supply | Cost push |
| *Rule on nominal average rent inflation*, $\pi_t^{\text{rent}}$ | | | | |
| Optimal coefficient, $\phi_r^*$ | $-0.49$ | $-0.49$ | $-0.18$ | $-0.48$ |
| Welfare gain at $\phi_r^*$ | 1.093 | 0.345 | 0.106 | 0.087 |
| *Rent-inclusive targets*, welfare change | | | | |
| Actual rentals, $\omega_r = 0.053$ | $-0.176$ | $-0.022$ | $-0.081$ | $-0.003$ |
| Rental equivalence, $\omega_o = 0.13$ | $-0.724$ | $-0.086$ | $-0.327$ | $-0.013$ |
| Net acquisitions in levels, $\omega_o = 0.09$ | 0.288 | 0.053 | $-0.458$ | 0.042 |
| *Cumulative deviation of the real average rent, quarters 1–20* | | | | |
| Benchmark, $\phi_r = 0$ | 16.70 | 1.96 | 6.01 | 0.392 |
| Optimized rule, $\phi_r^*$ | 6.00 | 0.90 | 5.22 | 0.203 |

**Table 5.** The Optimal Response to Rent Inflation

**Note.** Each column optimizes rule (27) against a one-standard-deviation shock, holding $\phi_\pi = 1.5$. Under the patience and government-spending shocks the optimum is the most negative admissible coefficient on our grid (Appendix C.2). Welfare changes are consumption equivalents relative to the $\phi_r = 0$ benchmark under the same shock, in basis points of permanent consumption. Rent-inclusive targets weight actual rentals at the model's rent share; rental equivalence and net acquisitions are the composites of Section 5.3, with actual rentals scaled by $1 - \omega_o$. Net acquisitions enters in levels: the owner-occupied term is the gap of the real house price from its steady state, $(p_t^h - p_{ss}^h)/p_{ss}^h$, in place of house-price inflation. Table F.10 in Appendix F.16 reports net acquisitions both in inflation and in levels, together with the optimum of a rule on the real rent level. The lower block reports the cumulative percent deviation of the real average rent from its steady state over quarters 1–20.

government-spending, and cost-push shocks, the house price falls as the rule tightens and recovers as rates come back down, so the level term eases throughout the tightening and offsets part of it. Net acquisitions in levels then improves on the index without housing costs, by between 0.04 and 0.29 basis points. Under the rental supply shock, in contrast, the house price rises with rents as households shift toward owning, and the level term tightens exactly when the rental market needs relief: it loses more welfare than any other index. Even in its most favorable form, the owner-occupied term falls short of the gain from a negative response to rent inflation under every shock (Table 5 and Appendix F.17).

A negative coefficient tells the central bank to ease when measured rent inflation rises. Easier policy works on the margins that raised rents in Section 4.3: it reduces landlords' mortgage costs and the return on the financial assets they could hold instead of a rental unit. Supplying rental housing becomes more attractive, so rents need to rise by less to clear the market. This holds whatever raised rents in the first place, whether the rule's own tightening, as under the demand and cost-push shocks, or landlords' higher financing costs, as under the rental supply shock.

The relief reaches the rent that tenants pay. A lower rent on new leases passes into the average rent as leases turn over, and the optimized rule lowers the cumulative rise in the real average rent over the first five years by between 48 and 64 percent under the demand and cost-push shocks, and by 13 percent under the rental supply shock (Table 5).

Accommodation also has costs. After a demand shock, the economy is already expanding, and easier policy pushes consumption, hours, and inflation higher still: under the patience shock, nondurable inflation on impact is 1.7 annualized percentage points under the optimized rule, against 0.7 under the benchmark. The welfare criterion counts these costs through labor disutility and wage-adjustment costs, so under the demand shocks the case for a negative coefficient rests largely on who gains from it, which Section 6.3 examines. After a cost-push shock, the economy is contracting, and accommodation softens the contraction: consumption falls by half as much on impact. Appendices F.11 to F.14 report the responses under both rules.

In practice, it is hard for a central bank to decompose the relative contribution of different shocks in real time. However, the four welfare curves overlap, so any coefficient between $-0.25$ and 0 improves welfare under each of the four shocks (Figure C.1).[45]

### 6.2.2 Is This Just a Less Hawkish Rule?

This negative coefficient admits two readings. The additive rule (27) can be rewritten identically as

$$i_t = r_{ss} + (\phi_\pi + \phi_r)\,\pi_t^c + \phi_r\left(\pi_t^{\text{rent}} - \pi_t^c\right), \tag{29}$$

a lower coefficient on nondurable inflation plus a negative response to real rent growth (25). The gain could come entirely from the first term, that is, from a less hawkish rule. To check, we compare the optimized rule with rules that have no rent term but a different coefficient on nondurable inflation.

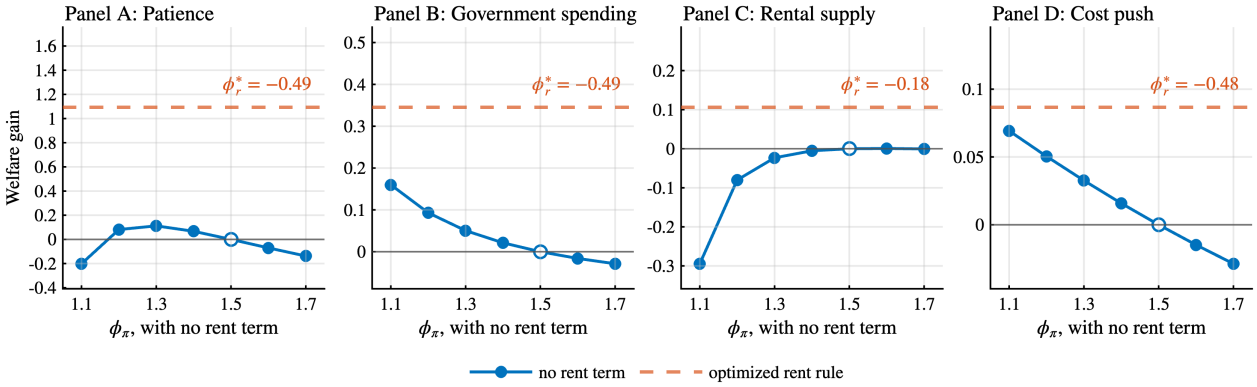



**Figure 6.** Welfare under Alternative Monetary Policy Rules

**Note.** The solid lines report welfare relative to the shock-specific benchmark $(\phi_\pi, \phi_r) = (1.5, 0)$ for rules without a rent term, varying $\phi_\pi$ over $[1.1, 1.7]$ with $\phi_r = 0$. The dashed lines report the welfare gain from optimizing the rent coefficient while holding $\phi_\pi = 1.5$. Both are consumption-equivalent gains, in basis points of permanent consumption.

Figure 6 shows that, independently of the shock under consideration and across a wide

[45] The same holds when welfare includes the expected taste-shock utility of tenure choices, which also leaves the four optimal coefficients unchanged. Under the patience shock, evaluating welfare at the steady-state discount factor gives the same optimal coefficient, with a gain of 1.12 basis points instead of 1.09.

range of potential $\phi_\pi$, the welfare gain implied by including a rent term is higher than that from a rule on nondurable inflation alone. However, whether this alternative rule can approximate the optimal rule, and how, varies across shocks. Under the government-spending and cost-push shocks, reducing $\phi_\pi$ reduces the tightening reaction of the central bank. The effect is similar to what $\phi_r < 0$ generates, and therefore recovers 46 and 80 percent, respectively, of the gain from the rent term. Under the patience shock, a similar argument is true, but the best rule without a rent term generates only 10 percent of the welfare gain from the rent term. In the case of the rental supply shock, given that it generates almost no nondurable inflation, the rule that ignores rents hardly improves welfare compared with the benchmark rule.

Alternatively, in Table F.8, we consider the case in which we keep $\phi_\pi + \phi_r$ constant and change only the response to real rent growth. For effective coefficients of 1.2, 1.3, and 1.5, welfare rises as $\phi_r$ falls from 0 to $-0.4$ under all four shocks. Under the rental supply shock, the best of these rules keeps the benchmark's effective coefficient of 1.5 and eases with rents, with $\phi_\pi = 1.9$ and $\phi_r = -0.4$. Optimizing both coefficients over all admissible rules points the same way under three of the four shocks (Appendix F.15). The rent term therefore does more than lower the overall response to inflation: it lowers the policy rate when real rents are rising, which allows it to respond to different types of shocks in a more targeted way by design.

The negative response to rent inflation does not rest on the rule's other features or on lease duration. With interest-rate smoothing, an output-gap term, a transfer proportional to labor productivity, shorter leases, or more frequent lease turnover, moving $\phi_r$ from 0 to $-0.20$ raises welfare under each of the four shocks (Table F.7 in Appendix F.15). When leases reprice more often, the gain is larger than in the baseline.

## 6.3 Distributional Decomposition across Cohorts and Channels

We now ask who gains from the optimal response, and through which channels. We focus on the patience shock, under which rents rise the most, and report the other shocks in Appendix F. We follow cohorts defined at the date of the shock and decompose their welfare changes with the inputs of Section 4.3: each channel replaces one input path under the benchmark rule with its path under the optimized rule. Figure 7 reports the channels and the net change for each cohort, in utility units, which is how the planner adds them up.

Lower rents (purple) move resources from landlords to tenants: the rent channel raises welfare in every renter cohort and lowers it for landlords. A lower rent is worth most to tenants with low consumption and high marginal utility, which gives a utilitarian planner a distributional reason to value rent relief.

The net gain is progressive within renters. Poor renters gain the most from lower housing costs and hold little wealth exposed to lower returns on savings. As wealth rises, the loss of interest income (light green) grows and eventually outweighs the rent relief, so the wealthiest renters lose. Renters as a group still gain. For the poorest renters, rent relief is the largest gain, about one and a half times the gain from higher transfers (gray-blue).

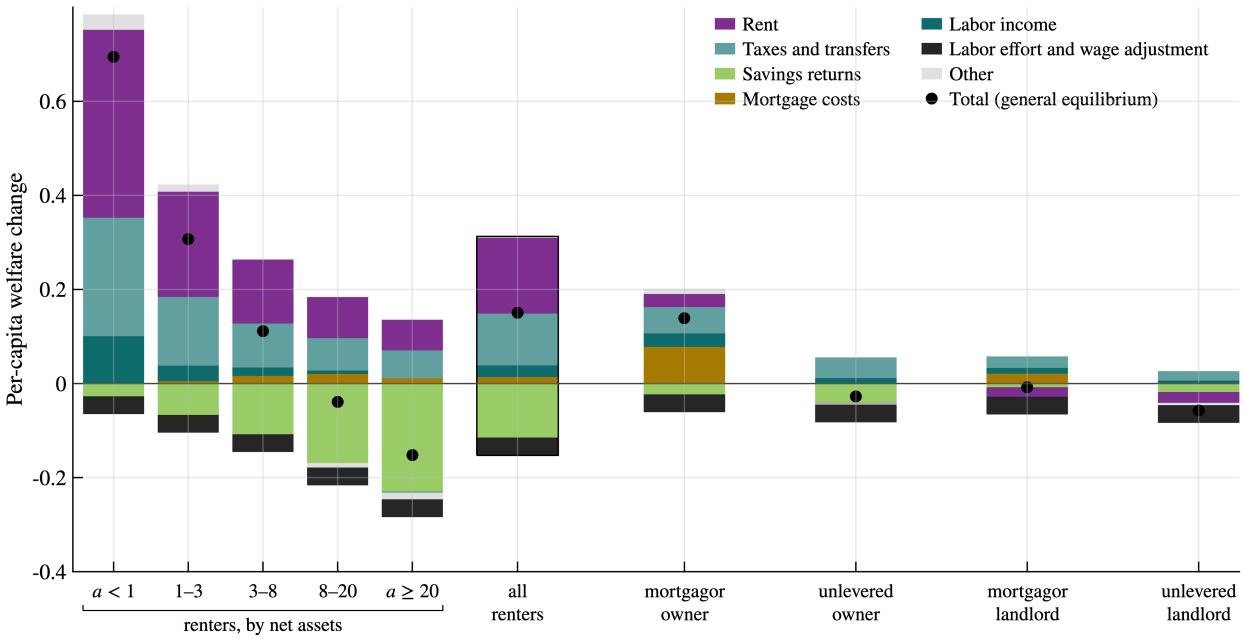



**Figure 7.** Distributional Decomposition of the Welfare Gain

**Note.** Per-capita welfare change from moving $\phi_r$ from 0 to $-0.49$ under the patience shock, by date-of-shock cohort, in utility units. Net financial assets $a$ are measured in units of quarterly steady-state income. Each channel holds the benchmark equilibrium and replaces one input with its path under the optimized rule; the rent channel moves the rent tenants pay and the rent landlords receive together. "Savings returns" groups the portfolio effects on renters, owners, and landlords of Figure 5; "Mortgage costs" groups the homebuyer and landlord mortgage rates; "Other" groups the house price and lease survival. Labor effort and wage adjustment is the change in labor disutility and wage-adjustment costs, common to all households. The all-renters bar aggregates the five renter cohorts. Black dots are the general-equilibrium total.

Mortgagor-owners gain mainly through lower mortgage costs (dark yellow). Unlevered owners and landlords lose: they earn less on their savings, landlords collect lower rents, and every household bears the cost of higher hours and inflation (black bars).

Higher transfers come from the government's lower interest bill, which first reduces public debt and reaches households over the following years through the fiscal rule (22) (Figure F.16 in Appendix F.15). Because households hold the public debt, the higher transfers are largely paid for by the interest income that savers lose. Taken together, the two channels account for two-fifths of the aggregate gain under the patience shock, half under the government-spending shock, almost none under the rental supply shock, and about an eighth under the cost-push shock. The negative coefficient does not rest on the progressivity of the transfer: when the transfer is distributed in proportion to labor productivity, moving $\phi_r$ from 0 to $-0.20$ still raises welfare under every shock, by between 53 and 119 percent of the baseline gain.

The other shocks share this pattern, with two differences. Under the supply shocks, higher transfers matter as much as rent relief for the poorest renters, and mortgagor-landlords gain, because lower mortgage costs outweigh the rent they lose. Measured in consumption equivalents,

renters and mortgagor-owners gain under all four shocks, while unlevered owners and unlevered landlords lose (Table F.9).

**Efficiency versus redistribution.** Following Dávila and Schaab (2025), we split the marginal welfare effect of the rent coefficient into efficiency, the households' total willingness to pay, and redistribution, which is positive when the change favors the households the planner weighs more (Figure 8). The planner weighs renters most, then owners, then landlords, and, among owners and landlords, those with a mortgage.

Under the patience shock, efficiency favors a less negative coefficient at every point of Figure 8: accommodation during a boom raises hours and inflation, and these costs outweigh what households would pay for easier policy. The negative optimum rests on redistribution. Under the government-spending and rental supply shocks, efficiency alone would call for a less negative coefficient, about $-0.18$ and close to zero, respectively, and redistribution accounts for the rest. Under the cost-push shock, both components favor a more negative coefficient: accommodation softens a contraction.

The rental market shapes this redistribution. Monetary policy moves the rent, a price that renters pay to landlords, two groups with very different marginal utilities. Leaning against rent increases moves resources toward renters.

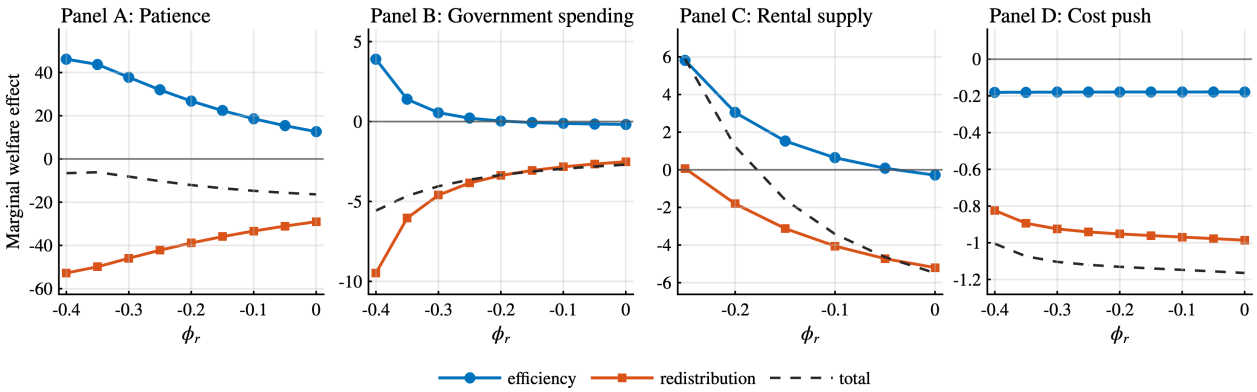



**Figure 8.** Efficiency and Redistribution

**Note.** Marginal welfare effect of the coefficient on rent inflation in rule (27), $d\mathcal{W}/d\phi_r$, holding $\phi_\pi = 1.5$, split into efficiency and redistribution components following Dávila and Schaab (2025). The dashed line is their sum. Positive values favor a higher coefficient. Effects are measured in $10^{-5}$ units of perpetual consumption, a unit of consumption to every household at every date. Appendix C.1 contains the derivations.

# 7 Ramsey Policy

Section 6 chooses the rent coefficient that maximizes utilitarian welfare from the date of the shock. This criterion mixes stabilization with redistribution across households whose marginal utilities differ, and part of that redistribution is available only because policy moves after households have

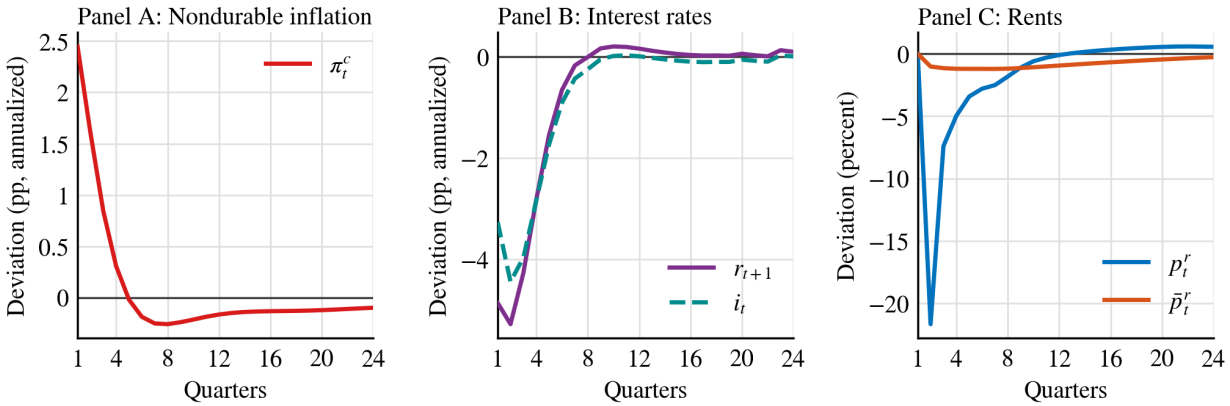



**Figure 9.** The Ramsey Plan without a Shock

**Note.** The standard Ramsey plan from the stationary equilibrium with no shock, in deviations from steady state. $r_{t+1}$ is the real rate set at date $t$ through (20), $p_t^r$ the rent on new leases and $\bar{p}_t^r$ the average rent.

made their decisions. To separate these motives, we solve for optimal policy unconstrained by any rule, with the computational methods of Barcons et al. (2026) and the timeless Ramsey theory of DS. Judged from the date it acts, optimal policy eases sharply, even when no shock has hit. From a timeless perspective, which sets that part aside, it is mostly stabilization, and after the patience shock the best simple rule tightens.

## 7.1 The Standard Ramsey Problem

In the standard Ramsey problem, as in DS, the central bank follows no rule and no earlier promise binds it. At the date of the shock, it chooses the whole path of the nominal interest rate, $\mathbf{i} = \{i_t\}_{t \geq 1}$, and commits to it, to maximize criterion (28). Let $V(a, h, y, s; \mathbf{i})$ be the welfare of a household that starts the transition in state $(a, h, y, s)$, as in Appendix C.1, and $g_{ss}(a, h, y, s)$ the stationary distribution; the criterion is then $\mathcal{W}(\mathbf{i}) = \sum_{h,s} \int\int g_{ss}(a, h, y, s)\, V(a, h, y, s; \mathbf{i})\, da\, dy$.

**Optimal policy in the absence of shocks.** A central bank that only stabilized would leave policy unchanged. The Ramsey plan instead eases by surprise (Figure 9): the policy rate and the real interest rate fall sharply in the first quarters, the policy rate to below zero, nondurable inflation rises, and the rent on new leases falls sharply in the second quarter, the first in which leases reprice, while the average rent moves much less. The surprise raises welfare by 5.81 basis points of permanent consumption,[46] and all of it is redistribution: in the decomposition of Section 6.3, redistribution accounts for more than the whole gain, and efficiency is negative. The poorest renters gain 107 basis points in consumption equivalents and mortgagor-owners 88, while the richest renters lose up to 110.

[46] Appendix G describes how we compute the plans.

A utilitarian central bank values this transfer because its largest gainers, the poorest renters, value an extra unit of consumption more than three times as much as the average household, and landlords about half as much. An easing announced in advance would have changed the saving, borrowing, and tenure decisions households made before it, a distortion that the committed central bank counts against it; an easing at the date of the shock comes after those decisions, so the central bank collects the transfer without that cost. The plan therefore front-loads the easing, which shows up as a burst of inflation (Panel A). This is what DS call the time-0 problem, and the burst of inflation is the counterpart of their short-run inflationary bias. As in DS, the incentive behind it is distributional, although here a labor-income tax also holds employment down. Acharya et al. (2023) find the same temptation in a tractable heterogeneous-agent economy: a surprise cut in rates at the initial date compresses consumption inequality through households' unhedged interest-rate exposures (Auclert, 2019), which an anticipated cut would not.

**Optimal policy after a patience shock.** For a shock of the size we study, the plan makes a similar but larger move and gains 5.62 basis points, about as much as without the shock. The incentive to surprise exists even without a shock, and the shock interacts with it, so, on this criterion, the gain cannot be split into a part due to the surprise and a part due to the response to the shock (Appendix G.2). Relative to the benchmark rule, the optimized rule of Section 6 moves the real interest rate and the rent on new leases the same way at first, but its largest moves are less than a tenth of the plan's. A rule moves only when inflation does: without a shock it cannot ease at all, and after one it eases only as far as the inflation the shock creates lets it. It captures 19 percent of the Ramsey gain, 23 percent when the coefficient on nondurable inflation is also chosen, and between 2 and 7 percent under the other three shocks, up to 15 with both coefficients.[47]

## 7.2 Timeless Policy

**Timeless penalties.** A central bank bound by its earlier promises cannot surprise households whose decisions are already made. Following Woodford (2003) and Marcet and Marimon (2019), DS formalize this timeless perspective by charging the planner, at the date of the shock, for breaking the promises it would have made had it always been in charge. In our economy, the promises are on three objects that households and unions relied on before the shock: the value $V_1(a, h, y, s; \mathbf{i})$ of a household in state $(a, h, y, s)$ at the date of the shock, as in (7), on which its choice of tenure rested; its derivative in wealth, $\partial V_1(a, h, y, s; \mathbf{i})/\partial a$, on which saving or borrowing rested; and wage inflation, $\pi_1^w$. Each promise carries a penalty, $\bar{\Psi}_V(a, h, y, s)$, $\bar{\Psi}_{V^a}(a, h, y, s)$, and $\bar{\Psi}_\pi$, equal to the Lagrange multiplier, in the stationary Ramsey plan, on the condition that embodies the promise: the household's Bellman equation, its Euler equation, and the wage Phillips curve,

[47] If the policy rate cannot fall below zero, the best plan we find without a shock holds it at zero for the first year and keeps more than half of the gain of the unconstrained plan, almost all of it through redistribution. Under the government-spending, rental supply, and cost-push shocks, repeating that surprise still raises welfare, and the optimized rules of Section 6 capture between 2 and 8 percent of its gain. Under the patience shock, a surprise close to it lowers welfare (Appendix G).

respectively. These penalties depend on neither the shock nor policy. Writing $(\cdot)$ for the household state $(a, h, y, s)$, the timeless problem maximizes

$$\mathcal{W}^{TL}(\mathbf{i}) = \mathcal{W}(\mathbf{i}) + \underbrace{\sum_{h,s} \iint \left[ \bar{\Psi}_V(\cdot)\, V_1(\cdot;\mathbf{i}) + \bar{\Psi}_{V^a}(\cdot)\, \frac{\partial V_1(\cdot;\mathbf{i})}{\partial a} \right] da\, dy}_{\text{distributional penalty}} \underbrace{- \bar{\Psi}_\pi\, \pi_1^w \left(1 + \pi_1^w\right)}_{\text{inflation penalty}}. \tag{30}$$

**Timeless weights.** Integrating the marginal-value promise by parts in wealth turns the distributional penalty into a reweighting of households: up to small terms (Appendix G.6), the timeless problem adds up the same household welfare as the standard one, with the weights $\hat{g}(\cdot) = g_{ss}(\cdot) + \bar{\Psi}_V(\cdot) - \partial_a \bar{\Psi}_{V^a}(\cdot)$ in place of $g_{ss}(\cdot)$. In the absence of shocks, a surprise brings the timeless planner no first-order gain, as in DS (their Proposition 6).

**Timeless policy after a patience shock.** After the patience shock, the timeless plan is mostly stabilization: it holds nondurable inflation close to zero after the second quarter, raises the real interest rate by less than the benchmark rule on impact but for longer, and lets the average rent rise further. The coefficient of Section 6 now lowers welfare, by 2.10 basis points. The best rule we find tightens instead, with $\phi_\pi = 2.10$ and $\phi_r = 0.20$, and captures four-fifths of the timeless plan's gain. Under the other three shocks, the coefficients of Section 6 change timeless welfare by a quarter of a basis point or less.

**Why the best rule tightens.** From the date of the shock, easing pays because it moves resources toward the households the planner weighs more; the promises weaken that motive by changing the weights (Figure G.4 in Appendix G.6). The marginal-value promise is owed to households that chose how much to save: a planner who has always been in charge has kept the value of an extra unit of wealth high, so that households save. A surprise easing, which raises the welfare of poorer households relative to richer ones, breaks that promise, so the promise moves weight within each tenure away from the poorer and indebted households that easing favors and toward the richer savers it hurts. Together with the value promise, it is our counterpart of DS's penalty on the welfare gains of indebted households, and it turns the coefficient of Section 6 into a loss. The value promise moves weight across tenures, toward landlords, who supply the rental market, but on its own it slightly favors the coefficient of Section 6. With both promises, renters count 0.8 times as much as landlords from the timeless perspective, against 3.7 times at the date of the shock.

**Redistribution and the persistence of shocks.** From the timeless perspective, the motive to redistribute is much weaker, but, as in DS, it does not vanish. Under the rental supply, government-spending and patience shocks, what remains still favors a negative response to rent inflation, but at the calibrated persistence the efficiency cost of that response is four to five times the redistributive gain. As shocks become less persistent, the efficiency cost falls much faster than the redistributive

gain, and the timeless-optimal rent coefficient turns negative once the rental supply or government-spending shock is short-lived enough: under the rental supply shock, it is 0.06 at the calibrated persistence of 0.95 and crosses zero near 0.77. Under the cost-push shock it is already negative at the calibrated persistence. The stakes are small, but the sign of the timeless response depends on the persistence of shocks (Appendix G.5).

From the date of the shock, the welfare criterion favors easing when rents start to rise, largely because easing moves resources toward the poorest renters; from the timeless perspective, the best rule tightens after a patience shock and barely improves on the benchmark after the others.

# 8 Conclusion

Rent inflation poses a distinct problem for monetary policy because interest rates affect both the demand for rental housing and the return required to supply it. In Spain, an ECB tightening lowers prices excluding rents and raises real rents, and the model reproduces the sign of this response without targeting it. In our model, a monetary tightening keeps would-be buyers renting while raising landlords' financing and opportunity costs. Rents rise until households are willing to supply the additional rental housing. About two-thirds of the effect of higher interest rates on the rental market comes from landlords' financing and portfolio incentives. Sticky leases delay the response of measured rents.

This combination changes the consequences of including rents in the inflation target. A rule that responds positively to measured rent inflation keeps reacting to rental pressure generated by its own earlier tightening. The additional response raises the cost of supplying rental housing and deepens the consumption contraction. Over the first two years after a monetary tightening, consumption falls by more than under a rule that ignores rents, with no additional nondurable disinflation. Under each of the four structural shocks we consider, the rent-inclusive rule also lowers welfare.

The welfare-optimal response is accommodative under the two general demand shocks and the two supply shocks we consider: the rule should ease when rent inflation rises, relative to a rule that ignores it. Accommodation reduces the financing and opportunity costs of supplying rental housing, relieves mortgage borrowers, and lowers the government's interest bill. The interest savings return to households as transfers, while savers earn less. The welfare gain also reflects changes in employment and inflation costs. Under three of the four shocks, a less hawkish response to nondurable inflation recovers at most part of the gain. The negative response survives interest-rate smoothing, an output-gap term, a transfer proportional to labor productivity, and shorter leases.

Accommodation also reaches the rents that tenants pay: the optimized rule lowers the cumulative rise in the average rent by between half and two-thirds over the first five years after demand and cost-push shocks, and by about an eighth after a rental supply shock. This rent relief moves resources from landlords to tenants. It particularly benefits renters with little financial

wealth and high marginal utility of consumption. Wealthier renters can nevertheless lose overall because lower interest income outweighs their rent relief, while indebted homeowners benefit through lower mortgage payments.

Solving for Ramsey policy shows what this welfare criterion rewards. A central bank that chooses policy when the shock hits gains from easing by surprise, even without a shock, and all of that gain is redistribution, largely toward the poorest renters; after a patience shock, the optimized rule captures about a fifth of it. A central bank that evaluates its rule as if it had always been in place does not value that easing: after a patience shock, its best rule responds more strongly to nondurable inflation, with a positive coefficient on rent inflation.

Our framework abstracts from institutional features that shape monetary transmission in housing and rental markets. Mortgages in our model are effectively adjustable-rate. With fixed-rate debt, a tightening would not immediately reprice existing mortgages, weakening the direct mortgage-cost channel. Higher rates would still affect new borrowing and raise the return on alternative financial investments, so the landlord portfolio channel would remain operative. Nor does our economy have a fiscal instrument targeted at renters, such as a housing allowance, which could take over part of the redistribution that monetary policy achieves here.

In our welfare analysis, we optimize a simple rule against one unanticipated shock at a time and evaluate welfare from the date of the shock. Our results should therefore be read as conditional on the shocks we study, although any coefficient on rent inflation between $-0.25$ and $0$ raises welfare under all four of them. Incorporating our framework into a macro model with aggregate risk is a natural, but computationally intensive, avenue for future research.

In our economy, a central bank that eases when measured rent inflation rises does better than one that looks through it, which in turn does better than one that gives rents their expenditure weight in the policy index. How owner-occupied housing would enter the index matters too: imputing rents to owners, as the U.S. CPI does, costs about four times as much welfare as actual rentals alone, while the net-acquisitions approach favored by the ECB costs less, and improves on an index without housing costs under most shocks once house prices enter in levels rather than in growth rates.

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# APPENDIX

## A Derivations and Model Details

### A.1 Tenure Choice and Lease Survival

At the beginning of each period, the household observes the vector of choice-specific taste shocks $\varepsilon = \{\varepsilon(h')\}_{h'}$ in the household problem (7) and then chooses its next-period housing state. The formulas below are ex-ante with respect to these shocks: they condition only on the state $(a, h, y, s)$.

For each next-period housing state $h' \in \{0, 1, 2\}$, we define the deterministic choice-specific value

$$\widetilde{V}_t(a, h, y, s; h') = \max_{c,\, a'} \mathcal{U}_t(c, h, n) + \beta_t \mathbb{E}_t \left[ V_{t+1}(a', h', y', s') \mid y, h, s, h' \right],$$

where the maximization is subject to the budget constraint (3) and the borrowing constraints (5), and the expectation over the next lease state $s'$ follows (A.4). The value $\widetilde{V}_t$ collects everything the household controls once the tenure decision is fixed: it is the value of choosing consumption and savings optimally, conditional on moving to housing state $h'$.

After the taste shocks are observed, the household problem (7) reduces to a deterministic discrete choice across the three shock-augmented values,

$$V_t^{\varepsilon}(a, h, y, s) = \max_{h' \in \{0,1,2\}} \widetilde{V}_t(a, h, y, s; h') + \sigma_\varepsilon \, \varepsilon(h').$$

Because the shocks $\varepsilon(h')$ are drawn independently from a type-I extreme-value distribution with unit scale and zero mean, integrating over them yields closed-form expressions (Iskhakov et al., 2017). The ex-ante probability that a household in state $(a, h, y, s)$ chooses housing state $h'$ is

$$\mathbb{P}_t(h' \mid a, h, y, s) = \frac{\exp\left(\widetilde{V}_t(a, h, y, s; h')/\sigma_\varepsilon\right)}{\sum_{j \in \{0,1,2\}} \exp\left(\widetilde{V}_t(a, h, y, s; j)/\sigma_\varepsilon\right)}, \tag{A.1}$$

and the expected value before the shocks are realized is the log-sum

$$V_t(a, h, y, s) = \mathbb{E}_\varepsilon\left[V_t^{\varepsilon}(a, h, y, s)\right] = \sigma_\varepsilon \log\left[\sum_{j \in \{0,1,2\}} \exp\left(\widetilde{V}_t(a, h, y, s; j)/\sigma_\varepsilon\right)\right]. \tag{A.2}$$

Equation (A.2) is the continuation value that enters the Bellman equation (7) one period earlier, and the probabilities in equation (A.1) govern the evolution of the distribution $g_t(a, h, y, s)$ across housing states and, through (A.3), the retention rates that determine lease survival. Castellanos et al. (2026) and Albuquerque et al. (2026) use the same extreme-value structure for the housing

tenure decision.

The scale parameter $\sigma_\varepsilon$ governs how sensitive tenure choices are to differences in the choice-specific values. As $\sigma_\varepsilon \to 0$, the probabilities in equation (A.1) concentrate on the housing state with the highest deterministic value, and the tenure decision reduces to choosing the $h'$ that maximizes $\widetilde{V}_t$. A larger $\sigma_\varepsilon$ smooths the transition probabilities across the renter, owner-occupier, and landlord states: households whose choice-specific values are close transition gradually rather than all at once. The shocks are idiosyncratic taste shocks over tenure states, capturing motives for moving that are orthogonal to the household's financial state.

**Lease survival and continuation value.** Lease survival depends on the decisions of both sides of the market. Let $\mathbb{P}_t(h' \mid a, h, y, s)$ denote the probability that a household in state $(a, h, y, s)$ chooses housing state $h'$, given in (A.1), and let $\xi^s_{h,t}$ denote the tenure-retention rate of households in rental tenure $h \in \{0, 2\}$ and lease state $s$, the share of them that keep their tenure between $t$ and $t+1$,

$$\xi^s_{h,t} = \frac{1}{\mu^{h,s}_t} \iint \mathbb{P}_t(h' = h \mid a, h, y, s)\, g_t(a, h, y, s)\, da\, dy. \tag{A.3}$$

Because partners are matched at random within each lease state, a renter with lease state $s$ who remains a renter keeps its lease with probability $\varphi\, \xi^s_{2,t}$, and a landlord who remains a landlord keeps its lease with probability $\varphi\, \xi^s_{0,t}$. Writing $2 - h$ for the rental tenure of the partner, the continuation value in (7) is

$$\begin{aligned} &\mathbb{E}_t\left[V_{t+1}(a', h', y', s') \,\middle|\, y, h, s, h'\right] \\ &= \begin{cases} \mathbb{E}_{y'|y}\Big[\varphi\, \xi^s_{2-h,t}\, V_{t+1}(a', h', y', s^+) \\ \qquad + \left(1 - \varphi\, \xi^s_{2-h,t}\right) V_{t+1}(a', h', y', 0)\Big], & h' = h \in \{0, 2\}, \\ \mathbb{E}_{y'|y}\left[V_{t+1}(a', h', y', 0)\right], & h' = 1 \text{ or } h' \neq h, \end{cases} \end{aligned} \tag{A.4}$$

where $s^+ = \min\{s + 1, J\}$ is the state of a lease that continues. Households take the paths of retention rates $\{\xi^s_{h,t}\}$ as given, and in equilibrium these paths coincide with the ones their choices generate.

## A.2 Evolution of Average Rents

**Lease masses.** Random matching implies that the two sides of the market hold the same contracts. The masses of leases of each age on each side evolve as

$$\begin{aligned} \mu^{h,s+1}_{t+1} &= \varphi\, \mu^{h,s}_t\, \xi^s_{0,t}\, \xi^s_{2,t}, && s < J - 1, \\ \mu^{h,J}_{t+1} &= \varphi \sum_{s \in \{J-1, J\}} \mu^{h,s}_t\, \xi^s_{0,t}\, \xi^s_{2,t}, && h \in \{0, 2\}, \end{aligned} \tag{A.5}$$

and new leases account for the remainder, $\mu^{h,0}_{t+1} = \mu^{h}_{t+1} - \sum_{s\geq 1} \mu^{h,s}_{t+1}$. Whenever renters and landlords coincide within each lease state at $t$, they coincide at $t+1$ as well, given rental-market clearing at $t+1$. Rental-market clearing, $\mu^{0}_{t} = \mu^{2}_{t}$, holds lease state by lease state, and the rent paid by renters equals the rent received by landlords in every period.

**Pooled rent.** The pool accumulates the leases that reach age $J$, which carry the rent at which they were signed, and the pooled rent is the average over its contracts. In this way, it reflects both the persistence of existing contracts and past market conditions, which enter the pool every period with the cohort of leases that reaches age $J$. Formally,

$$\begin{aligned} \widetilde{p}^{r}_{t} &= \lambda_{t-1}\, \widetilde{p}^{r}_{t-1} + (1-\lambda_{t-1})\, p^{r}_{t-J}, \\ \lambda_{t} &= \frac{\mu^{0,J}_{t}\, \xi^{J}_{0,t}\, \xi^{J}_{2,t}}{\mu^{0,J}_{t}\, \xi^{J}_{0,t}\, \xi^{J}_{2,t} + \mu^{0,J-1}_{t}\, \xi^{J-1}_{0,t}\, \xi^{J-1}_{2,t}}, \end{aligned} \tag{A.6}$$

where $\lambda_t$ is the share of the leases in the pool at $t+1$ that were already in it at $t$, and $1-\lambda_t$ is the share of leases that reach age $J$.

**Average-rent weights.** Iterating the pooled-rent recursion (A.6) backward expresses the pooled rent as a weighted average of the market rents at which its leases were signed,

$$\widetilde{p}^{r}_{t} = \sum_{j=0}^{\infty} \left( \prod_{\ell=1}^{j} \lambda_{t-\ell} \right) (1-\lambda_{t-j-1})\, p^{r}_{t-J-j},$$

with $p^{r}_{t} = p^{r}_{ss}$ for $t \leq 1$ and $\lambda_t = \lambda_{ss}$ for $t \leq 0$. Substituting this expression into (9) writes the average rent as a weighted average of current and past market rents,

$$\overline{p}^{r}_{t} = \sum_{k=0}^{\infty} \varpi_{t,k}\, p^{r}_{t-k}, \qquad \varpi_{t,k} = \begin{cases} \dfrac{\mu^{0,k}_{t}}{\mu^{0}_{t}}, & k < J, \\ \dfrac{\mu^{0,J}_{t}}{\mu^{0}_{t}} \left( \displaystyle\prod_{\ell=1}^{k-J} \lambda_{t-\ell} \right) (1-\lambda_{t-k+J-1}), & k \geq J. \end{cases}$$

For $k < J$, the weight on the market rent of $k$ quarters ago is the share of renters holding a lease signed $k$ quarters ago; the pool spreads its share over older vintages according to the survival of pooled leases. The weights are nonnegative and sum to one at every date, so a lasting movement in market rents passes into the average rent one for one: lease turnover delays the movement without shrinking it. Because a lease is a contract between one renter and one landlord, the same weights describe the rent that landlords receive.

The weights move along the transition only through the composition of leases by age and the survival of pooled leases, and both move little. At their stationary values, the average rent is a

fixed linear filter of market rents,

$$\overline{p}^r_t \approx \sum_{k=0}^{\infty} \omega_k \, p^r_{t-k}, \qquad \omega_k = \begin{cases} \mu^{0,k}_{ss} / \mu^0_{ss}, & k < J, \\ \left(\mu^{0,J}_{ss} / \mu^0_{ss}\right)(1-\lambda_{ss})\lambda^{k-J}_{ss}, & k \geq J, \end{cases} \tag{A.7}$$

which tracks the solved average rent to within 0.7 percent of its peak deviation after the monetary shock of Section 4.1, under the benchmark rule and under both rent-inclusive rules of Section 5. Leases younger than $J = 20$ quarters carry 59 percent of the weight, and their weights decline by about 5 percent per quarter, the rate at which renegotiations and changes of tenure end young leases. The pool carries the remaining 41 percent, spread geometrically at its survival rate $\lambda_{ss} = 0.96$. The index therefore lags the market rent by a mean of $\sum_k k\,\omega_k = 22$ quarters, the mean age of outstanding leases. Figure A.1 illustrates both properties.

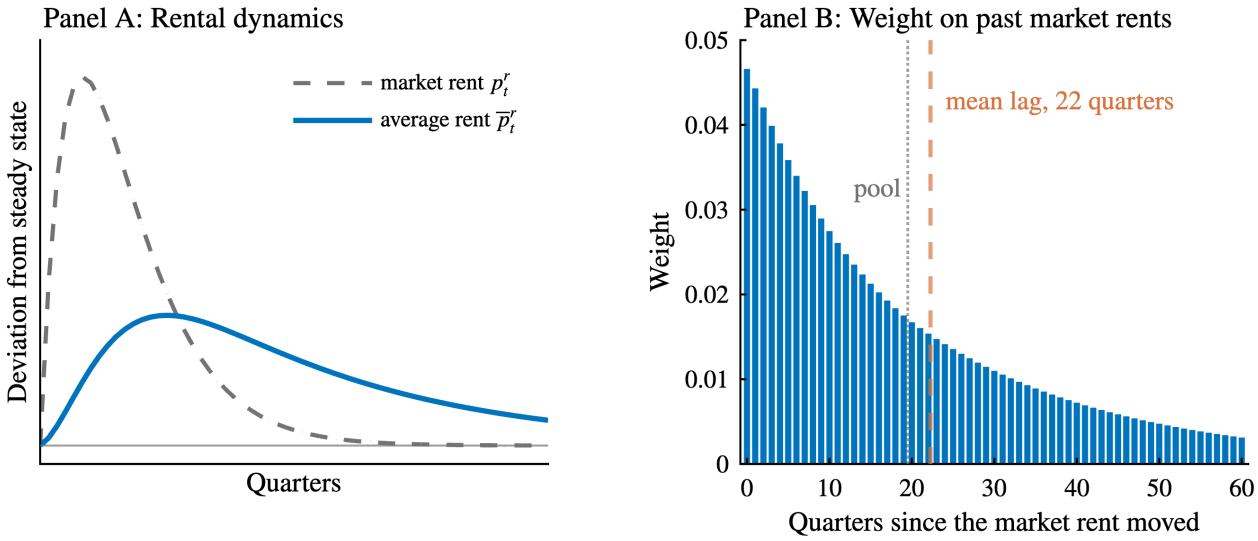



**Figure A.1.** Adjustment of Average Rents

**Note.** Panel A plots illustrative rental dynamics: a market rent that rises and returns to its steady state, and the average rent implied by the stationary weights in (A.7). Panel B shows the weight $\omega_k$ that the index places on the market rent of $k$ quarters ago, with the implied mean lag marked; leases aged $J = 20$ quarters or more belong to the pool.

**Nominal rents.** Let $P^c_t$ denote the nominal price of the consumption good and $\pi^c_t = P^c_t / P^c_{t-1} - 1$ nondurable inflation. In nominal terms, a lease signed in $t - s$, with $s < J$, pays

$$\widehat{P}^r_t(s) = P^c_t p^r_{t-s} = \widehat{P}^r_{t-s}(0) \prod_{k=t-s+1}^{t} (1 + \pi^c_k),$$

where $\widehat{P}^r_{t-s}(0) = P^c_{t-s} p^r_{t-s}$ is its nominal rent at signing. Thus, the lease is fully indexed: its nominal rent is updated every quarter by current nondurable inflation, and its real rent stays at its signing value. The pooled rent is indexed in the same way.

## A.3 Derivation of the Nonlinear New Keynesian Wage Phillips Curve

Each union introduced in Section 2.3 chooses its nominal wage to maximize the discounted utility of its members, internalizing the effect of its wage on their labor income and hours, and passing the Rotemberg adjustment cost through to them as a utility loss. Optimal wage setting delivers equation (12).

**Labor packer and union labor demand.** A continuum of unions indexed by $k \in [0,1]$ transforms household hours into differentiated labor services. Each household supplies hours to every union, so a household's total hours aggregate the services supplied through all unions, and each union rations hours equally across its members. A competitive labor packer combines the union services $N_{kt}$ into aggregate labor according to the CES technology

$$N_t = \left( \int_0^1 N_{kt}^{\frac{\epsilon_t - 1}{\epsilon_t}} \, dk \right)^{\frac{\epsilon_t}{\epsilon_t - 1}},$$

where $\epsilon_t$ is the elasticity of substitution across labor varieties, and sells the bundle to the consumption-good producer at the nominal wage $W_t$. Cost minimization by the packer yields the demand for union $k$'s service and the wage index,

$$N_{kt} = \left( \frac{W_{kt}}{W_t} \right)^{-\epsilon_t} N_t, \qquad W_t = \left( \int_0^1 W_{kt}^{1-\epsilon_t} \, dk \right)^{\frac{1}{1-\epsilon_t}},$$

where $W_{kt}$ is the nominal wage set by union $k$, so that $\partial N_{kt} / \partial W_{kt} = -\epsilon_t N_{kt} / W_{kt}$. Consistent with the definition of aggregate wage inflation in Section 2.3, union-specific gross wage inflation is $1 + \pi_{kt}^w = W_{kt} / W_{k,t-1}$. When varying its own wage, the union takes all aggregate objects as given, including $W_t$, $N_t$, prices, and the household distribution, as well as the income on which lenders assess its members' loan-to-income limits.

**Union's wage-setting problem.** Because the discount factor may vary over time, we define the compound discount $D_{t,j} = \prod_{\ell=t}^{j-1} \beta_\ell$, with $D_{t,t} = 1$. At date $t$, union $k$ chooses the sequence of its wages $\{W_{kj}\}_{j \geq t}$ to maximize the discounted utility of its members, augmented with the Rotemberg cost as in (1),

$$\sum_{j=t}^{\infty} D_{t,j} \left[ \int \Big( u(c_{ij}, h_{ij}) - v(n_{ij}) \Big) \, di - \frac{\theta_w}{2} \left( \pi_{kj}^w \right)^2 \right], \tag{A.8}$$

where $i$ indexes households. Union $k$ internalizes that its wage changes both its members' labor income and, through its labor demand, their total hours: $\partial n_{ij} / \partial W_{kj} = \partial N_{kj} / \partial W_{kj}$. In a symmetric equilibrium, hours rationing delivers $n_{ij} = N_j$, as in equation (11).

A member with productivity $y_{it}$ receives from union $k$ the after-tax labor income $(1-\tau_y)\,W_{kt}\,y_{it}\,N_{kt}/P_t^c$, where $P_t^c$ is the nominal price of the consumption good, together with the proportional employment subsidy $\tau_L(1-\tau_y)\,W_{kt}\,y_{it}\,N_{kt}/P_t^c$, which the union internalizes when setting its wage. The subsidy and the tax that finances it, $\tau_L(1-\tau_y)\,w_t\,y_{it}\,N_t$, both enter the transfer $T_t(y)$ described in Section 2.7; because the tax depends on aggregate employment, the individual union takes it as given. The derivative of the wage bill with respect to the union's wage is

$$\frac{\partial(W_{kt}N_{kt})}{\partial W_{kt}} = N_{kt} + W_{kt}\frac{\partial N_{kt}}{\partial W_{kt}} = (1-\epsilon_t)\,N_{kt},$$

so the direct change in member $i$'s resources is

$$\frac{\partial}{\partial W_{kt}}\left[(1+\tau_L)(1-\tau_y)\frac{W_{kt}\,y_{it}\,N_{kt}}{P_t^c}\right] = (1+\tau_L)(1-\tau_y)(1-\epsilon_t)\,y_{it}\,\frac{N_{kt}}{P_t^c}.$$

By the envelope theorem, the effect on member $i$'s value is its marginal utility of consumption times this change in resources, $u_c(c_{it},h_{it})\,(1+\tau_L)(1-\tau_y)(1-\epsilon_t)\,y_{it}\,N_{kt}/P_t^c$.

**First-order condition.** The date-$t$ wage enters the objective (A.8) through members' date-$t$ resources, their date-$t$ hours, and the wage-adjustment costs at $t$ and $t+1$. From the definition of union wage inflation,

$$\frac{\partial \pi_{kt}^w}{\partial W_{kt}} = \frac{1}{W_{k,t-1}}, \qquad \frac{\partial \pi_{k,t+1}^w}{\partial W_{kt}} = -\frac{W_{k,t+1}}{W_{kt}^2},$$

so the first-order condition with respect to $W_{kt}$ is

$$\int u_c(c_{it},h_{it})(1+\tau_L)(1-\tau_y)(1-\epsilon_t)y_{it}\frac{N_{kt}}{P_t^c}\,di - \nu'(n_t)\frac{\partial N_{kt}}{\partial W_{kt}} - \theta_w\pi_{kt}^w\frac{1}{W_{k,t-1}} + \beta_t\theta_w\pi_{k,t+1}^w\frac{W_{k,t+1}}{W_{kt}^2} = 0. \quad \text{(A.9)}$$

**Symmetry and the aggregate equation.** All unions face the same problem, so we impose the symmetric equilibrium $W_{kt}=W_t$, $N_{kt}=N_t$, and $\pi_{kt}^w=\pi_t^w$. Multiplying equation (A.9) by $W_t$, using $W_t/W_{t-1}=1+\pi_t^w$, $W_{t+1}/W_t=1+\pi_{t+1}^w$, the real wage $w_t=W_t/P_t^c$, and $\partial N_{kt}/\partial W_{kt} = -\epsilon_t N_t/W_t$, and rearranging yields

$$\pi_t^w(1+\pi_t^w) = \frac{\epsilon_t}{\theta_w}N_t\left(\nu'(N_t) - (1+\tau_L)(1-\tau_y)\frac{\epsilon_t-1}{\epsilon_t}\,w_t\int y_{it}\,u_c(c_{it},h_{it})\,di\right) + \beta_t\,\pi_{t+1}^w(1+\pi_{t+1}^w),$$

which is equation (12). Because consumption is conditional on the realized housing tenure choice, the integral is shorthand for the probability-weighted cross-sectional object

$$\int y_{it}\, u_c(c_{it}, h_{it})\, di = \sum_s \sum_{h',h} \iint y\, u_c\big(c_t(a,h,y,s;h'),\, h\big)\, \mathbb{P}_t(h' \mid a,h,y,s)\, g_t(a,h,y,s)\, da\, dy,$$

where $c_t(a,h,y,s;h')$ is the consumption policy conditional on choosing housing state $h'$ and $\mathbb{P}_t(h' \mid a,h,y,s)$ is the logit probability (A.1) of Appendix A.1.

## A.4 Aggregation, Government Budget, and Goods-Market Clearing

**Lemma A.1** (Walras's law)**.** *Any allocation and prices that satisfy conditions (i)–(viii) of Definition 1 satisfy goods-market clearing, $Y_t^c = C_t + S_t + G_t$, at every date.*

*Proof.* We aggregate the household budget constraint and combine it with the government budget constraint.

**Aggregate objects.** Aggregate mortgage debt $A_t^- < 0$, transaction-cost revenue $\mathcal{T}_t^h$, and the masses of households moving across housing states, $M_t(h,h')$, are defined in Section 2.7. Because the employment subsidy and its financing are both proportional to productivity and $n_t = N_t$, they cancel household by household, so the aggregate transfer $\int T_t(y)\, dg_t$ equals the lump-sum transfer $\overline{T}_t$.

**Aggregate housing outlays.** Each entry of the housing-cost function (6) is a change in the value of the housing position and a transaction cost on the units transacted. Summing over the transition masses, using $\sum_{h'} M_t(h,h') = \mu_t^h$ and $\sum_h M_t(h,h') = \mu_{t+1}^{h'}$, and recalling that the aggregate stock satisfies $H_t = \mu_t^1\, \tilde{h}_2 + \mu_t^2\, (\tilde{h}_1 + \tilde{h}_2)$, we can write aggregate housing outlays as

$$\sum_{h,h'} M_t(h,h')\, \psi_t(h,h') = p_t^h \big[H_{t+1} - (1-\delta_h) H_t\big] + \mathcal{T}_t^h = p_t^h Y_t^h + \mathcal{T}_t^h, \tag{A.10}$$

where the second equality uses the housing law of motion (17).

**Rental cash flows.** From the budget constraint (3), the rental cash flow of a household in state $(h,s)$ is $\widehat{p}_t^r(s)(h-1)$. Because a lease is a contract between one renter and one landlord, renters and landlords coincide within each lease state, $\mu_t^{0,s} = \mu_t^{2,s}$ (Section 2.2), so aggregate rent received by landlords equals aggregate rent paid by renters,

$$\sum_s \big(\mu_t^{2,s} - \mu_t^{0,s}\big)\, \widehat{p}_t^r(s) = 0.$$

**Aggregate budget.** Integrating the budget constraint (3), which holds with equality at the optimum, using the housing-outlay identity (A.10), the unit mean of labor productivity, and $w_t N_t = Y_t^c$

from equations (13) and (14), gives

$$C_t + A_{t+1} + p_t^h Y_t^h + \mathcal{T}_t^h = (1 + r_t) A_t + \kappa A_t^- + (1 - \tau_y)\, Y_t^c + \overline{T}_t.$$

The government budget constraint (21) gives the transfer,

$$\overline{T}_t = B_{t+1} - (1 + r_t)\, B_t - G_t + \tau_y Y_t^c + \mathcal{T}_t^h - \kappa A_t^- + p_t^L \overline{L}.$$

Substituting it and imposing financial-market clearing (19) at $t$ and $t+1$, the interest payments, the mortgage spread, the transaction costs, and the labor tax cancel, leaving

$$C_t + p_t^h Y_t^h = Y_t^c - G_t + p_t^L \overline{L}.$$

**Construction input and land.** The first-order conditions of the construction problem (16) imply $S_t = (1 - \alpha_L)\, p_t^h Y_t^h$ and $p_t^L \overline{L} = \alpha_L\, p_t^h Y_t^h$, so that payments to the construction input and to land exhaust construction revenue,

$$p_t^h Y_t^h = S_t + p_t^L \overline{L}.$$

The land revenue accrues to the government, which returns it to households through the transfer. Substituting gives the goods-market clearing condition

$$Y_t^c = C_t + S_t + G_t. \tag{A.11}$$

□

In the stationary equilibrium, government purchases equal the revenue from land permits, transaction costs, and the labor tax (Section 3.1), so the transfer that balances the budget is $\overline{T}_{ss} = -r_{ss}\overline{B} - \kappa A_{ss}^-$: households receive the revenue from the mortgage spread and pay the interest bill on public debt. In the monetary experiments, $G_t = G_{ss}$; $\tau_y$, $\tau_h$, and $\kappa$ remain fixed; and the transfer follows the fiscal rule (22). Under the rental supply shock of Section 6.1.2, which raises the spread on landlord debt to $\kappa_t^L$, replace $\kappa A_t^-$ throughout by $\kappa A_t^- + (\kappa_t^L - \kappa) A_{t,L}^-$, where $A_{t,L}^- = \sum_s \iint a\, \mathbb{1}_{a<0}\, g_t(a, 2, y, s)\, da\, dy$ is aggregate landlord mortgage debt; the landlord-specific spread revenue then enters the government budget.

# B Rent Decomposition: Derivations and Additional Results

Section 4.3 decomposes the rent response by asking how each equilibrium input changes rental-market excess supply, the gap between the mass of landlords willing to supply a rental unit and the mass of households demanding one. This appendix formalizes that construction and reports the cumulative contributions. Recall that $h \in \{0, 1, 2\}$ indexes renters, owner-occupiers, and landlords.

Given a path of household inputs $\mathbf{X}$, excess supply at $t+1$ is

$$\begin{aligned}\mathrm{ES}^r_{t+1}(\mathbf{X}) &= \mathcal{S}^r_{t+1}(\mathbf{X}) - \mathcal{D}^r_{t+1}(\mathbf{X}) \\ &= \sum_{s,h} \iint \left[\mathbb{P}_t(2 \mid a,h,y,s;\mathbf{X}) - \mathbb{P}_t(0 \mid a,h,y,s;\mathbf{X})\right] g_t(a,h,y,s;\mathbf{X})\, da\, dy. \end{aligned} \tag{B.1}$$

Thus, an input can change excess supply both by changing current tenure choices and by changing the distribution inherited from earlier choices. We keep both margins inside the economic channel generated by that input rather than reporting separate extensive and distribution channels.

The household block takes as inputs the paths

$$\mathbf{X}_t = \left\{ r^{\text{save}}_t,\ r^{\text{debt,own}}_t,\ r^{\text{debt,ll}}_t,\ w_t N_t,\ \tau_t,\ p^r_t,\ \widetilde{p}^r_t,\ p^h_t,\ \xi_t \right\}. \tag{B.2}$$

Here $\tau_t$ denotes the path of taxes and transfers, $T_t(\cdot)$, $\widetilde{p}^r_t$ the pooled rent (the rents of younger leases are past values of $p^r_t$), and $\xi_t$ the tenure-retention rates that govern lease survival. Retention rates move leases across lease states; to first order they change neither tenure choices nor excess supply. The three rates comove in equilibrium. We separate them only in the decomposition: $r^{\text{save}}_t$ applies to positive financial positions, while $r^{\text{debt,own}}_t$ and $r^{\text{debt,ll}}_t$ apply to the mortgage debt of owner-occupiers and landlords, respectively. Let $\mathbf{X}^{ss}$ and $\mathbf{X}^{eq}$ denote the steady-state and equilibrium paths. For input $j$, let $\mathbb{1}_j$ select its path and define the one-input experiment

$$\mathbf{X}^j = \mathbf{X}^{ss} + \mathbb{1}_j(\mathbf{X}^{eq} - \mathbf{X}^{ss}), \qquad d^j\mathbf{ES}^r = \mathbf{ES}^r(\mathbf{X}^j) - \mathbf{ES}^r(\mathbf{X}^{ss}). \tag{B.3}$$

**Result B.1** (Rental-market decomposition)**.** *To first order around the steady-state path, the equilibrium change in rental excess supply satisfies*

$$d\mathbf{ES}^r = d^{p^r,\widetilde{p}^r}\mathbf{ES}^r + \sum_{k=1}^{7} \chi^k + \boldsymbol{\epsilon}^{ES}. \tag{B.4}$$

*The seven indirect channels are*

$$\begin{aligned} \chi^{\text{ll mortg.}} &= d^{r^{\text{debt,ll}}}\mathbf{ES}^r, & \chi^{\text{buyer mortg.}} &= d^{r^{\text{debt,own}}}\mathbf{ES}^r, \\ \chi^{\text{portfolio},h} &= d^{r^{\text{save}},h}\mathbf{ES}^r, & & h \in \{2,1,0\}, \\ \chi^{\text{house price}} &= d^{p^h}\mathbf{ES}^r, & \chi^{\text{standard HANK}} &= d^{\bar{w}N}\mathbf{ES}^r + d^{\tau}\mathbf{ES}^r. \end{aligned}$$

*Here $d^{r^{\text{save}},h}\mathbf{ES}^r$ partitions the effect of the return on savings by the household's current tenure. The rental market clears in every quarter, so the market rent is pinned down date by date, and the rents of outstanding leases follow from it through (8) and (A.6). Let*

$$J_p = \left.\frac{\partial \mathbf{ES}^r_{2:T}}{\partial(\mathbf{p}^r_{2:T})^\top}\right|_{\mathbf{X}^{ss}}$$

*denote the response of excess supply to the market-rent path, including its effect through the rents of outstanding leases. The rent contribution of channel k is therefore*

$$d\mathbf{p}^{r,k}_{2:T} = -J_p^{-1}\boldsymbol{\chi}^k_{2:T}, \qquad dp^{r,k}_1 = 0, \tag{B.5}$$

*with* $d\mathbf{p}^r = \sum_k d\mathbf{p}^{r,k} + \boldsymbol{\epsilon}^p$*, where* $\boldsymbol{\epsilon}^{ES}$ *and* $\boldsymbol{\epsilon}^p$ *collect second-order terms.*

*Proof.* Differentiating equation (B.1) attributes the first-order change in tenure probabilities and in the inherited distribution to the input path that generates them. First-order additivity across the one-input experiments gives equation (B.4). Rental-market clearing then requires the direct rent effect to offset the combined indirect pressure. Linearizing the clearing conditions with respect to the market-rent path gives $J_p\, d\mathbf{p}^{r,k}_{2:T} = -\boldsymbol{\chi}^k_{2:T}$; inverting $J_p$ yields equation (B.5). □

The remaining discrepancy from exact additivity is reported as a first-order residual. The decomposition does not depend on the order in which the channels are introduced. Computed with the sequence-space Jacobian, as in Table B.1, each channel is linear in its input paths, so the channels add up exactly to the first-order response and coincide with a Shapley decomposition, which averages each channel's contribution over all orders of introduction. The one-input experiments in (B.3) evaluate each channel through the full nonlinear household block rather than its Jacobian. Over quarters 1–24, they differ from the first-order channels by at most 0.04 percentage-point-quarters, and their joint residual, which collects all interactions among channels and hence any dependence on the order, is 0.9 percent of the gross excess-supply pressure over quarters 1–24 and 2.0 percent over quarters 1–100.

**Cumulative contributions.** Figure 5 plots the direct rent effect and the indirect channels, grouped into supply, demand, the house price, and standard HANK, in excess-supply units, and the cumulative contribution of each channel to the average rent. A negative bar in excess-supply units means that, at the steady-state rent, the channel contracts rental supply relative to demand and therefore places upward pressure on the clearing rent. Table B.1 reports the cumulative effects of each of the seven channels on excess supply and on the market rent. Because the rent mapping is dynamic, it uses the entire time profile of each excess-supply path, so the sign of a channel's cumulative excess-supply effect need not determine the sign of its cumulative rent contribution when the channel changes sign over the transition.

In rent-response units, which measure the market rent, the two landlord channels together contribute more than the standard-HANK channel over the first 24 quarters, 1.30 units against 0.96, and less over 100 quarters, 0.80 against 1.62; individually, the standard-HANK channel is the largest. The house-price response remains the main offset over both horizons. The realized cumulative rent response rises from 1.33 to 1.70 units.

| Channel | Notation | Excess supply (pp-quarters) | Rent contribution (rent-response units) |
|---|---|---|---|
| Rent (direct effect) | $d^{p^r,\tilde{p}^r}\mathbf{ES}^r$ | 1.06 | — |
| Landlord mortgage cost | $\chi^{\text{ll mortg.}}$ | −0.65 | 0.59 |
| Homebuyer mortgage cost | $\chi^{\text{buyer mortg.}}$ | −0.37 | 0.31 |
| Portfolio effect on landlords | $\chi^{\text{portfolio,2}}$ | −0.98 | 0.71 |
| Portfolio effect on owners | $\chi^{\text{portfolio,1}}$ | 0.00 | −0.06 |
| Portfolio effect on renters | $\chi^{\text{portfolio,0}}$ | −0.52 | 0.49 |
| House price | $\chi^{\text{house price}}$ | 2.04 | −1.68 |
| Standard HANK | $\chi^{\text{standard HANK}}$ | −0.58 | 0.96 |
| First-order residual | $\epsilon^{ES}$ | 0.00 | 0.00 |
| Realized total | $d\mathbf{ES}^r$ | 0.00 | 1.33 |

**Table B.1.** Cumulative Rental-Market Decomposition, Quarters 1–24

**Note.** Excess supply is the landlord share minus the renter share. Its cumulative effects are measured in percentage-point-quarters (pp-quarters) of household mass. One rent-response unit is a one-percentage-point deviation of the market rent from $p^r_{ss}$ for one quarter. The rent column reports the cumulative $d\mathbf{p}^{r,k}$ of (B.5); its first-order residual, $\epsilon^p$, is 0.00 rent-response units, or 0.03 percent of the realized cumulative rent increase. In each column, the entries above the line sum to the realized total up to rounding. With the channels grouped as in Figure 5, supply accounts for 52.7 percent of the excess-supply contraction, demand for 28.7 percent, and the standard HANK channel for 18.6 percent, about a third of it through fewer landlords.

# C Numerical Solution

**Household problem and stationary equilibrium.** We solve the household problem on 287 asset nodes, evenly spaced between −20 and 45 times mean quarterly labor income, with a node at $a = 0$, and approximate the productivity process with a seven-state Rouwenhorst chain. Conditional on each tenure choice, we solve the consumption-saving problem with the endogenous grid method; because the tenure choice and the borrowing constraints make the continuation value non-concave, we add the upper-envelope step of Iskhakov et al. (2017). The distribution is updated with a lottery on the asset grid (Young, 2010). No price depends on the lease state in the stationary equilibrium, so we solve the household problem on $(a, h, y)$ and compute the split of renters and landlords across lease states as the fixed point of the lease transitions (A.5). The market-clearing conditions and the government budget hold in the stationary equilibrium up to residuals of order $10^{-12}$.

**Transitions.** We compute the perfect-foresight transition after each shock over $T = 200$ quarters in sequence space. The unknowns are the paths of the house price, employment, nondurable inflation, the real interest rate, the market rent, the lump-sum transfer, the pooled rent, and the retention rates of renters and landlords in each lease state. The equations are the housing law of motion (17), the wage Phillips curve (12), the Taylor rule (23) with the Fisher equation (20), financial-market clearing (19), rental-market clearing, the fiscal rule (22), the pooled-rent recursion (A.6), and the consistency of the retention rates with household choices (A.3). Rental-market clearing holds exactly in every quarter: the market rent is an unknown at every date after the impact quarter, and no restriction is placed on the shape of its path.

We solve this system with a quasi-Newton method. The initial matrix is the sequence-space Jacobian at the stationary equilibrium, which combines the Jacobian of the household block, computed with the fake-news algorithm of Auclert et al. (2021) and centered differences, with the derivatives of the remaining equilibrium conditions. Each accepted step updates the inverse matrix by Broyden's method, and a step is accepted only if it lowers the residual. The matrix affects only the speed of the solver: the equilibrium is the root of the nonlinear system, and we stop when the largest residual across all equations and dates falls below $10^{-10}$. The monetary transitions of Sections 4 and 5 converge in five iterations. The first-order decompositions of Appendix B and Section 5 use the same household Jacobian.

## C.1 Welfare Computation

**Consumption-equivalent units.** We express welfare differences as consumption equivalents. Let $\Lambda$ be the permanent proportional change in consumption under the benchmark rule ($\phi_r = 0$) that delivers the same aggregate utilitarian welfare (28) as the alternative rule. Split the benchmark objective into its consumption-utility part $\mathcal{W}_0^u = \sum_{t\geq 1} \widetilde{\beta}_t \int u(c_{it}, h_{it})\, di$ and the remainder $\mathcal{W}_0^n = \sum_{t\geq 1} \widetilde{\beta}_t \big[ -\nu(N_t) - \frac{\theta_w}{2}(\pi_t^w)^2 \big]$, so that benchmark welfare is $\mathcal{W}_0 = \mathcal{W}_0^u + \mathcal{W}_0^n$, both evaluated at $\phi_r = 0$ and summed over the transition plus a steady-state continuation. Because $u(c, h)$ is CRRA in $c$, scaling every household's consumption by $(1 + \Lambda)$ multiplies $\mathcal{W}_0^u$ by $(1 + \Lambda)^{1-\gamma}$ and leaves $\mathcal{W}_0^n$ unchanged. The consumption equivalent of a rule with welfare $\mathcal{W}$ then solves $(1 + \Lambda)^{1-\gamma}\mathcal{W}_0^u + \mathcal{W}_0^n = \mathcal{W}$, that is,

$$\Lambda = \left[ \frac{\mathcal{W} - \mathcal{W}_0^n}{\mathcal{W}_0^u} \right]^{\frac{1}{1-\gamma}} - 1,$$

and we report $10^4\Lambda$, in basis points.

**The $\phi_r$ grids.** We choose the rent coefficient by comparing welfare across policy rules that start from the same stationary equilibrium and differ only in $\phi_r$. We evaluate $\phi_r \in \{0, 0.05, 0.10\}$ and continue in steps of 0.05 in the direction in which welfare rises, until it falls. We then search in steps of 0.01 around the best point, over the rules under which the equilibrium is determinate, down to $\phi_r = -0.49$ at $\phi_\pi = 1.5$ (Appendix C.2); under the two demand shocks welfare is still rising at that point, which we report as the optimum. Figure C.1 reports the resulting welfare comparisons. Every transition used in Section 6 solves the equilibrium conditions with a largest residual below $10^{-10}$.

**Distributional decomposition of welfare gains.** Figure 7 splits the welfare change from moving to the optimized rule across initial cohorts and across the channels through which it reaches them. Consider a partition of the $(a, h)$ state space into groups indexed by $g$, with masses $m_g$ under the distribution at the date of the shock. Tracking each group forward under the transition's own

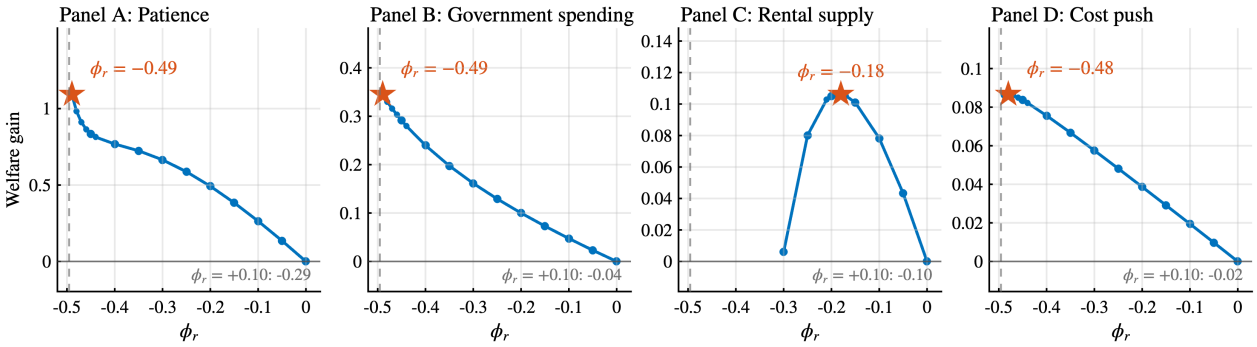



**Figure C.1.** Welfare Gains across Rent Coefficients

**Note.** Welfare gains relative to $\phi_r = 0$ under the same shock, with $\phi_\pi = 1.5$ throughout, in basis points of permanent consumption. Each point is a separately computed equilibrium transition, and the stars mark the optima. Welfare at $\phi_r = 0.10$ is reported by the annotations.

policies gives cohort welfare $\mathcal{W}_g(\phi_r)$, where the aggregate labor disutility and wage-adjustment cost in (28) are allocated across cohorts pro rata by mass and the steady-state continuation value by date-of-shock mass. The cohorts then add up, $\sum_g \mathcal{W}_g(\phi_r) = \mathcal{W}(\phi_r)$. Figure 7 reports the per-capita change $[\mathcal{W}_g(\phi_r^*) - \mathcal{W}_g(0)]/m_g$.

The channels are the one-input experiments of Appendix B, applied to welfare rather than to rental-market excess supply. Write $\mathcal{W}_g(\mathbf{X})$ for cohort $g$'s welfare when the household block is solved at the input path $\mathbf{X}$ of (B.2), and let $\mathbf{X}^0$ and $\mathbf{X}^*$ be the equilibrium paths under $\phi_r = 0$ and $\phi_r = \phi_r^*$. With the selector $\mathbb{1}_j$ of (B.3), channel $j$'s contribution to cohort $g$ is

$$\Delta_g^j = \frac{\mathcal{W}_g(\mathbf{X}^0 + \mathbb{1}_j(\mathbf{X}^* - \mathbf{X}^0)) - \mathcal{W}_g(\mathbf{X}^0)}{m_g}.$$

The change in aggregate labor disutility and wage-adjustment costs is common to all cohorts and is reported as its own bar. The difference between the general-equilibrium total and the sum of the channel contributions is the interaction across channels: the sum of the channel contributions exceeds the total by 18 percent under the patience shock, and by at most 1.3 percent under the other three shocks.

**Efficiency and redistribution.** Figure 8 splits the marginal welfare effect of the rent coefficient into efficiency and redistribution, following Dávila and Schaab (2025). Let $V(a, h, y, s; \phi_r)$ denote the welfare of a household that starts the transition in state $(a, h, y, s)$ under rule (27), measured as in (28): its expected discounted utility from consumption and housing, net of the labor disutility and wage-adjustment cost common to all households, plus a steady-state continuation. Aggregate welfare is $\mathcal{W}(\phi_r) = \sum_{h,s} \iint V(a, h, y, s; \phi_r)\, g_{ss}(a, h, y, s)\, da\, dy$. We express each household's gain in units of perpetual consumption, one unit of consumption at every date, which the household

values at

$$\lambda(a,h,y,s) = \sum_{t\geq 1} \widetilde{\beta}_t \, \mathbb{E}_1\left[u_c(c_{it},h_{it}) \,\middle|\, (a_{i1},h_{i1},y_{i1},s_{i1}) = (a,h,y,s)\right], \tag{C.1}$$

again with a steady-state continuation, evaluated along the transition at the assessed coefficient. Let $\bar{\lambda}$ be its population average and $\omega(a,h,y,s) = \lambda(a,h,y,s)/\bar{\lambda}$ the implied welfare weight, which averages to one. Writing $\mathrm{d}V^{\lambda}/\mathrm{d}\phi_r = \lambda^{-1}\,\mathrm{d}V/\mathrm{d}\phi_r$ for a household's willingness to pay for a higher coefficient, the marginal welfare effect in units of perpetual consumption splits into

$$\frac{1}{\bar{\lambda}}\frac{\mathrm{d}\mathcal{W}}{\mathrm{d}\phi_r} = \underbrace{\mathbb{E}_{ss}\left[\frac{\mathrm{d}V^{\lambda}(a,h,y,s;\phi_r)}{\mathrm{d}\phi_r}\right]}_{\text{efficiency}} + \underbrace{\mathbb{C}\mathrm{ov}_{ss}\left(\omega(a,h,y,s),\, \frac{\mathrm{d}V^{\lambda}(a,h,y,s;\phi_r)}{\mathrm{d}\phi_r}\right)}_{\text{redistribution}}, \tag{C.2}$$

where $\mathbb{E}_{ss}$ and $\mathbb{C}\mathrm{ov}_{ss}$ are taken over the stationary distribution $g_{ss}$. Efficiency adds up households' willingness to pay for a higher coefficient and does not depend on the planner's welfare weights. Redistribution is the covariance between the weights and that willingness to pay: it is positive when a higher coefficient favors the households that value consumption most at the margin, mostly renters and levered owners.

We evaluate the derivatives in (C.2) by centered differences across neighboring points of the 0.05 grids, holding $\lambda(a,h,y,s)$ at its value at the assessed coefficient, and obtain household welfare and the numeraire (C.1) by integrating the stored household policies forward from $g_{ss}(a,h,y,s)$. The coefficients that efficiency alone favors, quoted in Section 6.3, are the zeros of its line in Figure 8, interpolated between grid points.

## C.2 Determinacy and the Admissible Rules

The rules we compare in Section 6 are those with $0 \leq \phi_\pi \leq 4$ under which the equilibrium is determinate, which we check with the winding-number test of Auclert et al. (2025), applied to the linearized equilibrium conditions at the stationary equilibrium. Under the rent-inflation rule (27), the equilibrium is determinate if and only if $\phi_\pi + \phi_r > 1.0045$, up to an upper bound on $\phi_r$ that binds only for positive rent coefficients (Figure C.2). At $\phi_\pi = 1.5$ this requires $\phi_r > -0.4955$, so the grids of Appendix C.1 stop there, and the patience and government-spending optima of Table 5 are corner solutions. The rule on the real rent level of Appendix F.16 is determinate over the whole range of coefficients we consider.

The lower frontier, $\phi_\pi + \phi_r = 1.0045$, is where a permanent rise in inflation leaves the long-run real rate unchanged: in the long run rent inflation equals nondurable inflation, and with the Fisher equation (20) in levels the nominal rate must rise by $1 + r_{ss}$ per unit of inflation, less a small correction from the long-run response of asset demand. This frontier does not depend on $J$. The upper frontier, a pair of complex roots with a cycle of about ten years, is computed with leases pooled after eight quarters ($J = 8$), the largest $J$ for which the full test is feasible; it moves out as $J$

rises, except near $\phi_\pi = 0$, where the determinate strip above the lower frontier is narrow at every $J$.

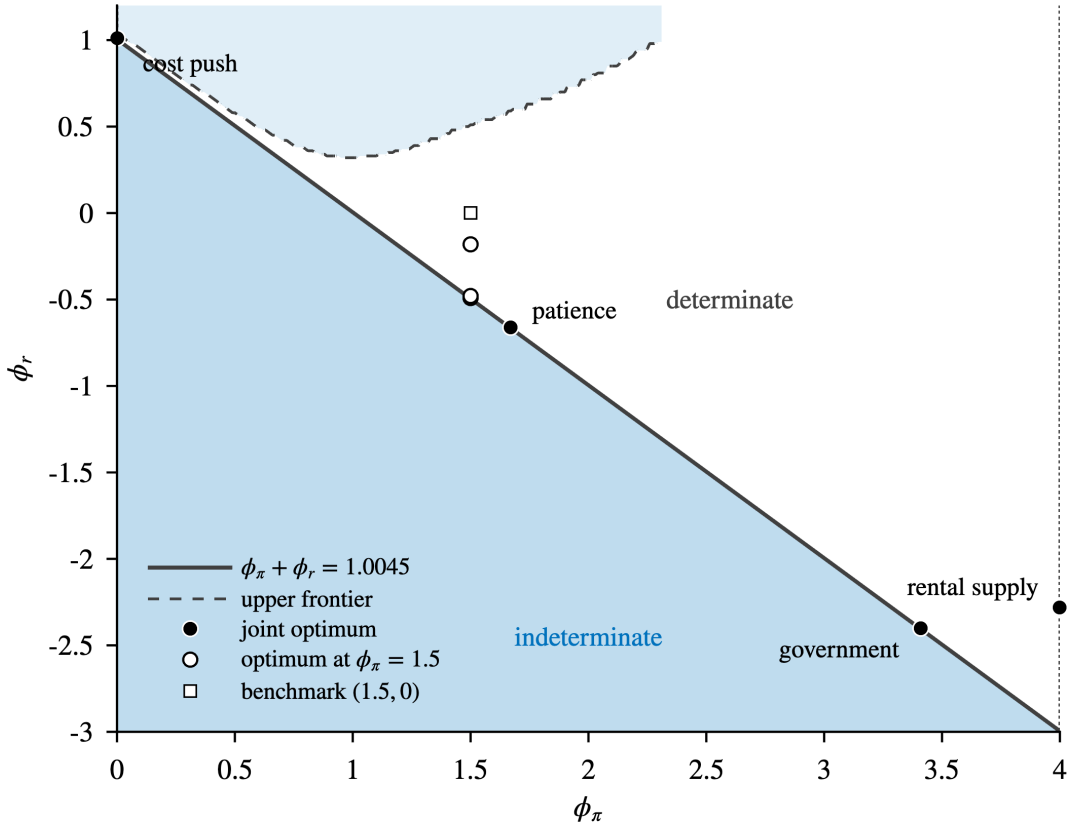



**Figure C.2.** Determinacy and the Admissible Rules

**Note.** Winding number of the determinant of the symbol of the sequence-space Jacobian (Auclert et al., 2025) under rule (27): zero in the white region (determinate), negative in the shaded regions (indeterminate). The solid line is the frontier $\phi_\pi + \phi_r = 1.0045$. The dashed line is the upper frontier, computed with leases pooled after eight quarters ($J = 8$). The dotted lines bound the admissible set at $\phi_\pi = 0$ and $\phi_\pi = 4$. Filled circles mark the jointly optimized rules by shock (Appendix F.15), open circles the optimized rent coefficients at $\phi_\pi = 1.5$ of Table 5, and the square the benchmark.

# D Data

This appendix documents the data sources and construction of the calibration targets and untargeted moments. We use Spanish observations unless otherwise stated. Sample periods differ across sources according to availability; averages are computed over the observations specified below.

## D.1 House Values and Rents

We construct the house-value-to-income ratio from the public statistical tables of the ECB Household Finance and Consumption Survey (HFCS), using the 2010, 2014, 2017, and 2021 waves. For each wave, we divide the mean value of the household's main residence among homeowners

(HFCS statistical table B4) by mean annual gross household income across all households (HFCS statistical table I2). Both series are expressed in the same monetary units. The calibration target is the simple average of these four ratios, equal to 5.25. This is a ratio of group means, rather than an average of household-level value-to-income ratios.

In the model, the target applies to a reference dwelling of unit quality. The two housing qualities, 0.905 and 1.095, have midpoint one; this normalization does not impose a population-weighted mean quality of one. Because owner-occupiers live in the higher-quality dwelling (1.095), an owner-occupier's dwelling has a value of approximately 5.75 times mean annual income.

The level of rents is not targeted. We compare it with the data through the rental burden, the mean across renters of rent relative to the household's own gross income. In the EFF, 2014–2022 waves, we compute it for renters as defined in Appendix D.2, as annual rent paid (twelve times the monthly rent, `p2_31`) over gross household income (`renthog`), averaging the five implicates with cross-sectional weights and the four waves with equal weights. The mean burden is 35.0 percent.

The EFF records the rent that tenants actually pay, as does EU-SILC, in which rent averaged 30.0 percent of tenant households' disposable income in Spain over 2004–2025 (Eurostat, `ilc_mded02`). Measures that assign each nonowner the local market rent are higher: Khametshin et al. (2024) report a mean burden of 38.7 percent in 2021 for households that do not own their main residence, valuing rent-free and reduced-rent accommodation at the average rent of its census tract. Our tenure classification excludes rent-free households.

In the model, every renter's gross income is labor plus interest income, and the steady-state rent $p^r_{ss}$ is 0.24 in units of mean quarterly labor income, which gives a mean burden of 30.4 percent. Renters' mean labor income is 0.83 of the population mean, against 0.74 for tenants in EU-SILC; the model does not target it.

The model does not generate the upper tail of the burden distribution. The median burden is 30.7 percent in the model and 24.1 percent in the EFF, where the mean exceeds the median by 11 points because of a tail of households whose rent is high relative to their income. The model's productivity process, whose lowest state is about two-fifths of mean labor income, does not generate that tail. The same compression makes aggregate rent a larger share of aggregate renter income in the model, 28.0 percent, than in the EFF, 20.4 percent, which is why the model's rent share of household spending exceeds Spain's HICP weight (Appendix D.8).

## D.2 Tenure Shares

We measure tenure shares in the 2014, 2017, 2020, and 2022 waves of the EFF, averaging the five implicates of each wave with cross-sectional weights and the four waves with equal weights. We classify a household as an owner-occupier if it owns its main residence (`p2_1`=2) and receives no rental income, as a landlord if it owns its main residence and receives income from renting real estate (`p7_1`=1 or `p7_2`>0), and as a renter if it does not own its main residence, receives no rental income, and pays rent (`p2_31`>0). Two groups have no counterpart in the model and are excluded: households that neither own their main residence nor pay rent, mostly in rent-free accommodation,

which account for 7.2 percent of all households, and households that receive rental income without owning their main residence, which account for 1.7 percent.

Among the remaining 91.1 percent of households, renters account for 17.0 percent, owner-occupiers for 72.9 percent, and landlords for 10.0 percent. The renter share rises from 12.2 percent in 2014 to 20.6 percent in 2022. A landlord household may rent out more than one dwelling, so the landlord share counts households rather than rental units; Section 3.2 discusses why the model's landlord share equals its renter share.

## D.3 Construction and Household Wealth

The residential investment share is gross fixed capital formation in dwellings divided by GDP at market prices. The numerator comes from Eurostat national accounts, dataset `nama_10_an6`, asset code `N111G`; the denominator comes from `nama_10_gdp`, item `B1GQ`. Both are annual current-price series in millions of euros. The average Spanish ratio over 1995–2025 is 6.9 percent. This measure covers investment in dwellings rather than total construction activity.

Using the same national accounts, we construct the nonresidential investment share as total gross fixed capital formation (`N11G`) less investment in dwellings (`N111G`), divided by GDP. The average over 1999–2025 is 15.6 percent. Comparing the residential and nonresidential shares helps assess the model's residential investment share when housing is its only produced illiquid asset.

Household wealth comes from the ECB Distributional Wealth Accounts (DWA). We sum the financial-asset components `F2M`, `F3`, `F5`, and `F62`, add housing wealth (`NUN`), and subtract liabilities. This construction excludes nonfinancial business wealth and therefore differs from the DWA's broader published net-wealth aggregate. The income denominator is the four-quarter rolling sum of gross disposable income (`B6G`) from the ECB Quarterly Sector Accounts.

We average the quarterly wealth-to-income ratios over 2011Q4–2025Q4 for Spain and 2009Q1–2025Q4 for the euro area, obtaining 7.33 and 6.13, respectively. The corresponding housing-wealth-to-income ratios are 5.66 and 4.37. The wealth statistics cover households (`S14`), whereas the available income denominator also includes nonprofit institutions serving households (`S1M`).

The top-decile net-wealth share is taken from the HFCS 2021 wave. The Spanish hand-to-mouth benchmarks are taken from Kaplan et al. (2014), who distinguish households with little liquid wealth according to whether they hold illiquid assets.

**Wealth by tenure.** In the population we study, the average landlord holds 3.0 times mean net worth in Spain and 1.7 times in the model, and the average renter 0.14 and 0.27 times; relative to the median, landlords hold 3.4 against 1.7 times and renters 0.01 against 0.22 times. The model ranks the groups as the data do and concentrates landlords even more at the top: 56 percent of them are in the top net-worth decile, against 40 percent in Spain, because its top decile consists mostly of landlords who own two dwellings. What the model compresses is the distance between landlords and tenants, which governs the gap in marginal utility that a utilitarian planner weighs.

**Hand-to-mouth households.** Kaplan et al. (2014) document that 15.2 percent of Spanish households hold illiquid wealth but little liquid wealth, and only 4.4 percent hold neither. In the model, owner-occupiers and landlords who hold housing wealth but no positive net financial wealth are 16.0 percent of households, while renters holding no more than a quarter of mean annual income in financial wealth are 1.7 percent. The shortfall at the poor end is structural rather than a matter of calibration, since renters cannot borrow in our model. Nearly all wealthy hand-to-mouth households in Spain own housing, so the illiquid asset that defines the group is the one the model has.

## D.4 Tenure-Transition Matrix

We estimate tenure transitions using the panel component of the EFF, linking households across the 2014–2017, 2017–2020, and 2020–2022 wave pairs. These pairs contain 3,634, 3,831, and 3,974 matched households, respectively, of which 3,300, 3,497, and 3,606 are renters, owner-occupiers, or landlords, as defined in Appendix D.2, in both waves. Households that live rent free or receive rental income without owning their main residence in either wave are dropped. We use all five imputed datasets supplied for each wave.

We build the matrix in three steps. First, for each wave pair $j$ and each of the five imputed datasets, we weight matched households by the panel weight `pesopan_1` (reported in the later wave and representative of the earlier wave's population), divide the weighted origin-to-destination counts by the weighted count in each origin state, and average the five resulting matrices. This gives $P_j$, the transition matrix over the pair's horizon. Second, we convert each $P_j$ into a quarterly matrix $Q_j$ from its principal matrix root $P_j^{1/q_j}$, where $q_j$ is the number of quarters between the two waves: $(q_1, q_2, q_3) = (12, 12, 8)$. Third, we average the three quarterly matrices row by row, weighting pair $j$'s row $h$ by $n_{jh}$, the unweighted number of matched households that start pair $j$ in state $h$:

$$Q_{hk} = \frac{\sum_j n_{jh} (Q_j)_{hk}}{\sum_j n_{jh}}.$$

The survey observes tenure every two or three years. Converting each wave-pair matrix to quarterly frequency assumes that tenure follows a time-homogeneous Markov chain between waves. The principal quarterly root of each matrix has small negative entries, all in the cells linking renters and landlords and none larger than 0.04 percentage points in absolute value; we set them to zero and renormalize the rows. This step moves the targeted entries by less than a hundredth of a standard error, and the resulting quarterly matrices, raised back to the survey horizon, reproduce the observed matrices to within 0.16 standard errors outside the renter–landlord cells.

The resulting matrix describes transitions among continuing households. It excludes household formation and dissolution, so its implied stationary distribution need not reproduce cross-sectional tenure shares. Two of its entries, the flows from renting to owner-occupation and from landlord to owner-occupation, are calibration targets (Section 3.2); the remaining entries test the

model's mobility across tenure states.

Table D.1 reports the nine entries of the pooled quarterly tenure-transition matrix in the data and in the model. Standard errors use the 1,000 replicate weights that the Banco de España distributes with the EFF panel weights. We recompute the full procedure, namely the matrix of each wave pair, its conversion to quarterly frequency, and the pooling, with each replicate weight, and combine the five implicates with Rubin's rules (Banco de España, 2021). The model reproduces every entry within sampling error except the flow from owner-occupation to renting, which it overstates, and hence the owner-occupier diagonal, which it understates. The flows between renting and being a landlord rest on eight and five observed transitions; the model nearly rules them out and lies within two standard errors of both.

A comparison at the survey horizon requires no conversion. We raise the model's quarterly matrix to the twelfth or eighth power and compare it with the observed matrix of each wave pair. Both sides then miss the moves that are reversed between waves, so this comparison is not affected by within-period churn. It confirms the misfit in the owner-occupier row. The flow from renting to owner-occupation fits in every wave pair, while the flow from owner-occupation to renting is overstated in every wave pair: over three years, 3.2 percent of owner-occupiers become renters in the model against 1.3 to 1.4 percent in the data. The flow from landlord to owner-occupation is stable at about 1.8 percent per quarter between 2014 and 2020 but more than doubles, to 4.0 percent, between 2020 and 2022, a difference significant at the 1 percent level. The pooled target therefore averages over periods with different mobility.

| Origin | Destination | Data | Model | Obs. |
|---|---|---|---|---|
| Renter | Renter | 98.797 (0.144) | 98.825 | 988 |
| | Owner-occupier† | 1.175 (0.145) | 1.175 | 118 |
| | Landlord | 0.028 (0.022) | 0.000 | 8 |
| Owner-occupier | Renter | 0.125 (0.021) | 0.303 | 81 |
| | Owner-occupier | 99.265 (0.053) | 99.038 | 6,218 |
| | Landlord | 0.610 (0.047) | 0.659 | 643 |
| Landlord | Renter | 0.006 (0.011) | 0.000 | 5 |
| | Owner-occupier† | 2.553 (0.240) | 2.553 | 390 |
| | Landlord | 97.441 (0.240) | 97.447 | 1,952 |

**Table D.1.** Quarterly Tenure Transitions: Data and Model

**Note.** Quarterly transition probabilities in percent. Data: EFF panel 2014–2022 (three wave pairs), households that are renters, owner-occupiers, or landlords in both waves of a pair, weighted by the panel weight; principal matrix root of each pair with negative entries set to zero and rows renormalized, pooled by origin-state counts. Standard errors in parentheses from the 1,000 Banco de España panel replicate weights, combined across the five implicates with Rubin's rules. Obs. is the unweighted number of matched transitions summed over the three pairs. †Targeted in the calibration.

## D.5 Lease Terms and Rental Turnover

**Lease terms and rent updates.** Article 9 of Ley 29/1994 de Arrendamientos Urbanos provides for a minimum term of five years when the landlord is an individual and seven years when it is a legal person, through mandatory extensions when the agreed term is shorter. These terms were introduced by Real Decreto-ley 7/2019. Article 18 permits rent updates at most annually when the contract provides for them. Temporary limits generally capped updates at 2 percent in 2022–2023 and 3 percent in 2024 (Real Decreto-ley 6/2022 and Ley 12/2023); from 2025, the INE reference index for annual rent updates, IRAV, applies to contracts within its scope. The model approximates these update provisions by full indexation to current nondurable inflation, so the real rents of continuing leases do not respond to inflation; it does not reproduce each statutory cap.

**EFF tenancy duration.** We use the 2014, 2017, 2020, and 2022 waves of the EFF to measure how long market-rent tenants have lived in their current dwelling. The calculation selects households renting at market prices (`p2_1`=1 and `p2_30`=1), uses the reported year in which they began living in the dwelling (`p2_29`) and the interview year and month, and weights households by the cross-sectional weight `facine3`. Because the survey reports the start year rather than the start month, we approximate starts in earlier years by the middle of the year and starts in the interview year by the midpoint of the elapsed part of that year. We give each wave equal weight. This calculation uses the first of the five imputed datasets and gives a recent-mover share of 13.7 percent, rounded to 14 percent in the main text, and a median tenancy duration of about four and a half years. Dwelling occupancy does not record a new contract signed with the same tenant, so this moment is a benchmark for lease turnover rather than an identical empirical counterpart.

**Other turnover evidence.** INE's *Índice de Precios de Vivienda en Alquiler*, table 59008, reports weights by contract age for dwellings rented out by individual landlords in the common tax territory. The new-contract category carries a weight of approximately 12 to 18 percent over 2021–2024. These are rent-expenditure weights, valued at the previous year's prices, rather than the fraction of dwellings with a new contract; we therefore do not treat them as a direct turnover target. The Observatori Metropolità de l'Habitatge de Barcelona's report *Els contractes de lloguer a la demarcació de Barcelona 2010–2023* (2025) finds that about a fifth of registered contracts ended each year in Barcelona city over 2010–2023. This local registry provides a higher-turnover benchmark.

**Model counterpart.** At the stationary equilibrium, an approximate quarterly lease-survival probability is $\varphi\xi_{0,ss}\xi_{2,ss}$, implying a recent-lease share of approximately $1-(\varphi\xi_{0,ss}\xi_{2,ss})^4$. With the calibrated retention rates, setting $\varphi=1$ already implies approximately 14 percent of outstanding leases signed within the previous year, because tenure changes end leases. The choice $\varphi=0.99$ also allows other terminations and generates a recent-lease share of 17.29 percent in the stationary lease-age distribution. We report this stock share, rather than the number of contracts signed during a year divided by the lease stock.

## D.6 Mortgage Market

We measure the mortgage spread using monthly ECB interest-rate data. The mortgage rate covers new Spanish loans to households for house purchase with a floating rate or an initial rate-fixation period of up to one year (MIR series `M.ES.B.A2C.F.R.A.2250.EUR.N`). We subtract the monthly average one-year Euribor rate (FM series `M.U2.EUR.RT.MM.EURIBOR1YD_.HSTA`). Both rates are quoted in annual percent, so their difference is measured in percentage points.

The mean spread over May 2001–April 2026 is 1.38 percentage points. We set the mortgage spread $\kappa$ to this mean at quarterly frequency, 0.00345, so that the mortgage rate exceeds the real rate on savings by 1.38 annualized percentage points at every date, as the margin over Euribor of a Spanish variable-rate mortgage does. The loan-to-value and loan-to-income limits and the amortization requirement are set as described in Section 3.1.

**Owner-occupier debt.** In the model, 9.9 percent of owner-occupiers are net debtors, against 40.7 percent in the EFF. Two-thirds of the households that become owner-occupiers in the model are landlords who sell their rental unit and repay their mortgage, and buyers who were renters borrow about half the value of the house and often repay ahead of the minimum schedule.

## D.7 Earnings Process Calibration

**Persistence.** We use the Swedish annual persistence estimate, $\rho_a = 0.8139$, of Flodén and Lindé (2001) as a benchmark for the time-varying productivity component. The wage autocorrelations reported by De la Roca (2014) for Spain decline slowly with the lag: 0.939, 0.908, 0.882, 0.861, 0.843, 0.828, 0.809 at one through seven years. To distinguish persistent individual heterogeneity from time-varying earnings risk, we fit two simple decompositions to these seven autocorrelations.

Without an independent transitory component, we fit

$$c_k = w + (1 - w)\rho^k,$$

obtaining $w = 0.782$ and $\rho = 0.765$. Allowing for an independent transitory component, we fit

$$c_k = a\left[w + (1 - w)\rho^k\right],$$

obtaining $a = 0.972$, $w = 0.739$, $\rho = 0.866$. In the second specification, the permanent share of total variance is $aw = 0.718$, while the independent transitory share is $1 - a = 0.028$. These illustrative fits place the annual persistence of the time-varying component between 0.77 and 0.87, around our benchmark. They are our calculations using the reported autocorrelations, rather than estimates reported by De la Roca or an independently identified decomposition.

**Innovation dispersion and frequency conversion.** Figure 4(c) of Arellano et al. (2022) reports a spread of approximately 0.52 between the 90th and 10th percentiles of men's residualized annual

log-earnings changes in 2005–2006. We use this individual-earnings statistic as a benchmark for productivity risk.

For a stationary Gaussian AR(1), the corresponding spread of four-quarter endpoint changes is

$$2\Phi^{-1}(0.9)\sqrt{\frac{2\sigma_a^2}{1+\rho_a}},$$

where $\Phi$ is the standard normal distribution function. Equating this expression to 0.52 gives $\sigma_a^2 = 0.03733$. We convert to quarterly frequency by preserving persistence over a year and matching stationary variance:

$$\rho_y = \rho_a^{1/4}, \qquad \sigma_y^2 = \sigma_a^2 \frac{1-\rho_y^2}{1-\rho_a^2}.$$

These expressions give $\rho_y = 0.949823$ and $\sigma_y = 0.104016$. The calculation matches endpoint changes in the Gaussian process; it does not establish an exact match to changes in earnings aggregated over a year.

## D.8 Actual Rents in the HICP

We obtain annual expenditure weights from Eurostat's HICP item-weight dataset (`prc_hicp_inw`), using actual rentals for housing (`CP041`). Published weights are expressed per thousand; we divide them by ten to report percentages. The euro-area series uses changing composition.

Spain's weight on actual rentals averages 3.1 percent over 2014–2022, the period of the EFF waves; Appendix F.7 reports the responses at this weight. Until 2022, Spain's HICP weights were built from the household budget survey, which places rents below their share in the national accounts, about 3.5 percent of household consumption excluding imputed rents (`nama_10_co3_p3`); since 2023, when Spain began building the weights from the national accounts, the weight has been about 3.6 percent. Figure D.1 reports the weight of actual rents in the euro-area HICP. Over 2000–2025, it averages approximately 6.3 percent, compared with 10.4 percent in Germany and 2.8 percent in Spain.

In the calibrated stationary equilibrium, rental expenditure represents

$$\omega_r^{\text{model}} = \frac{p_{ss}^r \mu_{ss}^0}{C_{ss} + p_{ss}^r \mu_{ss}^0} \simeq 5.3 \text{ percent}$$

of household spending on consumption and rent. This share follows from the calibrated rent and renter share, together with nondurable consumption, and it exceeds Spain's HICP weight because the model compresses the dispersion of rental burdens (Appendix D.1): at the EFF's aggregate burden of 20.4 percent, the model's share would be 3.9 percent. It describes the model's simplified consumption basket, whereas the empirical HICP weight reflects expenditure across a broader basket. Section 5 uses the model's share, and Appendix F.7 also reports Spain's weight.

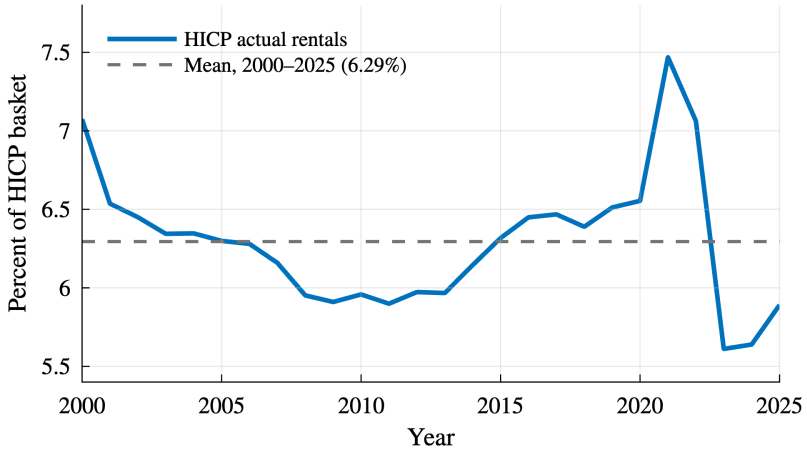



**Figure D.1.** Euro-Area HICP Weight of Actual Rents

**Note.** The figure reports the expenditure weight of actual rents paid by tenants in the euro-area HICP basket since 2000. The 2021 spike reflects pandemic reweighting, with no discrete change in rental-market coverage: an ECB Economic Bulletin box documents pandemic-induced shifts in HICP expenditure weights, with food and housing services increasing especially strongly (Gonçalves et al., 2021).

## D.9 Shock Processes

Each shock in Section 6 is unanticipated and moves one exogenous variable along a first-order autoregressive path: its deviation from the steady-state value is $\sigma\rho^{t-1}$ in quarter $t \geq 1$. We set $\sigma$ to one standard deviation of the innovation and $\rho$ to its estimated persistence, and we give each shock the sign that raises rents. Table D.2 reports the values.

The patience shock is the domestic risk-premium shock of the ECB's New Area-Wide Model, which enters the bond Euler equation as our discount factor does, so its estimated standard deviation, 0.161 percent of $\beta$, and its persistence carry over without rescaling (Christoffel et al., 2008, Table 1). The government-spending process is an AR(1) on the log of Spanish real general-government consumption net of a quadratic trend, 1995Q1–2019Q4; one standard deviation is 0.72 percent of purchases, or 0.14 percent of GDP. The spread on landlords' debt follows the series to which we calibrate $\kappa$ (Section 3.1): an AR(1) on its quarterly averages, 2003Q1–2026Q1, gives a persistence of 0.95 and a residual standard deviation of 22 annualized basis points. The cost-push shock lowers the elasticity across labor varieties, which raises the wage markup by 0.9 percent on impact. Its persistence is that of the New Area-Wide Model's wage-markup shock. Because that model normalizes the shock to its own wage Phillips curve, we set the size so that nondurable inflation rises by 0.15 annualized percentage points on impact, the response to a one-standard-deviation wage-markup shock in Christoffel et al. (2008).

| Shock | Exogenous variable | Impact, $\sigma$ | Persistence, $\rho$ |
|---|---|---|---|
| Patience | Discount factor, $\beta_t$ | $-0.0016$ | 0.917 |
| Government spending | Purchases, $G_t / G_{ss}$ | $+0.72\%$ | 0.96 |
| Rental supply | Spread on landlords' debt, $\kappa_t^L$ (annualized) | $+22$ bp | 0.95 |
| Cost push | Log labor-variety elasticity, $\ln \epsilon_t$ | $-0.08$ | 0.67 |

**Table D.2.** Shock Processes

**Note.** Each variable deviates from its steady-state value by $\sigma\rho^{t-1}$ in quarter $t \geq 1$. Sources: Christoffel et al. (2008), Table 1 (patience shock; persistence of the cost-push shock); Eurostat quarterly national accounts, `namq_10_gdp` (government spending); ECB Data Portal (spread); authors' calculations.

## D.10 A Spanish Central Bank in a Monetary Union

Spain does not set its own interest rate. The ECB sets it in response to euro-area inflation, so the rule that Spanish households face is

$$i_t = r_{ss} + \phi_\pi \, \pi_t^{\mathrm{EA}} + \varepsilon_t^m, \tag{D.1}$$

whereas rule (23) responds to Spanish inflation. Euro-area inflation is a weighted average of inflation in Spain and in the rest of the euro area,

$$\pi_t^{\mathrm{EA}} = \vartheta \, \pi_t^{\mathrm{ES}} + (1 - \vartheta) \, \pi_t^{\mathrm{xES}}, \tag{D.2}$$

where $\vartheta$ is Spain's weight in the euro-area HICP. The weight rose from 9.1 percent in 1999 to 12.8 percent in 2009 and stood at 11.9 percent in 2025, with an average of 11.4 percent over 1999–2025.[48] Substituting (D.2) into (D.1),

$$i_t = r_{ss} + \phi_\pi \vartheta \, \pi_t^{\mathrm{ES}} + \left[\phi_\pi (1 - \vartheta) \, \pi_t^{\mathrm{xES}} + \varepsilon_t^m\right]. \tag{D.3}$$

Taken alone, the response to Spanish inflation is small: with $\phi_\pi = 1.5$, $\phi_\pi \vartheta \approx 0.17$. The bracketed term, however, moves with Spanish inflation whenever inflation in the rest of the euro area does. If $\pi_t^{\mathrm{xES}} = \pi_t^{\mathrm{ES}}$, the bracket folds into the response to Spanish inflation and (D.3) coincides with rule (23).

Inflation in Spain and in the rest of the euro area shares one cycle. Over 1999–2025, quarterly year-on-year HICP inflation in Spain and in the euro area excluding Spain has a correlation of 0.82. Writing the linear projection $\pi_t^{\mathrm{xES}} = a + b \, \pi_t^{\mathrm{ES}} + u_t$, rule (D.3) responds to Spanish inflation with the effective coefficient $\tilde{\phi}_\pi = \phi_\pi[\vartheta + (1 - \vartheta) \, b]$; the constant $a$ shifts the intercept, and the residual

[48] Indices and country weights are from Eurostat (`prc_hicp_midx`, `prc_hicp_cow`) for 1999–2025, before the 2026 change in classification; the euro area has changing composition. We average monthly indices to quarters and compute year-on-year inflation. Inflation in the euro area excluding Spain follows from the chain-linked aggregation of the HICP: within each year, the euro-area index relative to the previous December is the weighted average of the member indices relative to that December, so the corresponding relative for the rest of the euro area is $(\mathrm{rel}_t^{\mathrm{EA}} - \vartheta \, \mathrm{rel}_t^{\mathrm{ES}})/(1 - \vartheta)$. Rebuilding the official euro-area index from its members in the same way reproduces its inflation rate to within 0.014 percentage points on average.

$u_t$, uncorrelated with Spanish inflation, acts as an additional policy shock. Over the same sample $b = 0.746$, so $\tilde{\phi}_\pi = 1.16$: seen from Spain, the ECB's rule satisfies the Taylor principle.

Figure D.2 shows the same co-movement in units of the policy rate. It plots the inflation term of the rule, $\phi_\pi \pi_t$ with $\phi_\pi = 1.5$, evaluated at Spanish and at euro-area inflation. The two prescriptions rise and fall together through the expansion before 2008, the disinflations of 2009 and 2014–2016, and the surge of 2021–2023, with a correlation of 0.86. The largest gap is in levels: before 2008, Spanish inflation averaged 3.2 percent against 2.2 percent in the euro area, which places the Spanish prescription about 1.6 percentage points higher, a difference absorbed by the intercept rather than by the dynamics. The argument concerns shocks common to the euro area, which move inflation in Spain and elsewhere together; the monetary shock of Section 4 is one by construction.

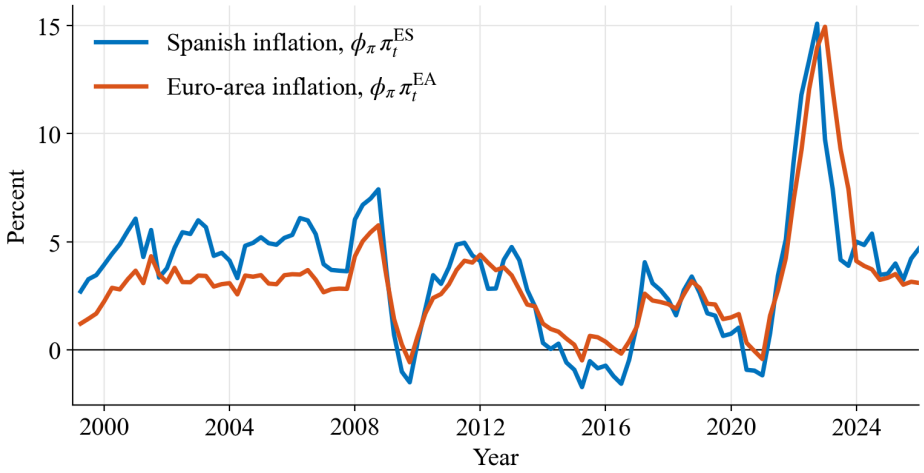



**Figure D.2.** Policy Prescriptions from Spanish and Euro-Area Inflation

**Note.** The figure plots $\phi_\pi \pi_t$ with $\phi_\pi = 1.5$, where $\pi_t$ is year-on-year HICP inflation, in quarterly averages, in Spain and in the euro area (changing composition), 1999Q1–2025Q4. The correlation between the two series is 0.86. Source: Eurostat (`prc_hicp_midx`).

The MEDEA model of the Spanish economy faces the same modeling choice (Burriel et al., 2010). In a model of Spain alone in which the ECB responds to Spanish inflation only through Spain's weight, the indeterminacy region is "so large that the model becomes nearly useless." MEDEA therefore uses two limiting cases: a weight of one on Spanish aggregates in the rule, equivalent to an independent Spanish monetary policy, in its estimation over 1986–2007; and an exogenous interest rate, on the grounds that Spain is too small to influence the ECB, in its estimation over 1997–2007. Rule (23) is the first case. The co-movement documented above connects the two: when inflation in the rest of the euro area moves with Spanish inflation, the euro-area rate that the second case takes as given is close to the rate a Spanish rule would set.

# E Monetary Policy and Rents in Spain: Empirical Evidence

This appendix describes the vector autoregression behind Figure 4: the data, the identification of the monetary shock, the full set of responses, and their robustness.

## E.1 Data

Table E.1 lists the series. The estimation sample is 2005:3–2019:12 ($T = 178$ months after six lags). The HICP excluding rents is built by chain-linking the all-items index and the index of actual rentals with Eurostat's item weights, and all price indices are seasonally adjusted. The sample ends in December 2019: after 2020, most of the surprises around the hiking cycle are information shocks in the decomposition of Jarociński and Karadi (2020), and the period contains the pandemic, the energy shock, and the 2022–2024 cap on updates of existing Spanish leases.

| Variable | Source | Units |
|---|---|---|
| ECB monetary surprise | EA-MPD (Altavilla et al., 2019) | basis points |
| One-year rate | ECB, euro-area AAA yield curve | percent |
| Industrial production | Eurostat | 100 log, seasonally adjusted |
| HICP excluding rents | Eurostat | 100 log, seasonally adjusted |
| HICP actual rentals | Eurostat | 100 log, seasonally adjusted |
| Spain–Germany ten-year spread | ECB | percentage points |

**Table E.1.** Data for the Spanish VAR

## E.2 Identification and Estimation

The surprise is the change in the one-year overnight index swap rate in a window around each ECB announcement, aggregated to monthly frequency as in Gertler and Karadi (2015). Because the ECB sets its rate for the euro area as a whole, the shock is plausibly exogenous to Spanish conditions (Appendix D.10). We estimate the monthly VAR in levels

$$y_t = c + \sum_{\ell=1}^{6} A_\ell \, y_{t-\ell} + u_t, \qquad y_t = \left(z_t,\ i_t,\ ip_t,\ p_t,\ q_t,\ s_t\right)', \qquad u_t = B\,\varepsilon_t,$$

where $z_t$ is the surprise, $i_t$ the one-year rate, $ip_t$ industrial production, $p_t$ the HICP excluding rents, $q_t$ the HICP index of actual rentals, $s_t$ the spread, and $B$ the lower-triangular Cholesky factor of the residual covariance matrix. The monetary shock is the innovation in the surprise, ordered first; using the surprise as an internal instrument identifies the effects of the shock without requiring the VAR to be invertible (Plagborg-Møller and Wolf, 2021). Responses are scaled so that the one-year rate rises by 25 basis points on impact, and real rents are $q_t - p_t$. The first-stage $F$-statistic of the one-year-rate residual on the surprise is 11.9. Information criteria select between one and eleven lags; we use six and report twelve below. Bands come from a residual moving-block bootstrap

with 24-month blocks and 2,000 draws (Jentsch and Lunsford, 2019), and the figures report the median of the draws.

## E.3 Responses

Figure E.1 reports every variable in the VAR. The one-year rate rises in a hump, peaking at 39 basis points after three months, and returns to zero after about seven months. Industrial production falls by 1.3 percent after a year, with the 68 percent band below zero, and the spread does not move significantly. Prices excluding rents fall by 0.7 percent after a year, with the 90 percent band below zero from month 9 to month 35. The rents that tenants pay do not respond significantly at any horizon, so real rents rise by 0.6 percent after a year, with the 90 percent band above zero from month 9 to month 18. The shock accounts for a modest share of fluctuations, as monetary shocks typically do: about a tenth of the forecast-error variance of prices excluding rents and a twentieth of that of real rents at two years (Table E.2).

| | 12 months | 24 months |
|---|---|---|
| One-year rate | 0.04 [0.02, 0.21] | 0.03 [0.02, 0.19] |
| Industrial production | 0.02 [0.01, 0.15] | 0.03 [0.01, 0.14] |
| Prices excluding rents | 0.05 [0.01, 0.19] | 0.11 [0.02, 0.23] |
| HICP rents | 0.02 [0.01, 0.08] | 0.03 [0.00, 0.12] |
| Real rents | 0.03 [0.01, 0.18] | 0.05 [0.02, 0.19] |

**Table E.2.** Share of Forecast-Error Variance Due to the Monetary Shock

**Note.** Point estimates and bootstrap 90 percent bands.

## E.4 Robustness

Figure E.2 reports the median responses of alternative specifications against the baseline bands. The first row varies the specification: twelve lags, INE's CPI rent instead of the HICP index, no spread, and the spread together with oil prices. Real rents are higher after a year in all 24 specifications that combine six or twelve lags, the two rent indices, two price indices, and three sets of controls, and the results are unchanged when the HICP excluding rents is seasonally adjusted with month dummies instead.

The second row varies the identification and the sample. When output, prices, and rents are restricted not to respond within the month of the shock, prices fall by 0.5 percent and real rents rise by 0.5 percent after a year. Press-conference surprises can carry information about the economic outlook as well as about policy (Jarociński and Karadi, 2020); keeping only the announcements at which the swap rate and stock prices move in opposite directions gives larger responses of the same sign, but a weaker instrument ($F = 6.0$). The surprises of 2008 and 2011, the two tightening cycles in the sample, account for 70 percent of the variance of the instrument. Setting the surprises of any single year to zero leaves the signs unchanged, with real rents between 0.16 and 0.72 percent

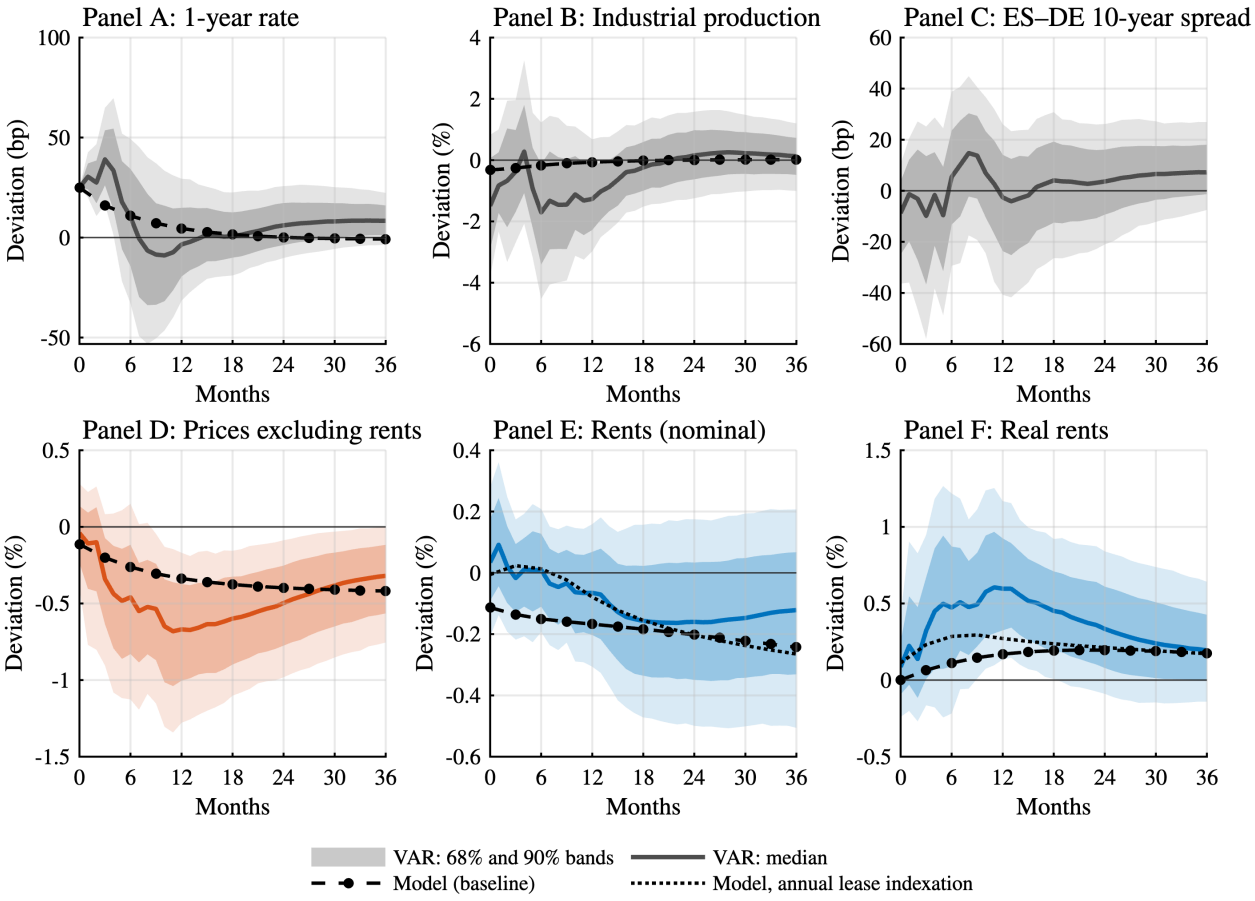



**Figure E.1.** Spanish VAR: All Responses

**Note.** Responses to a monetary tightening that raises the one-year rate by 25 basis points on impact; median of 2,000 bootstrap draws with 68 and 90 percent bands. Model lines as in Figure 4. The model's counterpart of industrial production is nondurable output $Y_t^c$, and that of the HICP rent index is the nominal average rent $P_t^c \overline{p}_t^r$. The spread has no model counterpart.

after a year; setting both to zero leaves too little variation to estimate the responses precisely ($F = 5.3$). Extending the sample to 2025, with monthly dummies for 2020:3–2021:7, gives the same signs with wider bands ($F = 8.9$).

## E.5 Model and Evidence

In Figures 4 and E.1, the model's counterpart of the one-year rate is its one-year nominal yield, the average of the policy rates over the next four quarters, whose annualized deviation is $\sum_{k=0}^{3}(i_{t+k} - r_{ss})$. We scale the model so that this yield rises by 25 basis points in the impact quarter, as the estimated one-year rate does in the impact month, and plot model quarter $t$ at month $3(t-1)$. The dotted line solves the model with the rents of continuing leases updated once a year by the previous year's inflation (Appendix F.10). Under this indexation the model's nominal average rent stays flat for three quarters and then declines with the price level, as the HICP rent index does.

Appendix F.1 compares the model with estimates for the United States and the euro area on the same nominal scale.

# F Additional Figures and Tables

## F.1 Model and Evidence on a Common Scale

Table F.1 compares the model's responses to a monetary tightening with estimates for the United States and the euro area. Each study normalizes its shock to a nominal interest rate, and we rescale the model to the same rise in the corresponding nominal rate: its one-year nominal yield for Dias and Duarte (2019) and Abramson et al. (2025), as in Figure 4, and its policy rate on impact for Dias and Duarte (2022) and Corsetti et al. (2022). The one-year-yield targets are the first-quarter averages of the estimated one-year-rate responses, 0.31 and 0.34 percentage points; the latter converts Abramson et al.'s 25-basis-point mortgage-rate shock into a one-year-yield change using our estimate from local projections with instrumental variables.[49] The policy-rate target is 25 basis points. The model's 25-basis-point innovation raises the nominal one-year yield by 0.056 percentage points and the policy rate by 9.8 basis points on impact, so the scaling factors are 5.5, 6.1, and 2.6.

On this scale, both rents rise by less than the point estimates but lie within their 68 percent bands. Part of the remaining gap reflects lease contracts: the estimates are for the United States, where leases typically run twelve months, whereas a lease keeps its signing rent for up to five years in our calibration, so the market rent rises by less and passes into the average rent more slowly. The decline in homeownership is about half as large as the estimates and lies within both of their 68 percent bands. The homeownership response is limited by the housing stock: because each landlord rents out a single, smaller unit, the renter share can rise only as the aggregate housing stock falls (equation (18)), and the stock falls by less than 0.003 percent over the transition. House prices fall on impact by more than in the estimates, within both 68 percent bands, and recover within two years, whereas the estimated decline persists. Consumption falls by more than in the Spanish estimates of Corsetti et al. (2022) on impact and recovers within two years, whereas in the data it keeps falling for three to four years.

[49] The conversion estimates the response of the one-year Treasury yield, averaged over the first quarter, at 1.37 times the impact response of the 30-year fixed mortgage rate, using the surprises of Bauer and Swanson (2023) over 2011–2022. The first stage is weak ($F = 2.8$), so the reported bands do not capture uncertainty in this conversion.

| | Horizon | Evidence | Model |
|---|---|---|---|
| Homeownership rate (pp) | quarters 8–12 | $-0.058$ to $-0.070$ | $-0.035$ |
| Nominal house price (%) | impact | $-0.18$ to $-0.28$ | $-0.35$ |
| Real average rent (%) | mean, months 0–24 | 0.27 [0.12, 0.43] | 0.17 |
| Real market rent, new leases (%) | mean, months 0–24 | 1.04 [0.52, 1.56] | 0.85 |
| Private consumption, Spain (%) | impact | $-0.08$ | $-0.17$ |
| Private consumption, Spain (%) | quarters 12–15 | about $-0.55$ | 0.01 |

**Table F.1.** Model and Evidence on a Common Scale

**Note.** Evidence: homeownership and house prices from Dias and Duarte (2022), point estimates under their two instruments; average rent from Dias and Duarte (2019), converted from their CPI deflator to prices excluding shelter, as in the model; new-lease rent from Abramson et al. (2025), built from listings; consumption from Corsetti et al. (2022), the Spanish response to a euro-area shock, read from their figure, which reports no bands for individual countries. Brackets give the averages over months 0–24 of the pointwise 68 percent bands. Model responses are rescaled as described in the text. The model's new-lease rent is the market rent $p_t^r$, the rent at which leases are signed.

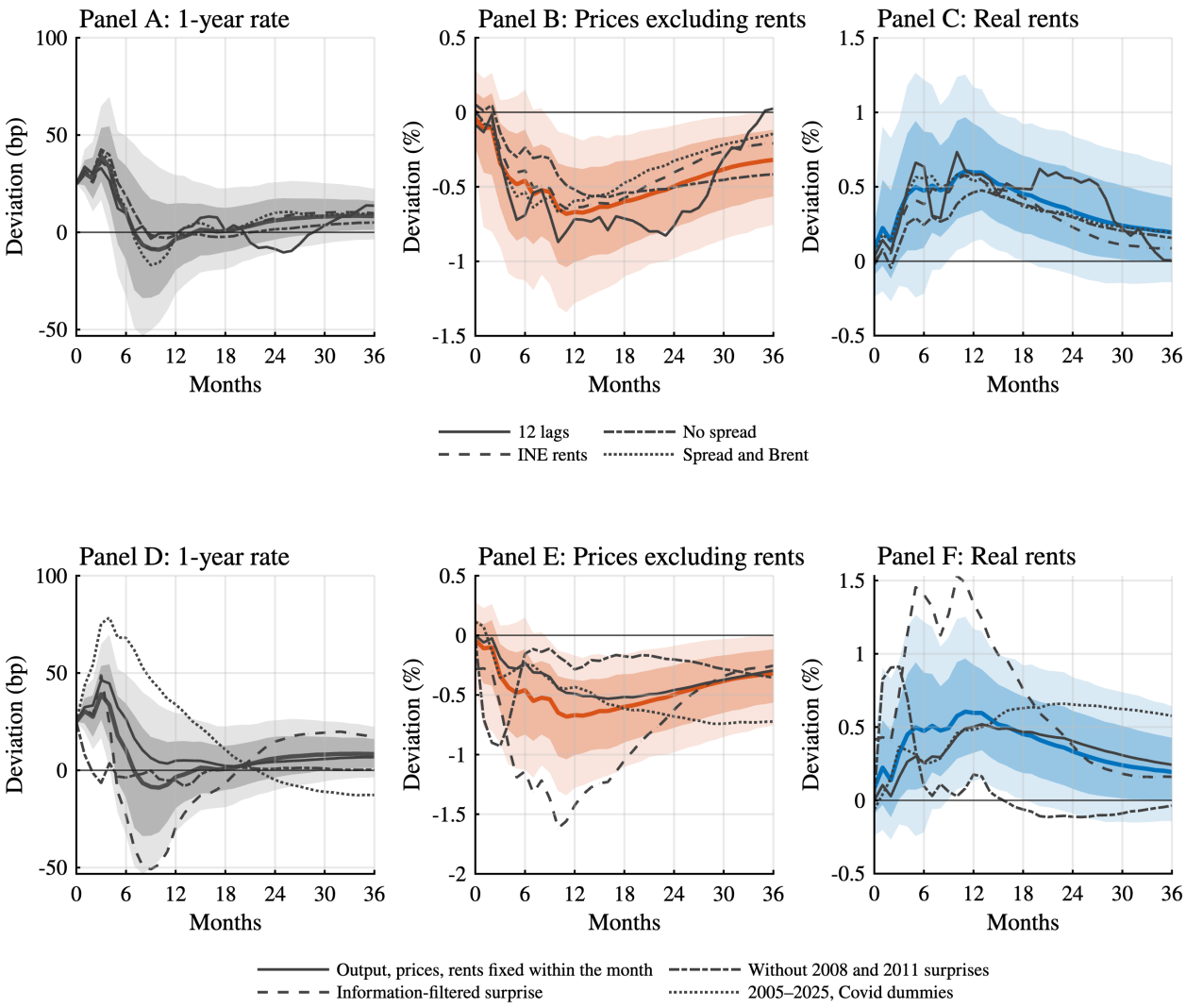



**Figure E.2.** Spanish VAR: Robustness

**Note.** Bands: baseline specification, 68 and 90 percent. Lines: bootstrap medians of the alternative specifications, each normalized to raise the one-year rate by 25 basis points on impact. Information-filtered surprise: announcements at which the one-year swap rate and the EURO STOXX 50 move in the same direction are set to zero.

## F.2 Rents and Lease Length

Figure F.1 repeats the monetary experiment of Section 4.1 for lengths $J$ of the period at the signing rent between one quarter and five years, holding $\varphi$ and all other parameters at their calibrated values. Every lease pays the same rent in the stationary equilibrium, so $J$ does not affect steady-state prices or allocations, only how leases are grouped by age. It shapes the response of the market rent. With $J = 1$, a new lease pays the market rent for a single quarter and the pooled rent afterward, so the market rent must move by more to clear the rental market: it rises by 1.68 percent in the second quarter and falls below its steady state from the seventh, and the average rent peaks at 0.15 percent in the fifth quarter and is back to 43 percent of its peak in the twelfth quarter. As $J$ rises, a new lease pays its own signing rent for longer, and the market rent rises by less and declines more slowly; its fall below steady state shrinks to 3.4 percent of its peak with $J = 8$ and disappears with $J = 20$. The average rent peaks later and lower, and its deviation in the twelfth quarter rises from 43 percent of its peak with $J = 1$ to 85 percent with $J = 8$ and 93 percent with $J = 20$ (Panel C). The aggregate responses are nearly identical: consumption falls by between 0.062 and 0.066 percent on impact for every $J$. The mean lag of the rent index changes by less than two quarters across $J$, from 20.5 quarters with $J = 1$ to 22.3 with $J = 20$: the persistence of the average rent comes from the path of the market rent.

The length $J$ matters more for the rent-inclusive rule of Section 5, which responds to the path of the average rent. With $J = 1$, the average rent rises faster and returns sooner, so under the model's rent share the rule adds about seven times as much tightening over the first two years, a cumulative 10.4 against 1.4 annualized basis points on the policy rate, and gives it back from the eleventh quarter. Consumption then falls by about as much as under the benchmark rule over two years and by 11 percent less over five, whereas with $J = 20$ it falls by 3 percent more over two years and 1 percent more over five. For every $J$, the rule delivers less nondurable disinflation per unit of output lost.

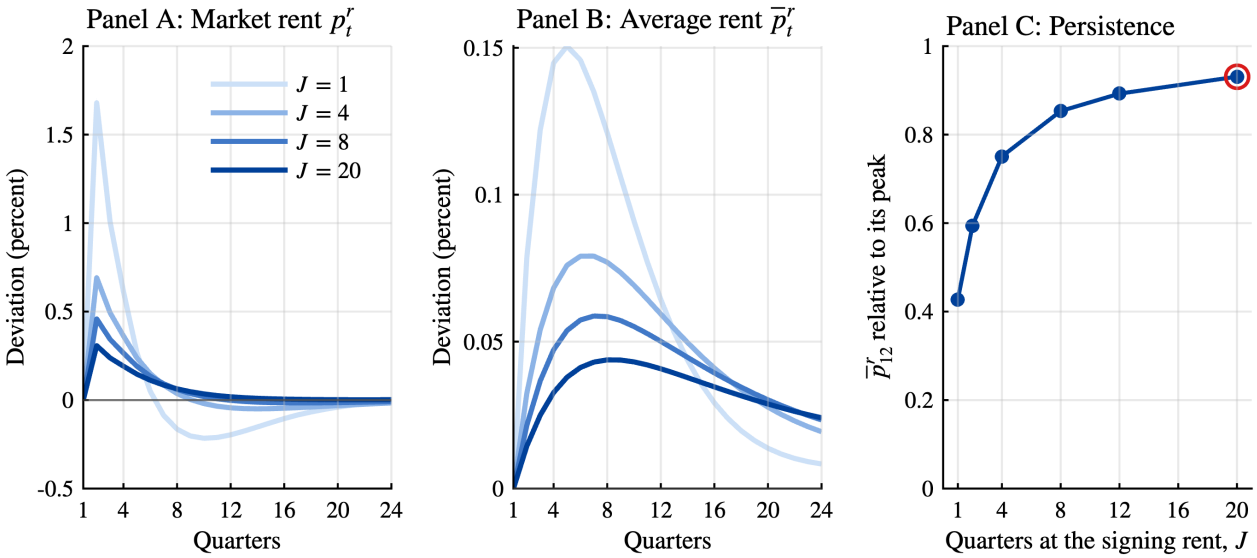



**Figure F.1.** Rents and the Length of the Period at the Signing Rent

**Note.** Responses of the real market rent $p_t^r$ (Panel A) and the real average rent $\overline{p}_t^r$ (Panel B) to the monetary shock of Section 4.1, in percent deviations from steady state, for four values of $J$, the number of quarters during which a lease pays its own signing rent. Panel C reports the average rent in quarter 12 relative to its peak for $J \in \{1, 2, 4, 8, 12, 20\}$; the circled point is the baseline, $J = 20$. All other parameters are held at their calibrated values.

## F.3 Rents and the Rental Yield

Figure F.2 repeats the monetary experiment of Section 4.1 in economies calibrated to different gross rental yields. We replace the residential-investment target of Section 3.2 with a target for the annual yield, $4p_{ss}^r/(p_{ss}^h \tilde{h}_1)$, between 3.5 and 6.3 percent, and recalibrate all internal parameters to the remaining targets. Because the house-value-to-income ratio pins down the house price, the yield fixes the steady-state market rent. Depreciation falls from 0.0073 to 0.0011 per quarter as the yield falls from 6.3 to 3.5 percent, and residential investment from 10.9 to 1.6 percent of GDP. Lower yields also move the rental burden away from its Spanish value: at a yield of 3.8 percent it is 22.7 percent, against 30.4 percent in our calibration and 35 percent in the data.

The sign of the rent response changes with the yield. Above a yield of about 3.8 percent, a tightening raises both the market rent and the average rent, as in the baseline; below it, both fall. Over the first three years, the real average rent is on average 3.6 basis points above its steady state at a yield of 6.3 percent and 1.6 basis points below it at 3.5 percent. In the second quarter, the market rent goes from 40 basis points above steady state to 14 basis points below (Panels A and B).

The switch follows the response of the house price. Its fall on impact is approximately inversely proportional to the yield, from 8.6 basis points at a yield of 6.3 percent to 15.4 basis points at 3.5 percent (Panel C). The price–rent ratio is the inverse of the yield, so at low yields more of a dwelling's value comes from rents in the distant future, and the same rise in the real rate lowers its price by more. A larger fall makes housing cheaper for prospective landlords and homebuyers,

which strengthens the channel that offsets the rise in rents in Section 4.3. The adjustment runs through the price rather than through quantities. At lower yields fewer households move into renting: 0.4 basis points of households in the eighth quarter at a yield of 3.5 percent, against 1.4 at 6.3 percent. The housing stock is within 0.2 basis points of its steady state ten years after the shock in every calibration.

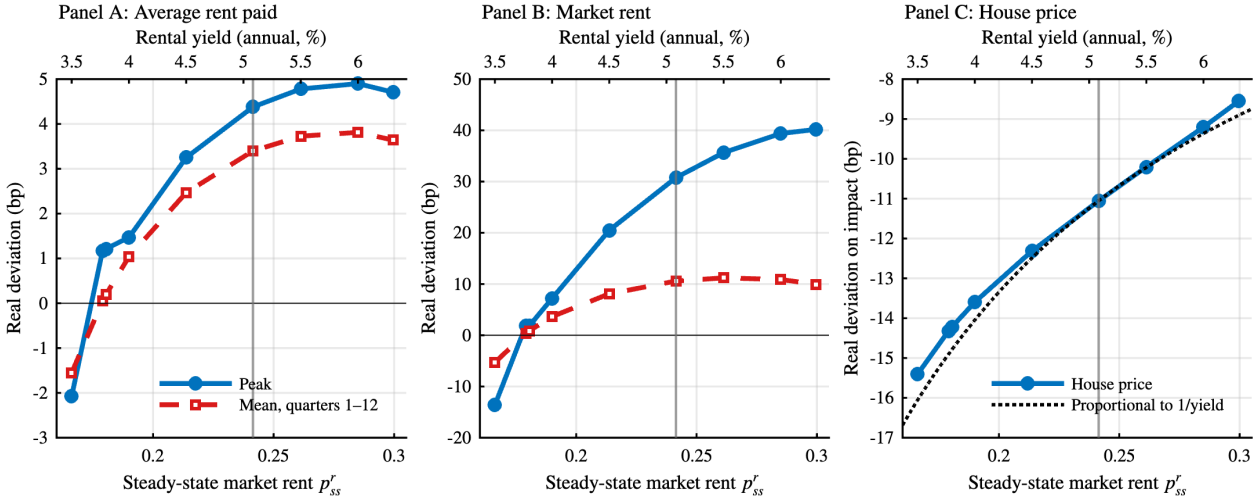



**Figure F.2.** Rents and the Rental Yield

**Note.** Responses to the monetary shock of Section 4.1 in economies recalibrated to the annual gross rental yield on the top axis, with residential investment untargeted. Panels A and B report the real average rent $\overline{p}^r_t$ and the real market rent $p^r_t$: the peak is the deviation largest in absolute value over the first ten years, and the mean is the average over quarters 1 to 12, in basis points. Panel C reports the real house price on impact, in basis points; the dotted line is proportional to the inverse of the yield and passes through the baseline. The vertical line marks the baseline calibration.

## F.4 Rental-Supply Elasticity

When we perturb the market-rent path around the calibrated steady state, the model's rental-supply elasticity is 0.37 over the impact quarter and the subsequent three quarters, just below the short-run range of 0.4 to 0.7 discussed below, and 1.06 over the following four quarters, between the short- and long-run benchmarks and above the 0.81 implied by the estimated model of De Llanos (2026).[50]

The elasticity rises with the rental yield. In the recalibrated economies of Appendix F.3, it is 0.80 in quarters 5 to 8 at a yield of 3.5 percent and 1.22 at 6.3 percent. Because the renter share is a target, the number of additional landlords per unit increase in the rent falls slightly across these economies. The elasticity rises because the same increase is a smaller percentage of a higher rent.

Rotberg and Steinberg (2024) infer a long-run rental-supply elasticity of 1.4 from the share

[50]The elasticity is a partial-equilibrium object: we move the market rent with persistence 0.97, close to the 0.98 that De Llanos (2026) estimates for his housing-demand shifter, let the rents of outstanding leases follow (8) and (A.6), and hold other prices, labor income, taxes, and tenure-retention rates at steady state. Quarters 5–8 are his year-one horizon.

of property taxes borne by renters in the United States, and obtain a short-run elasticity of 0.4 to 0.7 from the rise in rents in Warsaw and Kraków after the 2022 inflow of Ukrainian refugees documented by Trojanek and Głuszak (2022). The long-run estimate allows the aggregate housing stock to adjust; the short-run one refers to a horizon at which the stock is nearly fixed (their footnote 11). Our partial-equilibrium experiment measures the willingness to supply rental housing behind both estimates: it moves the market rent and records how many households choose to rent out a dwelling, holding other prices and incomes fixed.

A related object is the tenure-supply curve of Greenwald and Guren (2025), the price–rent ratios at which landlords supply different amounts of rental housing, which they estimate to be steep. In the model, this curve is also steep. The landlord margin gives it its slope, but the number of rental units can change only as construction changes the housing stock (equation (18)): after the monetary tightening of Section 4.1, the stock moves by less than 0.3 basis points while the house price falls by 11. A shift in rental demand therefore moves rents and house prices more than the number of rental units. The two objects play complementary roles. The partial-equilibrium elasticity measures how much the rent must rise to induce entry when holding a rental unit becomes more costly; the slow adjustment of the stock limits how much tenure reallocation the economy can absorb in the short run.

## F.5 Tenure-Transition Responses

The tenure flows behind the rent channels of Section 4.3 are visible in the transition probabilities. Figure F.3 plots, for each current tenure state $h \in \{0, 1, 2\}$, the deviation $\mathbb{P}_t(h' \mid h) - \mathbb{P}_{ss}(h' \mid h)$ for next-period choices $h'$, where $\mathbb{P}_t(h' \mid h)$ is the share of households in tenure state $h$ at date $t$ that choose $h'$. The homebuyer mortgage-cost channel appears in the renter panel: on impact, current renters are 0.03 percentage points more likely to remain renters and as much less likely to become owner-occupiers, so the tightening raises rental demand by keeping marginal renters in the rental market.[51]

The landlord channels appear in the owner and landlord panels. On impact, the probability that an owner-occupier becomes a landlord rises from 0.66 to 0.68 percent per quarter, and the probability that a landlord sells its rental unit rises from 2.55 to 2.60 percent. Owner-occupiers outnumber landlords almost four to one, so the additional entry more than offsets the additional exit, and the landlord share rises with the renter share, as clearing requires. Exits recede after the first year, and from the seventh quarter landlords are less likely to sell than in the stationary equilibrium.[52]

[51] The model holds the household unit fixed, so all adjustment runs through tenure choices. In the data, downturns can also reshape household composition: household size is countercyclical because young adults move back into, or delay leaving, the parental home (Dyrda et al., 2024).

[52] The discrete housing ladder shapes these flows: households cannot adjust the scale of a rental portfolio, so a landlord who reduces rental-property holdings must exit landlord status, and rental supply that existing landlords would otherwise provide by expanding their portfolios comes here through entry by new landlord households.

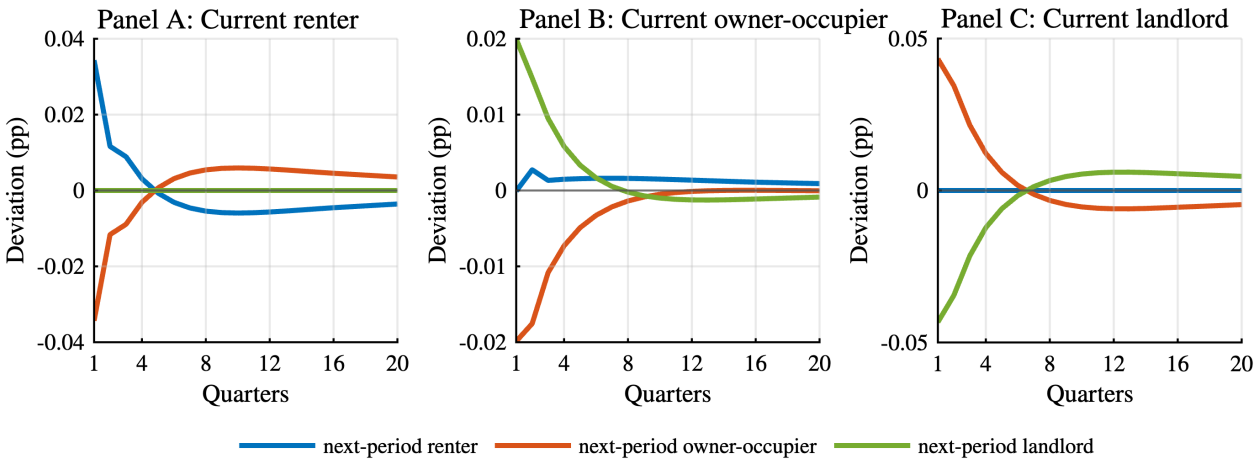



**Figure F.3.** Tenure-Transition Response to a Contractionary Monetary Shock

**Note.** Each panel conditions on the household's current tenure state and reports deviations in next-period tenure-choice probabilities from the stationary equilibrium, in percentage points. Conditional probabilities are computed by normalizing expected transition masses within each current tenure state.

## F.6 Consumption Incidence of a Monetary Shock across Households

We fix each household group, defined below, at the date of the shock and follow its members forward as they move across tenure and balance-sheet states. We compare each initial cohort's per-capita consumption along the transition with that of the identical initial population propagated under steady-state policies. This comparison avoids conflating the consumption response of a given cohort with changes in the composition of contemporaneous tenure groups.

We report two partitions: borrowers and savers, defined by net financial assets; and renters, mortgagor-owners, unlevered owners, mortgagor-landlords, and unlevered landlords, defined by tenure and leverage. Table F.2 reports, for both partitions, the impact response and the discounted sum of all consumption deviations over the transition. Figure F.4 reports impact responses across net-financial-asset bins in Panel A and the consumption paths of three representative initial cohorts in Panel B.

**Balance-sheet exposure.** Mortgagor-owners experience the largest impact consumption contraction, about a third larger than that of unlevered owners, and their consumption falls further in the second quarter, when the higher debt service comes due; mortgagor-landlords contract slightly more than unlevered landlords.[53] Tenure also matters within net-financial-asset bins: renters holding small positive balances contract by more than the average borrower.

Mortgage repricing connects these consumption responses to the landlord mortgage-cost and homebuyer mortgage-cost channels of Section 4.3. Higher borrowing costs discourage the provision

[53]The impact response is anticipatory rather than contemporaneous. The real return on assets carried into the first period is set before the shock, so debt service at $t = 1$ remains at its steady-state rate. Indebted households adjust consumption in anticipation of higher debt service from the following quarter onward.

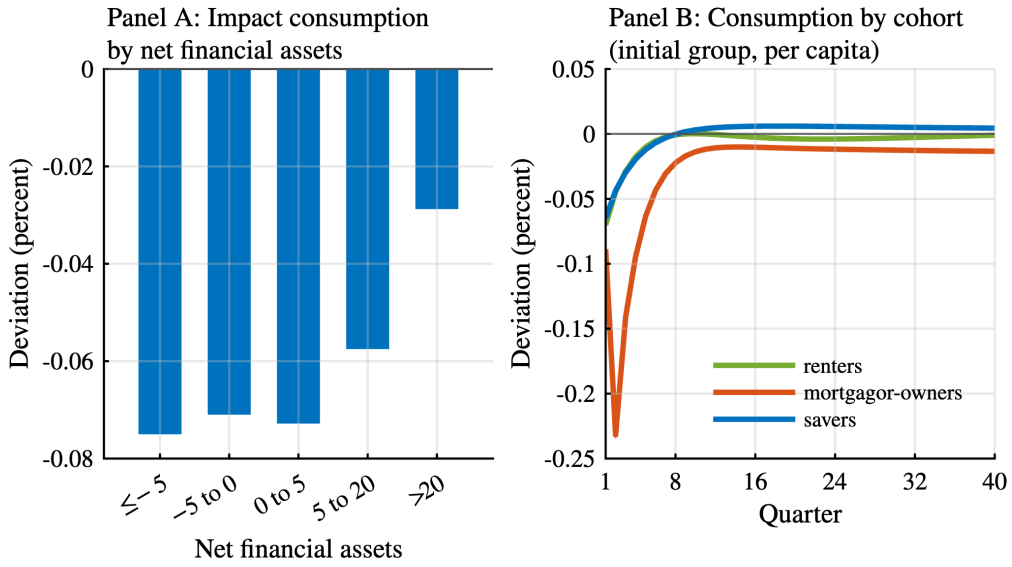



**Figure F.4.** Consumption Incidence of the Monetary Shock

**Note.** Panel A reports impact consumption responses by date-of-shock net-financial-asset bin (in units of quarterly steady-state income), in percent deviations from steady state. Panel B reports per-capita consumption paths for three representative initial cohorts (renters, mortgagor-owners, and savers), measured as percent deviations from each cohort's no-shock counterfactual; these span the exposure range but are not a partition, since savers ($a \geq 0$) include the renters and the unlevered owners and landlords.

of rental housing and entry into homeownership; they also reduce the resources available for consumption among households already carrying mortgage debt. The cohort responses reported here incorporate all equilibrium channels, rather than isolating these mortgage effects.

**Rent exposure.** Renters experience a smaller impact contraction than mortgagor-owners but a larger one than unlevered owners and landlords, and, unlike them, they consume less over the transition as a whole. Higher rental payments reduce the resources available for consumption and offset part of the benefit from higher returns on financial savings.

Lease stickiness determines when the higher rental bill arrives. Households signing new leases face the market rent, while renters on older leases keep paying the rent at which they signed (Section 2.2). The would-be buyers who remain renters therefore generally absorb the increase gradually. All rents remain at their steady-state values on impact, so the initial consumption response reflects changes in other equilibrium conditions and anticipation of future payments. As rents subsequently rise, rental exposure adds a burden that a classification based only on financial assets does not capture.

| | SS share | Impact | Discounted sum |
|---|---|---|---|
| *By net financial assets* | | | |
| Borrowers ($a < 0$) | 0.121 | −0.073 | −1.250 |
| Savers ($a \geq 0$) | 0.879 | −0.065 | 0.117 |
| *By housing tenure* | | | |
| Renters | 0.170 | −0.070 | −0.134 |
| Mortgagor-owners | 0.066 | −0.089 | −1.593 |
| Unlevered owners | 0.594 | −0.066 | 0.155 |
| Mortgagor-landlords | 0.056 | −0.061 | −0.968 |
| Unlevered landlords | 0.114 | −0.057 | 0.177 |

**Table F.2.** Consumption Incidence of the Monetary Shock across Initial Cohorts

**Note.** Each cohort is fixed at the date of the shock and tracked forward; entries are per-capita consumption deviations from the cohort's no-shock counterfactual, in percent. *Impact* is the $t$=1 response. *Discounted sum* adds all deviations over the first 100 quarters, discounted at the steady-state real rate; a positive value means that the cohort consumes more over the transition than in the counterfactual. Owner-occupiers and landlords are split by net financial assets into mortgagor ($a < 0$) and unlevered ($a \geq 0$) groups. Renters are not split by net financial assets: they cannot borrow (the third case of (5) imposes $a' \geq 0$ whenever $h' = 0$), so every renter is a saver. SS share is the steady-state population share.

## F.7 Rent-Inclusive Targets: Responses

Figure F.5 compares the responses to the monetary shock of Section 4.1 under the benchmark rule and under the model's rent share. Figure F.6 does the same for the composites with owner-occupied housing of Section 5.3. Table F.4 reports the impact, cumulative, and root-mean-square responses under the benchmark rule, the model's rent share, and the two composites; Table 4 in Section 5.2 summarizes them relative to the benchmark rule.

| | | Actual rentals | | Owner-occupied housing | |
|---|---|---|---|---|---|
| | Benchmark (no rents) | Spain's weight | Model's share | Rental equivalence | Net acquisitions |
| *First two years (quarters 1–8)* | | | | | |
| Real interest rate | + | +1.3 | +2.3 | +8.3 | −0.2 |
| Real house price | − | −1.8 | −3.1 | −11.5 | −8.6 |
| Real average rent | + | +1.9 | +3.4 | +12.5 | +14.4 |
| Consumption | − | −1.6 | −2.7 | −9.9 | −9.4 |
| Nondurable inflation | − | 0.0 | 0.0 | +0.3 | −5.6 |
| *First five years (quarters 1–20)* | | | | | |
| Real interest rate | + | +1.3 | +2.3 | +8.4 | +0.6 |
| Real house price | − | −1.4 | −2.4 | −8.8 | −5.5 |
| Real average rent | + | +1.8 | +3.1 | +11.5 | +15.5 |
| Consumption | − | −0.8 | −1.4 | −5.0 | −8.6 |
| Nondurable inflation | − | +1.5 | +2.7 | +10.7 | −1.0 |

**Table F.3.** Rent-Inclusive Targets: Responses Relative to the Benchmark Rule, Both Weights and Horizons

**Note.** Responses to the monetary policy shock of Section 4.1, cumulated over quarters 1–8 and 1–20. The first column gives the direction of the cumulative deviation from steady state under the benchmark rule, whose index excludes rents ($\omega_r = 0$), the same at both horizons. Each other entry is the difference between the cumulative deviation under the rule in the column and under the benchmark rule, in percent of the benchmark's absolute cumulative deviation; a negative entry means the variable is lower than under the benchmark rule. Actual rentals enter the index at Spain's HICP weight, 3.1 percent, or at the model's rent share, 5.3 percent. Rental equivalence weights owner-occupied housing at 13 percent and net acquisitions at 9 percent, with actual rentals at the model's share scaled by $1 - \omega_o$. Table 4 in Section 5.2 reports the first two years.

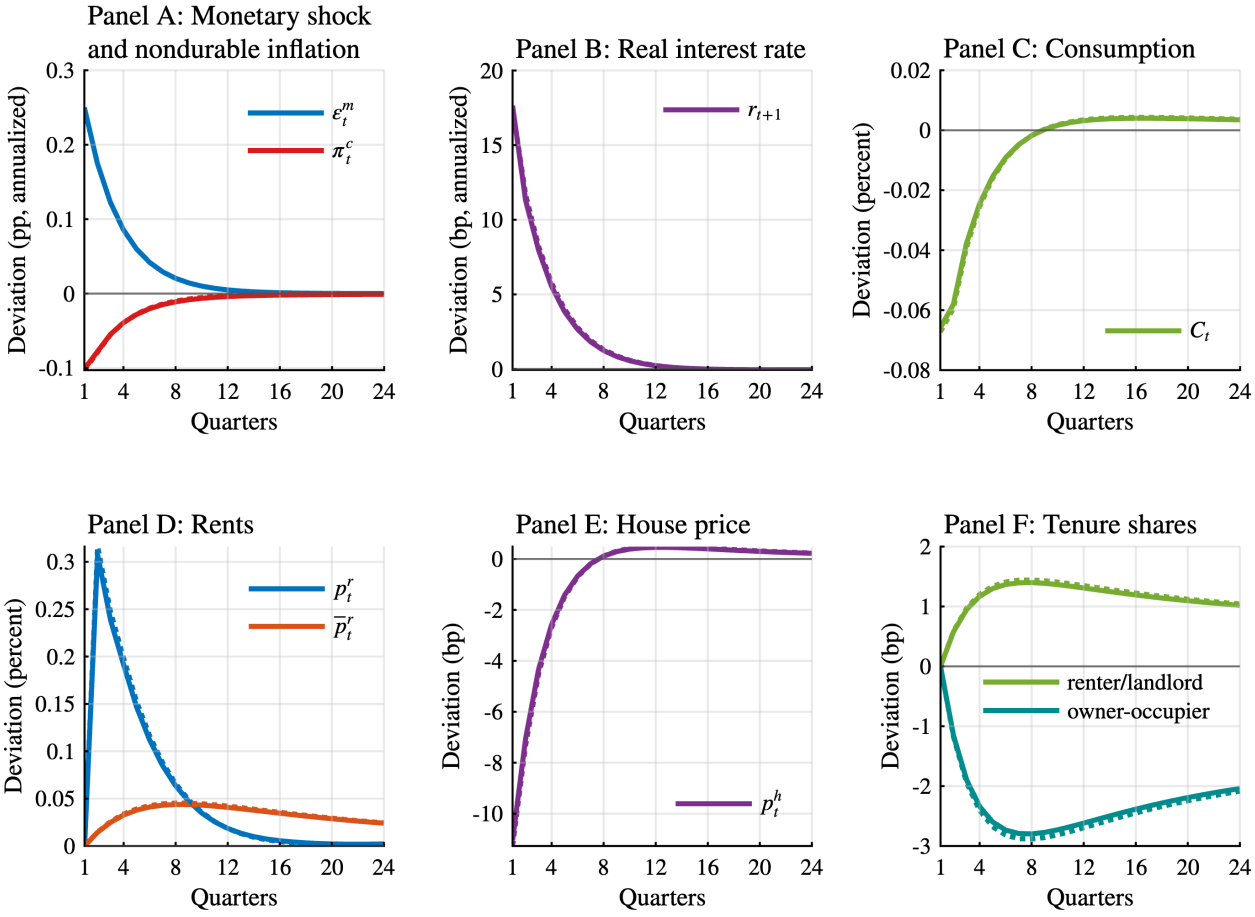



**Figure F.5.** Monetary Policy Shock under Alternative Policy Rules: Model's Rent Share

**Note.** Responses under $\omega_r = 0$ (solid lines) and $\omega_r = 0.053$, the model's rent share (dotted lines). Panel A reports the monetary policy innovation and nondurable inflation in annualized percentage points; Panel B reports the real interest rate in annualized basis points; Panel C reports consumption in percent deviations from steady state. Panel D reports real market and average rents in percent deviations; Panel E reports house-price deviations in basis points; Panel F reports tenure-share deviations in basis points.

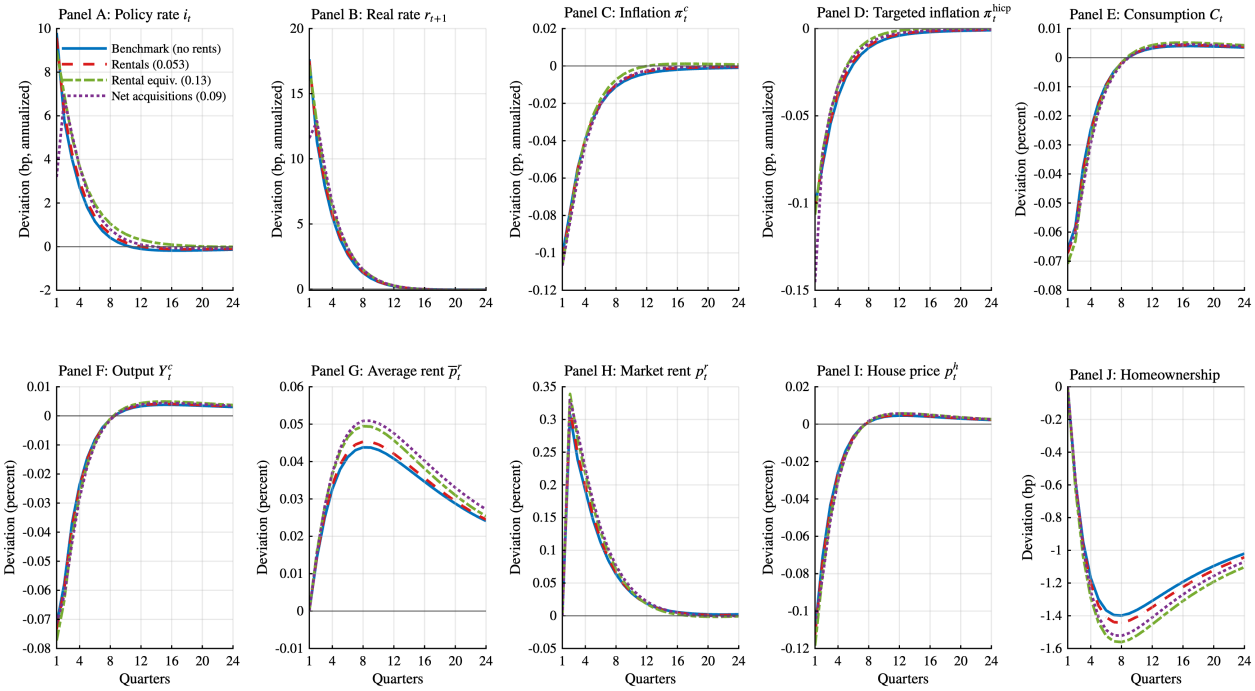



**Figure F.6.** Owner-Occupied Housing in the Inflation Target

**Note.** Responses to a 25-basis-point annualized monetary shock under the benchmark rule, the model's rent share on actual rentals ($\omega_r = 0.053$), rental equivalence ($\omega_o = 0.13$), and net acquisitions ($\omega_o = 0.09$), with the composite index of Section 5.3 and actual rentals at the model's share scaled by $1 - \omega_o$. Interest and inflation rates in annualized units; quantities and prices in percent deviations from steady state; homeownership in basis points.

| | Rule | Impact | Cumulative | | Root mean square | |
|---|---|---|---|---|---|---|
| | | | Q1–8 | Q1–20 | Q1–8 | Q1–20 |
| Consumption $C$ (%) | Benchmark (no rents) | −0.066 | −0.218 | −0.179 | 0.036 | 0.023 |
| | Rentals (0.053) | −0.067 | −0.224 | −0.182 | 0.036 | 0.023 |
| | Rental equiv. (0.13) | −0.071 | −0.239 | −0.188 | 0.039 | 0.025 |
| | Net acquisitions (0.09) | −0.066 | −0.238 | −0.194 | 0.038 | 0.024 |
| Nondurable output $Y^c$ (%) | Benchmark (no rents) | −0.072 | −0.218 | −0.180 | 0.037 | 0.023 |
| | Rentals (0.053) | −0.073 | −0.224 | −0.183 | 0.038 | 0.024 |
| | Rental equiv. (0.13) | −0.077 | −0.241 | −0.191 | 0.040 | 0.026 |
| | Net acquisitions (0.09) | −0.071 | −0.238 | −0.194 | 0.039 | 0.025 |
| Nondurable inflation $\pi^c$ (pp, annualized) | Benchmark (no rents) | −0.101 | −0.347 | −0.387 | 0.053 | 0.034 |
| | Rentals (0.053) | −0.103 | −0.348 | −0.377 | 0.053 | 0.034 |
| | Rental equiv. (0.13) | −0.107 | −0.346 | −0.346 | 0.055 | 0.035 |
| | Net acquisitions (0.09) | −0.107 | −0.367 | −0.391 | 0.056 | 0.036 |
| Real rate $r_{t+1}$ (bp, annualized) | Benchmark (no rents) | 17.6 | 52.0 | 54.0 | 8.4 | 5.3 |
| | Rentals (0.053) | 17.5 | 53.2 | 55.3 | 8.5 | 5.4 |
| | Rental equiv. (0.13) | 17.2 | 56.3 | 58.6 | 8.7 | 5.5 |
| | Net acquisitions (0.09) | 11.6 | 51.9 | 54.4 | 7.7 | 4.9 |
| Average rent $\overline{p}^r$ (%) | Benchmark (no rents) | 0.000 | 0.238 | 0.679 | 0.033 | 0.036 |
| | Rentals (0.053) | 0.000 | 0.246 | 0.700 | 0.034 | 0.037 |
| | Rental equiv. (0.13) | 0.000 | 0.268 | 0.757 | 0.037 | 0.040 |
| | Net acquisitions (0.09) | 0.000 | 0.272 | 0.785 | 0.038 | 0.041 |

**Table F.4.** Owner-Occupied Housing in the Inflation Target

**Note.** Responses to a 25-basis-point annualized monetary shock, in deviations from steady state, under the benchmark rule, a rule on actual rentals at the model's rent share (0.053), and HICP-plus-owner-occupied-housing composites in which owner-occupied housing has weight $\omega_o$ and every other item, actual rentals at the model's share included, is scaled by $1 - \omega_o$: rental equivalence ($\omega_o = 0.13$, priced with the average rent) and net acquisitions ($\omega_o = 0.09$, the nominal house price). Cumulative entries sum the quarterly deviations; the root mean square is taken over deviations from steady state.

## F.8 The Cost of the Rent-Inclusive Rule across Weights

Figure F.7 extends Table 4 to rent weights $\omega_r \in \{0, 0.01, \ldots, 0.15\}$, holding the economy and the monetary policy shock fixed. The responses of the real interest rate, the house price, the average rent, and consumption move monotonically with the weight, and almost in proportion to it. At $\omega_r = 0.15$, over the first two years the rise in the real interest rate, the fall in house prices, the rise in the average rent, and the fall in consumption are 7, 10, 10, and 8 percent larger than under the benchmark rule. Nondurable inflation over the same two years is nearly unchanged at every weight. Over five years, the rule delivers less disinflation as the weight rises, 9 percent less at $\omega_r = 0.15$. The costs of responding to rents therefore scale with the weight on rents, and the disinflation they buy does not.

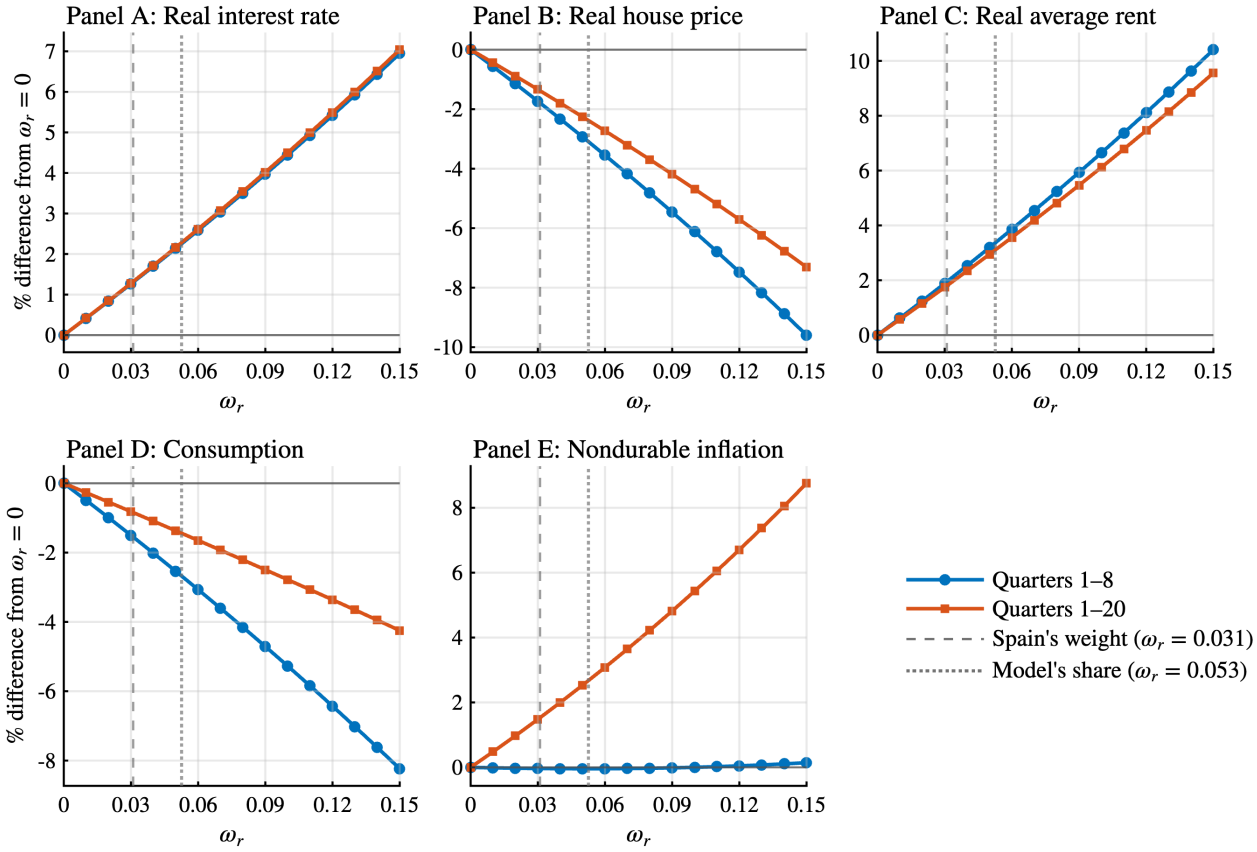



**Figure F.7.** Rent-Inclusive Targets across Rent Weights

**Note.** Responses to the monetary policy shock of Section 4.1. Each panel reports the difference between the cumulative deviation under weight $\omega_r$ and under the benchmark rule ($\omega_r = 0$), in percent of the benchmark's absolute cumulative deviation, over quarters 1–8 and 1–20, as in Table 4; a negative value means the variable is lower than under the benchmark rule. The dashed and dotted lines mark Spain's HICP weight and the model's rent share.

| | | Benchmark rule | | Additional, $\omega_r = 0.031$ | | Additional, $\omega_r = 0.053$ | |
|---|---|---|---|---|---|---|---|
| | SS share | Impact | Losses | Impact | Losses | Impact | Losses |
| Borrowers ($a < 0$) | 0.121 | −0.073 | −1.474 | −0.0009 | −0.0162 | −0.0016 | −0.0278 |
| Savers ($a \geq 0$) | 0.879 | −0.065 | −0.180 | −0.0008 | −0.0030 | −0.0013 | −0.0052 |
| Renters | 0.170 | −0.070 | −0.258 | −0.0008 | −0.0017 | −0.0014 | −0.0028 |
| Mortgagor-owners | 0.066 | −0.089 | −1.862 | −0.0011 | −0.0208 | −0.0019 | −0.0357 |
| Unlevered owners | 0.594 | −0.066 | −0.194 | −0.0008 | −0.0033 | −0.0014 | −0.0058 |
| Mortgagor-landlords | 0.056 | −0.061 | −1.151 | −0.0008 | −0.0124 | −0.0013 | −0.0213 |
| Unlevered landlords | 0.114 | −0.057 | −0.133 | −0.0007 | −0.0023 | −0.0011 | −0.0040 |

**Table F.5.** Consumption Incidence of Rent-Inclusive Rules across Initial Cohorts

**Note.** Cohorts are fixed at the date of the 25-basis-point monetary shock, as in Table F.2. Entries are per-capita consumption deviations from the cohort's no-shock counterfactual, in percent. *Losses* sums negative deviations over the first 100 quarters. *Additional* columns report the difference between the rent-inclusive rule and the benchmark rule; a negative entry is a larger loss.

## F.9 Distributional Incidence of the Rent-Inclusive Rule

Table F.5 applies the fixed-cohort method of Appendix F.6 to the two monetary policy rules. The last four columns measure the additional consumption loss generated by including rent inflation in the policy index, at each of the two weights. The impact contraction increases for every cohort. Although the real rate chosen on impact is slightly lower under the rent-inclusive rule, the return on assets carried into that period is predetermined. Households instead adjust to the anticipated path: indebted households face higher future debt service, while renters anticipate higher rental payments.

The additional loss is largest for mortgagor-owners. Measured relative to each cohort's no-shock consumption, it is about 12 times that of renters. Mortgage exposure therefore remains central to the incidence of a rule that responds to rent inflation. Renters also lose through higher rental payments and the broader contraction. The sum of losses increases for every tenure cohort. These consumption comparisons describe exposure to the policy-induced equilibrium adjustment; welfare additionally depends on households' marginal utilities.

## F.10 Alternative Lease Indexation

In the baseline, leases are fully indexed: the real rent of a continuing lease does not respond to inflation, because its nominal rent is updated every quarter by current nondurable inflation (Section 2.2). Spanish leases are instead updated once a year, on the anniversary of the contract, by the change in consumer prices over the previous year. We consider three alternatives with less indexation: lagged indexation, a quarterly update by the previous quarter's inflation; annual indexation, an update on each anniversary by nondurable inflation over the four quarters ending in the previous quarter; and no indexation, with nominal rents fixed for the life of the lease. A new lease pays the market rent in every case, and the stationary equilibrium is unchanged, since

steady-state inflation is zero.

Under the three alternatives, the nominal rents of continuing leases do not respond to inflation in the impact quarter. The tightening lowers prices, so their real value rises by the fall in the price level, and measured rent inflation reflects only the new leases, 4.7 percent of the total: it is $-0.005$ annualized percentage points on impact, against $-0.10$ for nondurable inflation. Remark 1 therefore no longer holds. The policy index exceeds nondurable inflation on impact by 0.3 basis points under Spain's weight and 0.5 under the model's rent share, and the rent term raises the policy rate even on impact, by 0.6 to 0.7 basis points under the model's rent share, rather than from the second quarter.

Figure F.8 shows that the real average rent rises on impact and peaks higher and earlier than in the baseline. Under annual updates the increase is shorter-lived, because each anniversary passes the lower price level into the nominal rent. With nominal rents fixed for the life of the lease, the market rent rises by half as much, because a lease whose nominal rent is fixed gains real value as prices keep falling, so a lower rent at signing delivers the same real rent over the life of the contract. Table F.6 repeats Table 4 under each indexation. Under both weights on actual rentals, the additional tightening stays below one basis point, house prices fall by more than under the benchmark rule, and the rent term buys no additional disinflation over two years. The consumption differences are smaller than in the baseline under lagged indexation, and larger over two years without indexation; under annual indexation, consumption falls slightly less than under the benchmark rule over five years.

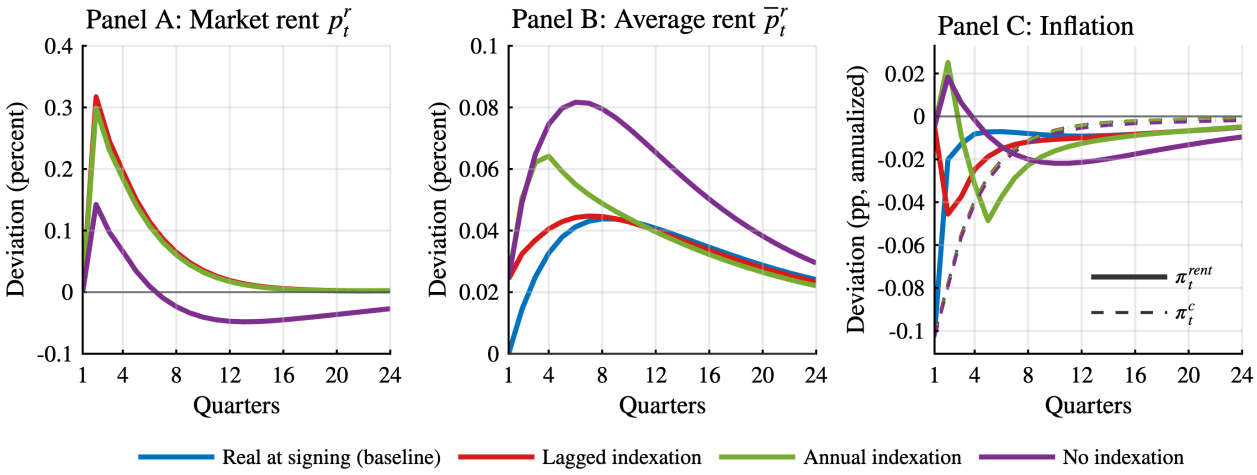



**Figure F.8.** Rents under Alternative Lease Indexation

**Note.** Responses to the monetary policy shock of Section 4.1 under the benchmark rule. Panels A and B report percent deviations from steady state; Panel C reports annualized percentage points.

| | Benchmark | | Actual rentals | | | | Owner-occupied housing | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | (no rents) | | Spain's weight | | Model's share | | Rental equivalence | | Net acquisitions | |
| | Q1–8 | Q1–20 | Q1–8 | Q1–20 | Q1–8 | Q1–20 | Q1–8 | Q1–20 | Q1–8 | Q1–20 |
| *Real rents fixed at signing (baseline)* | | | | | | | | | | |
| Real interest rate | + | + | +1.3 | +1.3 | +2.3 | +2.3 | +8.3 | +8.4 | −0.2 | +0.6 |
| Real house price | − | − | −1.8 | −1.4 | −3.1 | −2.4 | −11.5 | −8.8 | −8.6 | −5.5 |
| Real average rent | + | + | +1.9 | +1.8 | +3.4 | +3.1 | +12.5 | +11.5 | +14.4 | +15.5 |
| Consumption | − | − | −1.6 | −0.8 | −2.7 | −1.4 | −9.9 | −5.0 | −9.4 | −8.6 |
| Nondurable inflation | − | − | 0.0 | +1.5 | 0.0 | +2.7 | +0.3 | +10.7 | −5.6 | −1.0 |
| *Lagged indexation: every quarter, by the previous quarter's nondurable inflation* | | | | | | | | | | |
| Real interest rate | + | + | +1.5 | +1.5 | +2.6 | +2.5 | +9.2 | +8.9 | +0.2 | +0.8 |
| Real house price | − | − | −1.4 | −1.1 | −2.4 | −1.9 | −8.4 | −6.7 | −7.8 | −5.1 |
| Real average rent | + | + | +0.8 | +0.6 | +1.4 | +1.1 | +4.7 | +3.7 | +11.1 | +12.8 |
| Consumption | − | − | −1.1 | −0.3 | −1.8 | −0.5 | −6.3 | −1.3 | −8.5 | −7.7 |
| Nondurable inflation | − | − | +0.5 | +1.9 | +0.8 | +3.3 | +3.2 | +11.9 | −4.8 | −0.4 |
| *Annual indexation: on each anniversary, by nondurable inflation over the previous year* | | | | | | | | | | |
| Real interest rate | + | + | +2.0 | +1.8 | +3.5 | +3.0 | +11.4 | +10.0 | +1.0 | +1.3 |
| Real house price | − | − | −1.1 | −0.9 | −1.9 | −1.6 | −5.9 | −4.8 | −7.3 | −4.6 |
| Real average rent | + | + | −0.1 | −0.5 | −0.2 | −0.9 | −1.4 | −3.9 | +8.6 | +9.9 |
| Consumption | − | − | −0.6 | +0.5 | −1.0 | +0.8 | −2.7 | +3.4 | −7.6 | −6.5 |
| Nondurable inflation | − | − | +1.4 | +2.7 | +2.3 | +4.6 | +8.3 | +15.3 | −3.2 | +0.8 |
| *No indexation: nominal rents fixed for the life of the lease* | | | | | | | | | | |
| Real interest rate | + | + | +3.1 | +2.8 | +5.3 | +4.8 | +19.5 | +16.9 | +2.6 | +2.7 |
| Real house price | − | − | −2.4 | −1.0 | −4.2 | −1.6 | −14.0 | −3.8 | −9.4 | −4.6 |
| Real average rent | + | + | +1.7 | +0.8 | +2.9 | +1.2 | +7.7 | −0.3 | +11.3 | +10.5 |
| Consumption | − | − | −2.3 | −0.7 | −3.8 | −1.1 | −12.0 | +0.4 | −10.5 | −8.9 |
| Nondurable inflation | − | − | +1.6 | +4.4 | +2.9 | +7.7 | +13.8 | +29.5 | −2.6 | +3.7 |

**Table F.6.** Rent-Inclusive Targets under Alternative Lease Indexation: Responses Relative to the Benchmark Rule

**Note.** Responses to the monetary policy shock of Section 4.1 under four indexation schemes of the rents of continuing leases; a new lease pays the market rent $p_t^r$ in every case. Nondurable inflation, which excludes rents, approximates the update provisions of Spanish leases (Appendix D.5). The first two columns give the direction of the cumulative deviation from steady state under the benchmark rule, whose index excludes rents ($\omega_r = 0$), with the same indexation, over quarters 1–8 and 1–20. Each other entry is the difference between the cumulative deviation under the rule in the column and under that benchmark rule, in percent of the benchmark's absolute cumulative deviation; a negative entry means the variable is lower than under the benchmark rule. Under the benchmark rule, the cumulative real average rent over quarters 1–8 is +0.24, +0.31, +0.42 and +0.54 percent-quarters under the four indexations, in the order of the panels, and the cumulative fall in consumption 0.22, 0.22, 0.21 and 0.20 percent-quarters. Actual rentals enter the index at Spain's HICP weight, 3.1 percent, or at the model's rent share, 5.3 percent. Rental equivalence weights owner-occupied housing at 13 percent and net acquisitions at 9 percent, with actual rentals at the model's share scaled by $1 - \omega_o$.

## F.11 Patience Shock

Figure F.9 compares the responses to the patience shock under the benchmark (solid lines) and the optimized rule (dotted lines). The optimized rule removes most of the rise in the real rate after the first quarter. Rents, the house price, and the tenure shares move much less, while consumption, output, and nondurable inflation rise by more.

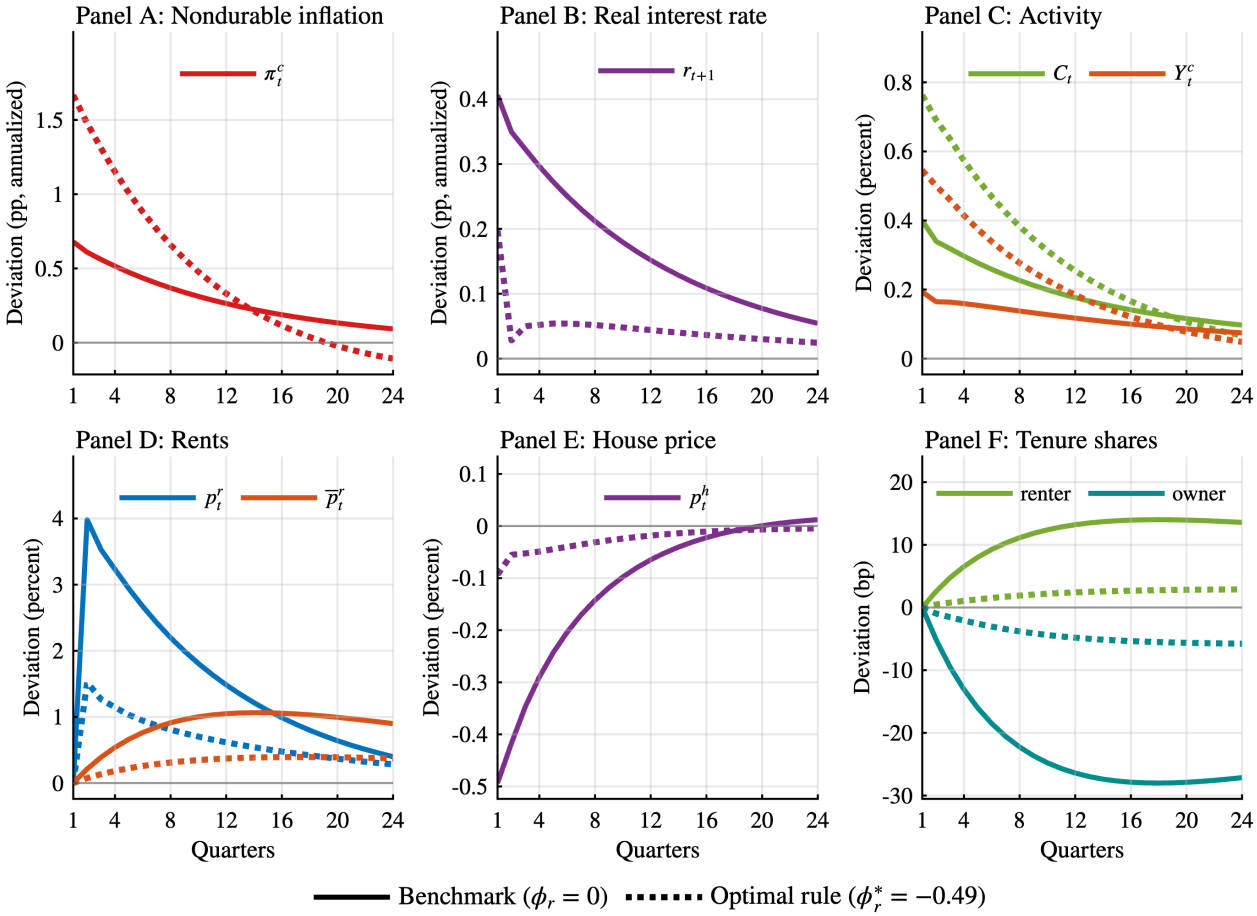



**Figure F.9.** Patience Shock under Alternative Policy Rules

**Note.** Responses under $\phi_r = 0$ (solid lines) and the optimized rule, $\phi_r = -0.49$ (dotted lines), with $\phi_\pi = 1.5$.

## F.12 Government Spending Shock

Figure F.10 compares the responses to the government-spending shock. As under the patience shock, the optimized rule removes most of the rise in the real rate. Consumption stays close to its steady state, rents rise by half as much, and nondurable inflation is higher. Figure F.11 reports the distributional decomposition.

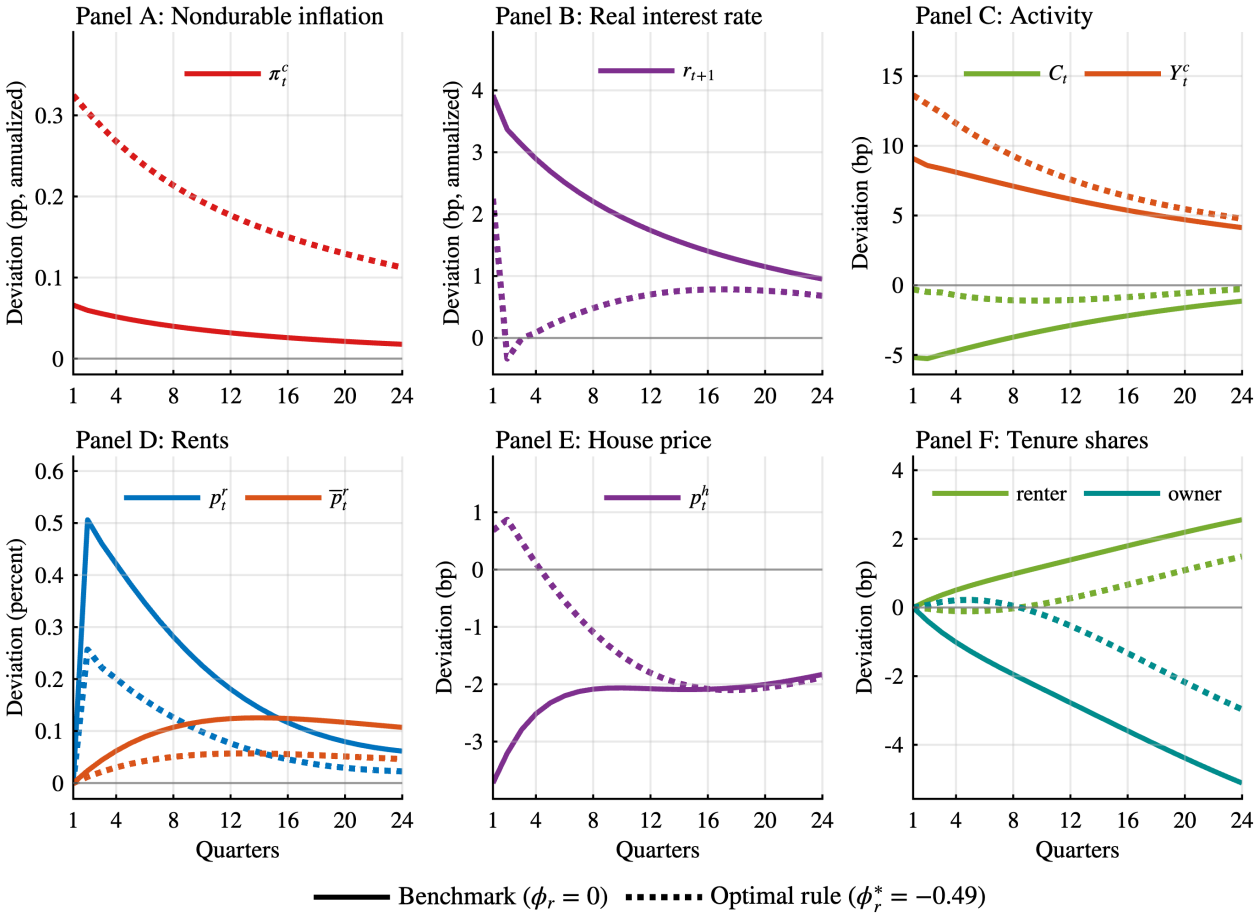



**Figure F.10.** Government-Spending Shock under Alternative Policy Rules

**Note.** Responses under $\phi_r = 0$ (solid lines) and the optimized rule, $\phi_r = -0.49$ (dotted lines), with $\phi_\pi = 1.5$.

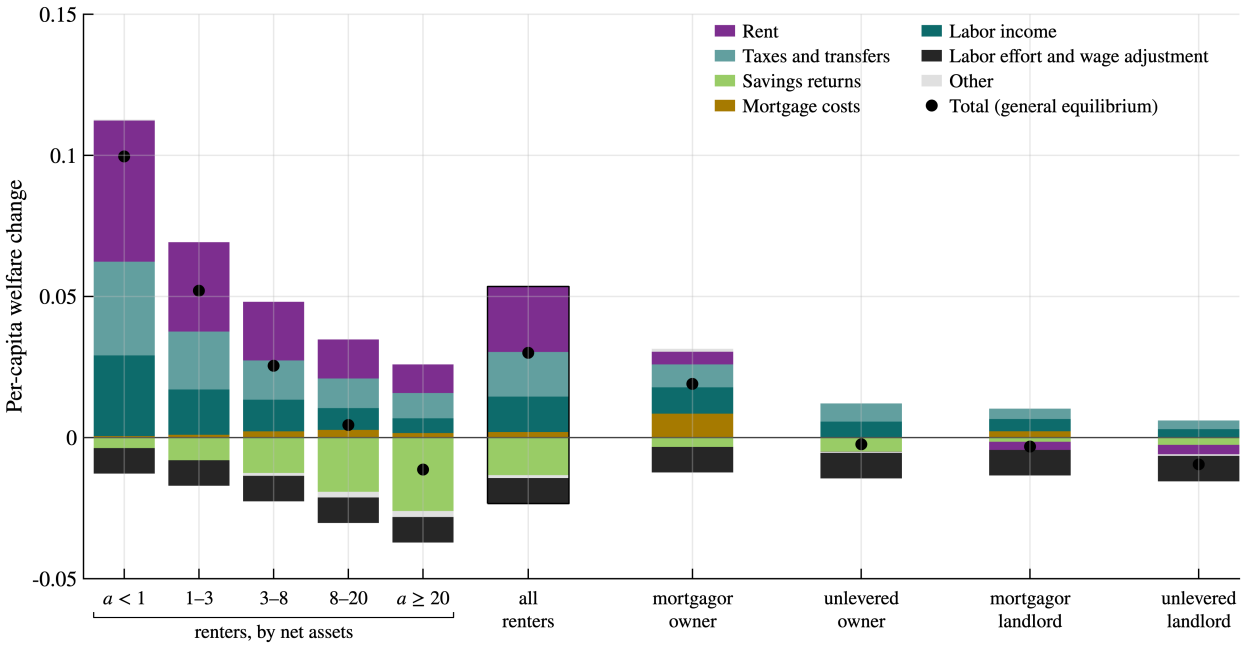



**Figure F.11.** Distributional Decomposition of the Welfare Gain: Government-Spending Shock

**Note.** Per-capita welfare change from moving $\phi_r$ from 0 to $-0.49$, by date-of-shock cohort, in utility units. Cohort definitions, channels, colors, and decomposition conventions follow Figure 7.

## F.13 Rental Supply Shock

Figure F.12 compares the responses to the rental supply shock. The shock raises rents with almost no movement in nondurable inflation, so the benchmark rule barely moves and the optimized rule eases because of its rent term. The optimized rule lowers the real rate by up to five basis points; rents rise less, the house price rises more, and more households become owners. Figure F.13 reports the distributional decomposition.

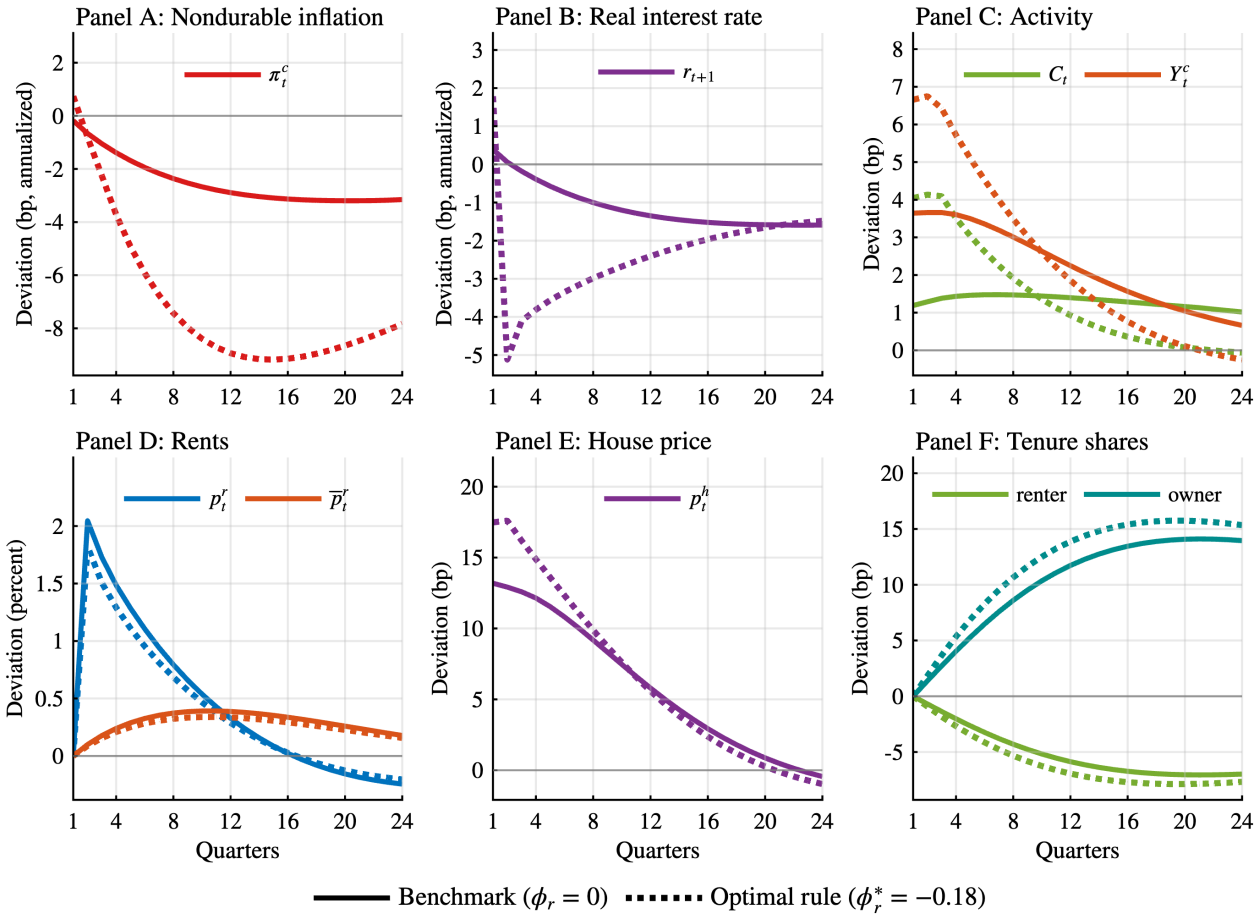



**Figure F.12.** Rental Supply Shock under Alternative Policy Rules

**Note.** Responses under $\phi_r = 0$ (solid lines) and the optimized rule, $\phi_r = -0.18$ (dotted lines), with $\phi_\pi = 1.5$.

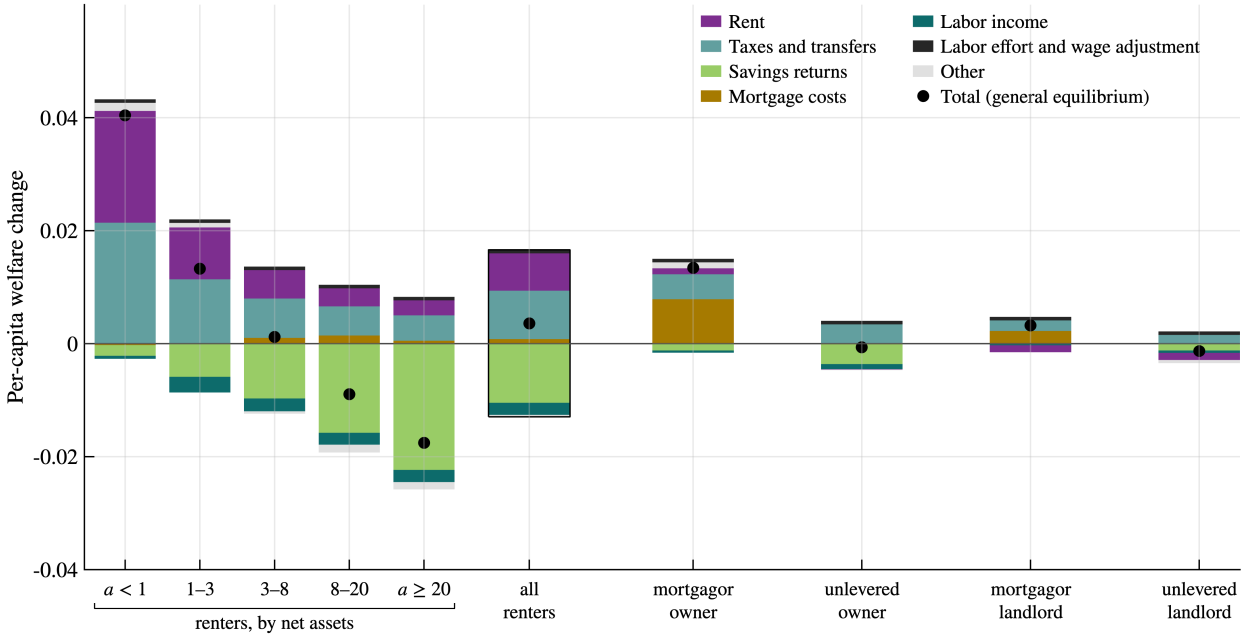



**Figure F.13.** Distributional Decomposition of the Welfare Gain: Rental Supply Shock

**Note.** Per-capita welfare change from moving $\phi_r$ from 0 to $-0.18$, by date-of-shock cohort, in utility units. Cohort definitions, channels, colors, and decomposition conventions follow Figure 7.

## F.14 Cost-Push Shock

Figure F.14 compares the responses to the cost-push shock. The optimized rule halves the rise in the real rate on impact. Rents rise by half as much, the house price falls by less, and consumption falls by half as much on impact; nondurable inflation is slightly higher on impact and slightly lower afterward. Figure F.15 reports the distributional decomposition.

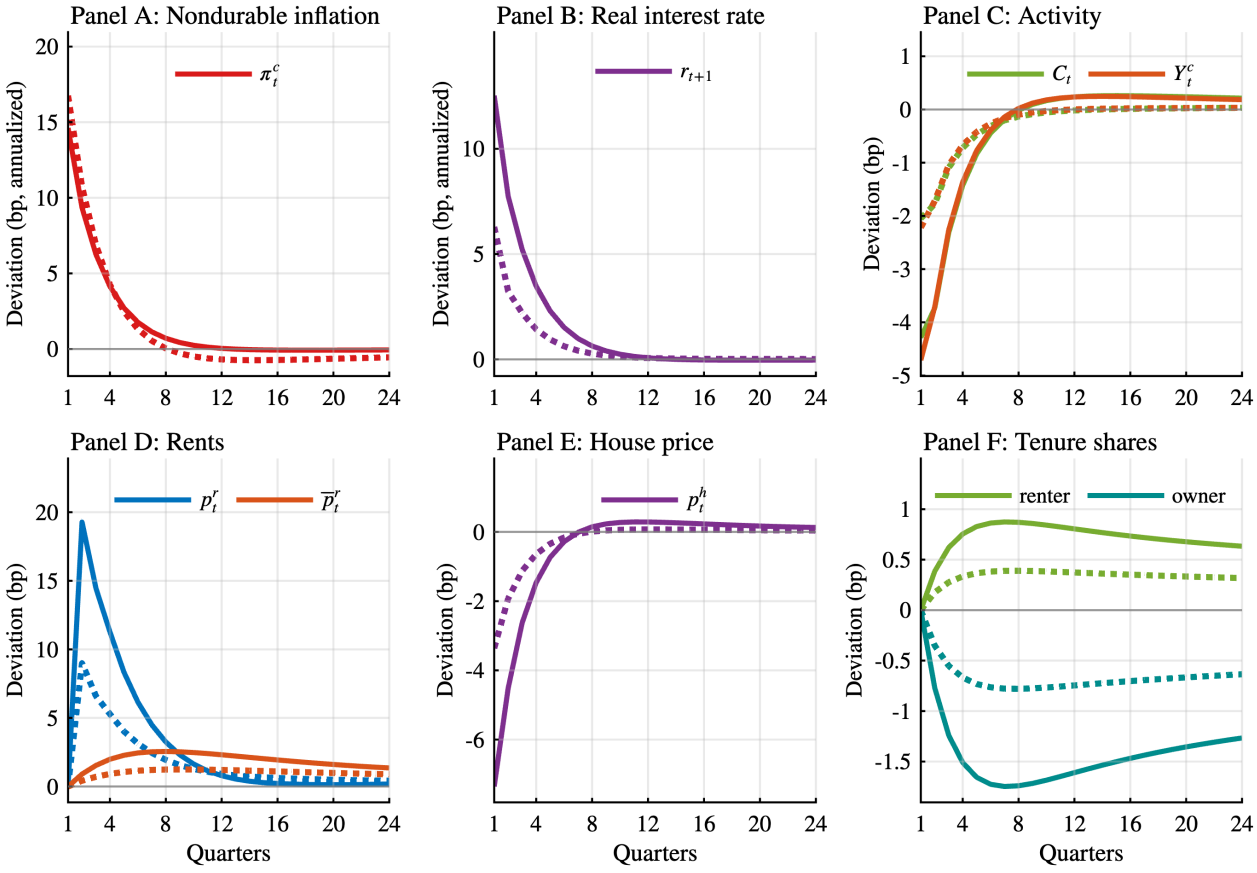



**Figure F.14.** Cost-Push Shock under Alternative Policy Rules

**Note.** Responses under $\phi_r = 0$ (solid lines) and the optimized rule, $\phi_r = -0.48$ (dotted lines), with $\phi_\pi = 1.5$.

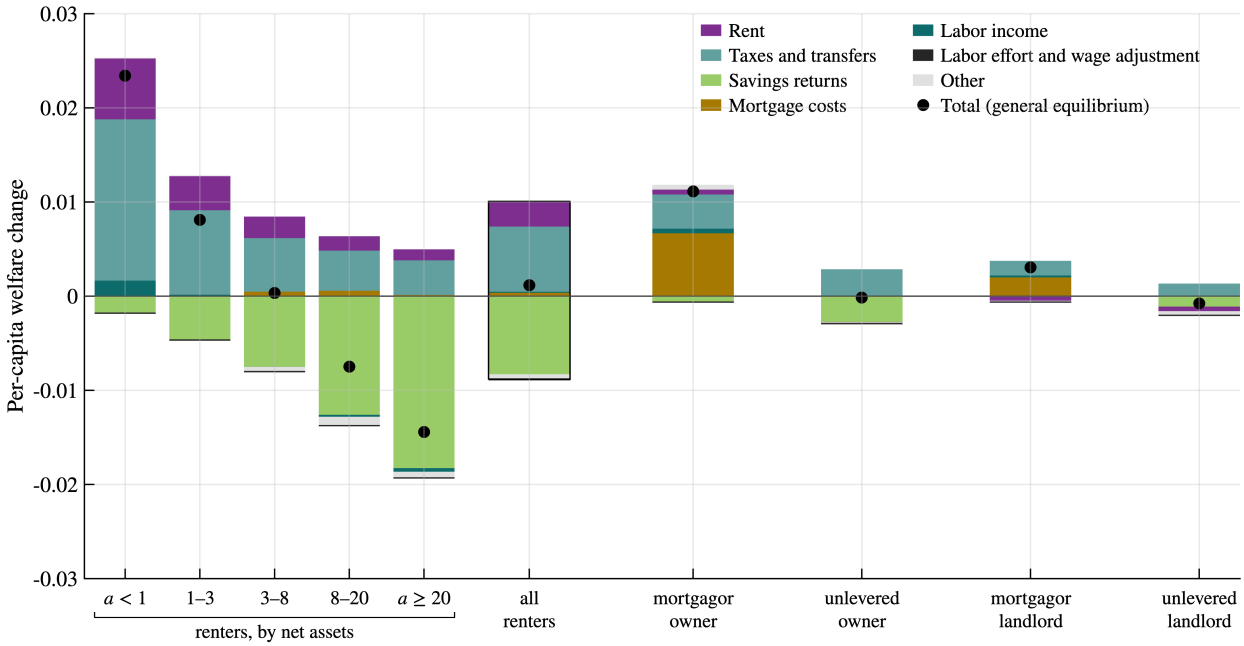



**Figure F.15.** Distributional Decomposition of the Welfare Gain: Cost-Push Shock

**Note.** Per-capita welfare change from moving $\phi_r$ from 0 to $-0.48$, by date-of-shock cohort, in utility units. Cohort definitions, channels, colors, and decomposition conventions follow Figure 7.

## F.15 Alternative Rules and Fiscal Closures

Table F.8 reports the welfare gain of rules that vary $\phi_\pi$ and $\phi_r$ jointly. Each row holds the effective coefficient on nondurable inflation in (29), $\phi_\pi + \phi_r$, fixed, so that along a row only the response to real rent growth changes. The best rule in each panel lies on the grid's edge at $\phi_r = -0.4$, so we also optimize the two coefficients jointly over the admissible rules of Appendix C.2, those with $0 \leq \phi_\pi \leq 4$ under which the equilibrium is determinate (Figure C.2). Under the patience and government-spending shocks welfare rises up to the determinacy frontier, and we report the rules just above it, $(\phi_\pi, \phi_r) = (1.67, -0.66)$ and $(3.41, -2.40)$, with gains of 1.27 and 0.49 basis points against 1.09 and 0.35 for the best rule with $\phi_\pi = 1.5$. Under the rental supply shock it is $(4, -2.28)$, at the upper bound on $\phi_\pi$ and well inside the determinate region, with a gain of 0.76 basis points against 0.11. In all three cases the rule responds negatively to the real rent growth of (29), with coefficients of $-0.66$, $-2.40$, and $-2.28$, while its overall response to nondurable inflation, $\phi_\pi + \phi_r$, is 1.01, just above the determinacy frontier, under the two demand shocks and 1.72 under the rental supply shock. Under the cost-push shock the optimal rule is $(0, 1.01)$, the corner at which the rule responds to rent inflation alone, with a gain of 0.11 basis points against 0.09; the differences across rules are small under this shock. Table F.9 reports the gains of the optimized rules by tenure group, in consumption equivalents.

Table F.7 reports the gain from moving $\phi_r$ from 0 to $-0.20$ under alternative rules, fiscal closures, and lease blocks, each against its own $\phi_r = 0$ benchmark. The gain is positive in every case. With interest-rate smoothing, the gain under the rental supply shock peaks at a smaller coefficient, near $-0.13$, where it is about 30 percent of the baseline gain.

| | Demand | | Supply | |
|---|---|---|---|---|
| | Patience | Government spending | Rental supply | Cost push |
| Baseline | 0.492 | 0.100 | 0.105 | 0.039 |
| Interest-rate smoothing, $\rho_i = 0.8$ | 0.536 (1.09) | 0.090 (0.90) | 0.018 (0.17) | 0.032 (0.82) |
| Output gap in the rule, $\phi_y = 0.125$ | 0.528 (1.07) | 0.034 (0.34) | 0.055 (0.53) | 0.034 (0.87) |
| Transfer proportional to productivity | 0.395 (0.80) | 0.119 (1.19) | 0.056 (0.53) | 0.034 (0.89) |
| Shorter leases, $J = 4$ | 0.774 (1.57) | 0.119 (1.19) | 0.109 (1.04) | 0.041 (1.06) |
| Non-renegotiation probability, $\varphi = 0.95$ | 0.728 (1.48) | 0.117 (1.17) | 0.115 (1.10) | 0.041 (1.06) |

**Table F.7.** Robustness of the Negative Response

**Note.** Welfare gain from moving $\phi_r$ from 0 to $-0.20$, with $\phi_\pi = 1.5$, after a one-standard-deviation shock, in basis points of permanent consumption, each against its own $\phi_r = 0$ benchmark. In parentheses, the ratio to the baseline gain. With smoothing, the rule is $i_t = r_{ss} + \rho_i(i_{t-1} - r_{ss}) + (1 - \rho_i)(\phi_\pi \pi_t^c + \phi_r \pi_t^{\text{rent}})$; the output-gap term adds $\phi_y(Y_t^c / Y_{ss}^c - 1)$ to rule (27).

Figure F.16 decomposes the change in the transfer under the four optimized rules. The lower real rate cuts the government's debt service, debt falls, and the fiscal rule returns the saving to households as a transfer that builds up over several years. Summed over the transition, lower interest payments drive the change in the transfer; tax and spread revenues move much less.

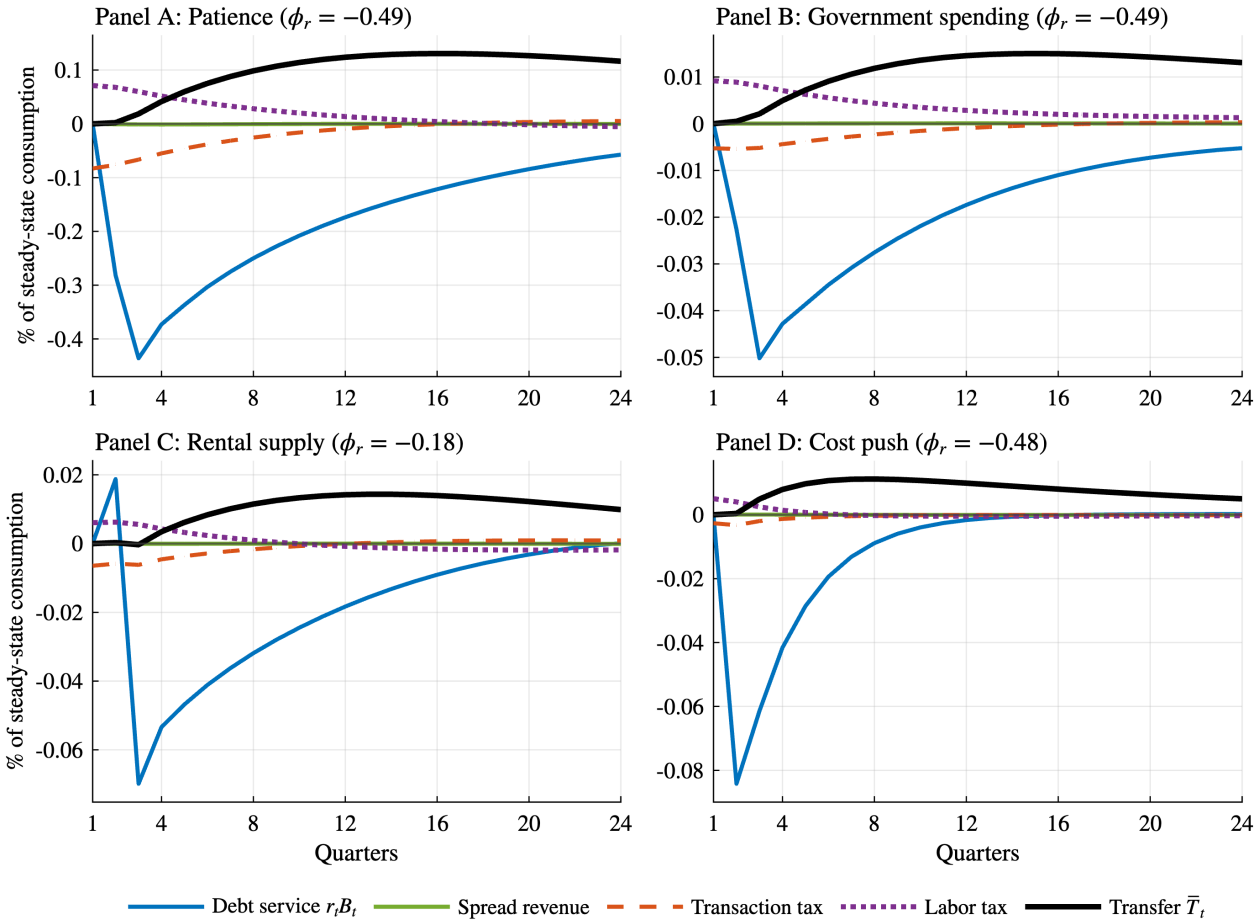



**Figure F.16.** Decomposition of the Fiscal Adjustment under the Optimized Rules

**Note.** Differences between each optimized rule and the benchmark $\phi_r = 0$ under the same shock, in percent of steady-state consumption. Higher revenues and lower debt service raise the transfer $\overline{T}_t$.

## F.16 A Rule on the Real Rent Level

Section 6 studies a rule on rent inflation, the object a price index measures. Here the rule responds instead to the gap of the real average rent from its steady state,

$$i_t = r_{ss} + \phi_\pi \pi_t^c + \phi_r \frac{\overline{p}_t^r - p_{ss}^r}{p_{ss}^r},$$

with $\phi_\pi = 1.5$. By (26), the real-rent component of the inflation rule sums to the rent gap: with a negative coefficient, it eases while real rents rise and tightens by the same cumulative amount as they fall back, so it shifts the timing of policy. A level term instead responds to the gap for as long as the gap lasts. The two coefficients are on different scales and are not directly comparable. Table F.10 reports the optimal coefficients, found on the grids of Appendix C.1 over $\phi_r \geq -2$, together with the net-acquisitions composite in inflation and in levels, which Appendix F.17 discusses.

Under the rental supply shock, the rent gap is created by the higher landlord spread, and

| $\phi_\pi + \phi_r$ | $\phi_r = 0$ | $-0.1$ | $-0.2$ | $-0.3$ | $-0.4$ |
|---|---|---|---|---|---|
| *Panel A: Patience shock* | | | | | |
| 1.1 | $-0.201$ | 0.154 | 0.420 | 0.619 | 0.768 |
| 1.2 | 0.081 | 0.329 | 0.518 | 0.665 | **0.779** |
| 1.3 | 0.112 | 0.325 | 0.492 | 0.626 | 0.733 |
| 1.4 | 0.067 | | | | |
| 1.5 | 0.000 | 0.187 | 0.341 | 0.469 | 0.577 |
| 1.6 | $-0.070$ | | | | |
| 1.7 | $-0.138$ | | | | |
| *Panel B: Government-spending shock* | | | | | |
| 1.1 | 0.160 | 0.181 | 0.202 | 0.221 | **0.240** |
| 1.2 | 0.093 | 0.118 | 0.140 | 0.161 | 0.180 |
| 1.3 | 0.050 | 0.077 | 0.100 | 0.121 | 0.141 |
| 1.4 | 0.021 | | | | |
| 1.5 | 0.000 | 0.026 | 0.048 | 0.069 | 0.088 |
| 1.6 | $-0.016$ | | | | |
| 1.7 | $-0.029$ | | | | |
| *Panel C: Rental supply shock* | | | | | |
| 1.1 | $-0.295$ | $-0.352$ | $-0.410$ | $-0.467$ | $-0.522$ |
| 1.2 | $-0.080$ | $-0.044$ | $-0.016$ | 0.006 | 0.023 |
| 1.3 | $-0.023$ | 0.047 | 0.105 | 0.153 | 0.194 |
| 1.4 | $-0.005$ | | | | |
| 1.5 | 0.000 | 0.088 | 0.162 | 0.226 | **0.281** |
| 1.6 | 0.001 | | | | |
| 1.7 | 0.000 | | | | |
| *Panel D: Cost-push shock* | | | | | |
| 1.1 | 0.069 | 0.071 | 0.072 | 0.074 | **0.076** |
| 1.2 | 0.050 | 0.053 | 0.055 | 0.058 | 0.060 |
| 1.3 | 0.033 | 0.036 | 0.039 | 0.041 | 0.044 |
| 1.4 | 0.016 | | | | |
| 1.5 | 0.000 | 0.004 | 0.008 | 0.012 | 0.015 |
| 1.6 | $-0.015$ | | | | |
| 1.7 | $-0.029$ | | | | |

**Table F.8.** Rent Term or a Less Hawkish Rule?

**Note.** Welfare gain of the rule $(\phi_\pi, \phi_r)$ over the benchmark $(1.5, 0)$, in basis points of permanent consumption, after a one-standard-deviation shock. Each row holds $\phi_\pi + \phi_r$ fixed; the column $\phi_r = 0$ holds the rules with no rent term. Bold: the best rule in each panel of the grid; every one lies on the grid's edge, and the optimum over all admissible rules is reported in the text.

a negative coefficient keeps the policy rate low for as long as rents stay high: in the fourth and fifth years, the real interest rate is lower than under the benchmark rule by about half of the remaining increase in the spread, whereas the inflation rule eases only while rents rise. Under the government-spending shock, the gain is negligible. Under the patience shock, real rents peak in quarters 12 to 16, after the expansion, so easing on the level would arrive once the boom has passed, and the optimal coefficient is slightly positive. Under the cost-push shock, welfare keeps rising up to the edge of the grid, $-2.00$, but flattens: $-1.00$ already delivers 84 percent of the gain.

| | Share | Demand | | Supply | |
|---|---|---|---|---|---|
| | | Patience $\phi_r^* = -0.49$ | Government spending $\phi_r^* = -0.49$ | Rental supply $\phi_r^* = -0.18$ | Cost push $\phi_r^* = -0.48$ |
| All households | 1.000 | 1.093 | 0.345 | 0.106 | 0.087 |
| Renters | 0.170 | 9.926 | 1.949 | 0.232 | 0.076 |
| Mortgagor-owners | 0.066 | 12.321 | 1.667 | 1.171 | 0.971 |
| Unlevered owners | 0.594 | −2.759 | −0.236 | −0.064 | −0.014 |
| Mortgagor-landlords | 0.056 | −0.939 | −0.377 | 0.385 | 0.365 |
| Unlevered landlords | 0.114 | −7.481 | −1.214 | −0.165 | −0.099 |

**Table F.9.** Welfare Gains of the Optimized Rules by Tenure

**Note.** Consumption-equivalent gain of the optimized rule over the benchmark $\phi_r = 0$ after a one-standard-deviation shock, in basis points of permanent consumption, for all households and for cohorts fixed at the date of the shock. Share: the cohort's mass in the stationary distribution. The all-households gain weights the cohort gains by marginal utility times consumption, not by share.

| | Demand | | Supply | |
|---|---|---|---|---|
| | Patience | Government spending | Rental supply | Cost push |
| *Rule on the real average-rent gap*, $(\overline{p}_t^r - p_{ss}^r)/p_{ss}^r$ | | | | |
| Optimal coefficient, $\phi_r^*$ | 0.03 | −0.05 | −0.16 | −2.00 |
| Welfare gain at $\phi_r^*$ | 0.270 | 0.004 | 0.681 | 0.122 |
| *Rule on nominal average rent inflation*, $\pi_t^{\text{rent}}$ | | | | |
| Optimal coefficient, $\phi_r^*$ | −0.49 | −0.49 | −0.18 | −0.48 |
| Welfare gain at $\phi_r^*$ | 1.093 | 0.345 | 0.106 | 0.087 |
| *Net acquisitions*, $\omega_o = 0.09$, welfare change | | | | |
| In inflation | −0.172 | −0.018 | −0.075 | −0.011 |
| In levels, $(p_t^h - p_{ss}^h)/p_{ss}^h$ | 0.288 | 0.053 | −0.458 | 0.042 |

**Table F.10.** Rules on Price Levels

**Note.** The first two blocks optimize each rule against a one-standard-deviation shock, holding $\phi_\pi = 1.5$. The third block reports the net-acquisitions composite of Section 5.3, with actual rentals at the model's rent share scaled by $1 - \omega_o$, and with the owner-occupied term in nominal house-price inflation or as the gap of the real house price from its steady state. Welfare changes are consumption equivalents relative to the $\phi_r = 0$ benchmark under the same shock, in basis points of permanent consumption. Under the cost-push shock, the optimum of the rule on the rent level lies at the edge of the grid.

## F.17 Net Acquisitions in Levels

Table 5 reports the net-acquisitions composite with its owner-occupied term in levels,

$$(1-\omega_o)\left[(1-\omega_r)\pi_t^c + \omega_r \pi_t^{\text{rent}}\right] + \omega_o \frac{p_t^h - p_{ss}^h}{p_{ss}^h},$$

with $\omega_o = 0.09$ and actual rentals at the model's rent share, so that only the owner-occupied term differs from Section 5.3. The gap is real because the nominal price level does not return to its steady state under inflation targeting, so a nominal gap would never close.

Under the monetary shock of Section 4.1, the house price falls because the rule tightens, and the level term eases for as long as it stays low, through the recovery as well. Over the first two years, the real interest rate is 14 percent lower than under the benchmark rule, while under net acquisitions in inflation it is unchanged. House prices fall by 10 percent less, the real average rent rises by 6 percent less, and consumption falls by 6 percent less, with as much additional nondurable disinflation as under the inflation composite (Table F.11). Welfare rises by 0.02 basis points, whereas net acquisitions in inflation lowers it by as much. Under the structural shocks of Section 6, the level term eases after the patience, government-spending, and cost-push shocks, which lower house prices, and tightens after the rental supply shock, which raises them as households shift toward owning (Table F.10). At Spain's HICP weight on actual rentals, the welfare changes have the same signs and similar magnitudes.

| | Net acquisitions in inflation | Net acquisitions in levels |
|---|---|---|
| Real interest rate | −0.2 | −14.1 |
| Real house price | −8.6 | +9.6 |
| Real average rent | +14.4 | −6.3 |
| Consumption | −9.4 | +6.2 |
| Nondurable inflation | −5.6 | −5.8 |
| Welfare change | −0.020 | 0.019 |

**Table F.11.** Net Acquisitions in Inflation and in Levels: Monetary Shock

**Note.** Responses to the monetary policy shock of Section 4.1, cumulated over quarters 1–8, under the convention of Table 4. Both composites weight owner-occupied housing at $\omega_o = 0.09$ and actual rentals at the model's rent share scaled by $1 - \omega_o$. The welfare change is the consumption equivalent relative to the benchmark rule, in basis points of permanent consumption.

# G Ramsey Policy and Two Welfare Perspectives

This appendix gives the details behind Section 7: the Ramsey plans, the simple rules evaluated from both perspectives, the timeless weights, and how the timeless rent coefficient depends on the persistence of shocks. Welfare changes are consumption equivalents relative to the benchmark rule $(\phi_\pi, \phi_r) = (1.5, 0)$ under the same shock, in basis points of permanent consumption.

## G.1 Two Welfare Perspectives

Section 7.2 defines the timeless criterion (30). Its penalties $\bar{\Psi}_V$, $\bar{\Psi}_{V^a}$, and $\bar{\Psi}_\pi$ are the multipliers of the stationary Ramsey plan. We compute the timeless Ramsey plan to first order, which DS

show is valid under the timeless penalties (their Proposition 11), and evaluate it, like every rule, along its exact nonlinear transition, so its gain is a lower bound on that of the timeless Ramsey plan, and a rule's share of it an upper bound. Appendix G.6 shows that the timeless criterion is, up to three small terms, utilitarian welfare with the weights $\hat{g}$ of Section 7.2, and argues that the calibrated steady state, with zero inflation, is a stationary Ramsey plan, so that the argument of DS's Proposition 6 applies.

## G.2 The Ramsey Plan from the Date of the Shock

Let $\mathbf{X}$ collect, for every date $t \geq 1$, the objects listed in Definition 1: the value and policy functions of a household in state $(a, h, y, s)$, the distribution $g_t(a, h, y, s)$, prices, including the policy rate $i_t$, aggregate quantities, and lease-survival objects. Given the path of the shock, the standard Ramsey problem is

$$\max_{\mathbf{X}} \ \mathcal{W}(\mathbf{X}) \quad \text{subject to (i)–(viii) of Definition 1 and (20), for all } t \geq 1, \tag{G.1}$$

where $\mathcal{W}(\mathbf{X})$ is criterion (28) evaluated at the allocation in $\mathbf{X}$. The economy starts at the stationary equilibrium: $g_1(a, h, y, s) = g_{ss}(a, h, y, s)$, public debt is $B_1 = \overline{B}$, balances carried into the date of the shock earn $r_1 = r_{ss}$, and every lease was signed at $p^r_{ss}$ (Section 2.9). The problem omits the Taylor rule, so the policy rate is the one free variable at each date. Fiscal policy is not chosen: the transfer follows the fiscal rule (22), and tax rates are fixed. As in DS, no condition dated before the shock constrains inflation or households' values at the date of the shock. This is the source of the time-0 problem, and the penalties of Section 7.2 apply to these objects.

Under the patience shock, the plan is a surprise easing. In the first two quarters the policy rate is set 9.7 and 9.3 annualized percentage points below what the benchmark rule would prescribe, a gap that closes within five quarters; in the second quarter, the policy rate is about 6 and the real interest rate set in that quarter 7.3 annualized percentage points below their steady-state levels. On impact, consumption rises by 3.4 percent, nondurable inflation by 3.25 annualized percentage points, and the house price by 4.8 percent. The rent on new leases, which can first move in the second quarter (Section 2.9), falls by 26.7 percent in that quarter, while the average rent falls by 1.1 percent. The plan therefore moves the real interest rate and rents in the same direction as the optimized rule of Section 6, and much further.

The gain of the plan is at least 5.62 basis points, about as much as without a shock: starting from the stationary equilibrium, with no shock at all, a surprise easing of the same kind raises welfare by at least 5.81 basis points. The incentive to surprise exists at the steady state, independently of the shock, and the shock interacts with it, so a share of the Ramsey gain attributable to the shock is not defined. In the decomposition of Section 6.3, redistribution accounts for between 114 and 120 percent of the gain without a shock, depending on whether marginal utilities are evaluated at the steady state or along the plan, and efficiency for between $-14$ and $-20$ percent. These are the two ends of the line integral that Dávila and Schaab (2025) use for non-marginal changes; at both,

redistribution exceeds the whole gain.

Under the government-spending, rental supply, and cost-push shocks, the patience plan, scaled down, gains at least 5.17, 5.20, and 5.21 basis points, respectively.

**The zero lower bound.** The plans of Section 7.1 let the policy rate fall below zero. We also compute the standard Ramsey plans with the policy rate kept at or above zero. Every number in this paragraph, for the plans and for the rules alike, is computed in the version of the economy with leases pooled after eight quarters, in which the unconstrained plan without a shock gains at least 6.50 basis points. These plans are not exact optima either, but every plan we report is an equilibrium with the policy rate at or above zero, so its gain over the benchmark rule is a lower bound on the Ramsey gain, and a rule's share of that gain is an upper bound on its share of the Ramsey gain; every rule we evaluate keeps the policy rate above zero.

Without a shock, the best plan we find holds the policy rate at zero for the first four quarters and gains 3.67 basis points, more than half of the unconstrained gain; redistribution accounts for 95 to 98 percent of it, depending on the marginal values at which we weigh households. Under each shock we apply the same policy: the same rates for the first twelve quarters, and the same adjustments to the benchmark rule after that. It gains 4.90 basis points under the government-spending shock, 2.64 under the rental supply shock, and 4.08 under the cost-push shock, of which the optimized rent coefficients of Section 6 capture at most 7.8, 4.5, and 2.1 percent. Under the patience shock, a policy close to the same one lowers welfare. The gain of the optimized rent coefficient in this economy, 1.50 basis points, is then the best lower bound we have, and it places no useful bound on a rule's share.

**Numerical limitations.** The plans in this appendix rest on six computational shortcuts.

1. We compute the plans in terms of the deviations of the policy rate from the benchmark rule, $\varepsilon_t^m = i_t - r_{ss} - 1.5\,\pi_t^c$, which reach the same equilibria as the rate itself but give much better-conditioned transitions.

2. We optimize in the economy with leases pooled after eight quarters, where each iteration already takes about twenty minutes, and evaluate the plans in the calibrated economy; the plans with the policy rate at or above zero remain in the eight-quarter economy.

3. The plan with the shock sets the policy rate only in the first 40 quarters and follows the benchmark rule afterward, which keeps it from exploiting the end of the 200-quarter horizon; at the first-order step, this costs 0.05 basis points.

4. We stop the optimizations short of an optimum: welfare is stiff and, through kinks in the household problem, not smooth in the path, so each iteration accepts only a short step.

5. Without a shock, we report three-quarters of a Newton step from the stationary equilibrium, and under the government-spending, rental supply, and cost-push shocks, three-quarters of

the patience plan's deviations; in both cases the full path did not solve.

6. The timeless plan is computed to first order, and only under the patience shock (Appendix G.1).

Every path we report solves the equilibrium conditions, with a largest residual below $10^{-9}$, and respects the zero bound where it is imposed, so each gain is attainable: the Ramsey gains are lower bounds, and a rule's share of them is an upper bound.

## G.3 Simple Rules under the Two Perspectives

Panel A of Table G.1 optimizes rule (27) from the date of the shock in three ways, over the admissible rules of Appendix C.2: the rent coefficient alone, as in Section 6; the coefficient on nondurable inflation alone; and both, as in Appendix F.15. Under the patience shock, the rent coefficient alone captures 86 percent of the gain of the best two-coefficient rule, and the coefficient on nondurable inflation alone captures 9 percent: from the date of the shock, the rent coefficient is almost ten times as effective. Under the government-spending shock, the two shares are 71 and 46 percent. Under the rental supply shock, the rent coefficient alone captures 14 percent, and the best rule, at the upper bound $\phi_\pi = 4$, responds strongly to nondurable inflation and, with a negative coefficient about half as large, to rent inflation. Under the cost-push shock, the best rule without a rent term, at $\phi_\pi = 1.01$, does better than the rent coefficient alone (0.091 against 0.087 basis points), and the best two-coefficient rule responds to rent inflation alone, with a positive coefficient.

From the timeless perspective (Panel B), the coefficients that are optimal from the date of the shock lower welfare after both demand shocks and change it by a tenth of a basis point at most after the two supply shocks.

Figure G.1 shows the patience shock in detail: the two perspectives push both coefficients in opposite directions. From the date of the shock, welfare rises as the rule eases relative to the benchmark, through a negative coefficient on rent inflation or a weaker response to nondurable inflation. From the timeless perspective, welfare rises as the rule tightens. A positive coefficient on rent inflation, $\phi_r = 0.45$, captures about two-thirds of the gain of the timeless plan, a stronger response to nondurable inflation alone somewhat more, and the two together four-fifths, with $\phi_\pi = 2.10$ and $\phi_r = 0.20$; from the date of the shock, the same rules lose. From the timeless perspective, rent inflation therefore enters the rule with a positive sign, next to a stronger response to nondurable inflation, while the timeless plan holds nondurable inflation close to zero and lets rents rise. Every rule in this paragraph is determinate on the maps of Appendix C.2.

In the one-asset HANK economy of DS, a strict zero-inflation target implements the timeless Ramsey plan in the absence of shocks, and a flexible target around zero inflation can implement optimal stabilization policy; their inflation is wage inflation, the one that is costly here too. Our conclusions rest on one shock and on a timeless plan computed to first order.

The timeless plan raises the real interest rate by less on impact but for longer, and lets the average rent peak at 1.33 percent above steady state, against 1.06 percent under the benchmark

| | Demand | | Supply | |
|---|---|---|---|---|
| | Patience | Government spending | Rental supply | Cost push |
| *Panel A: Evaluated from the date of the shock* | | | | |
| Rent coefficient alone, $\phi_r^*$ | $-0.49$ | $-0.49$ | $-0.18$ | $-0.48$ |
| welfare gain | 1.093 | 0.345 | 0.106 | 0.087 |
| No rent term, $\phi_\pi^*$ | 1.27 | $1.01^{\mathrm{F}}$ | 1.57 | $1.01^{\mathrm{F}}$ |
| welfare gain | 0.115 | 0.224 | 0.001 | 0.091 |
| Both coefficients, $(\phi_\pi^*, \phi_r^*)$ | $(1.67, -0.66)^{\mathrm{F}}$ | $(3.41, -2.40)^{\mathrm{F}}$ | $(4, -2.28)$ | $(0, 1.01)^{\mathrm{F}}$ |
| welfare gain | 1.273 | 0.488 | 0.763 | 0.108 |
| Ratio, rent alone / both (%) | 85.9 | 70.7 | 13.9 | 80.1 |
| Ratio, no rent term / both (%) | 9.0 | 45.9 | 0.1 | 84.1 |
| *Panel B: The same rules, evaluated from a timeless perspective* | | | | |
| Rent coefficient alone | $-2.104$ | $-0.214$ | $-0.028$ | 0.001 |
| No rent term | $-0.661$ | $-0.338$ | 0.001 | $-0.001$ |
| Both coefficients | $-1.948$ | $-0.178$ | $-0.099$ | $-0.008$ |
| Best rule evaluated | 0.475 | 0.005 | 0.002 | 0.002 |
| coefficients | $(2.10, 0.20)$ | $(1.5, 0.50)$ | $(1.65, 0)$ | $(1.5, -0.49)^{\mathrm{F}}$ |
| Timeless plan | 0.589 | – | – | – |

**Table G.1.** Optimized Simple Rules under Two Welfare Perspectives

**Note.** Welfare gains over the benchmark rule $(\phi_\pi, \phi_r) = (1.5, 0)$ after a one-standard-deviation shock, in basis points of permanent consumption. Panel A optimizes rule (27) on criterion (28) over the admissible rules of Appendix C.2, with $\phi_\pi = 1.5$ in the first row and $\phi_r = 0$ in the second; rules are rounded to two decimals. $^{\mathrm{F}}$Just above the determinacy frontier, at $\phi_\pi + \phi_r = 1.01$. The ratios divide the gain of a one-coefficient rule by that of the two-coefficient rule. The Ramsey plan from the date of the shock gains at least 5.62, 5.17, 5.20, and 5.21 basis points, so the rules capture at most 19.4, 6.7, 2.0, and 1.7 percent of it with the rent coefficient alone, and at most 22.6, 9.4, 14.7, and 2.1 percent with both coefficients. Panel B evaluates the rules of Panel A from the timeless perspective of Appendix G.1. The best rule evaluated is the rule with the highest timeless gain among those we solve under each shock; it is not an optimum. The timeless plan is computed to first order, for the patience shock only.

rule. The rule with $\phi_r = 0.45$ moves housing the same way and further. Evaluated from the date of the shock, the timeless plan loses 0.74 basis points and that rule 1.37.

## G.4 Efficiency and Redistribution from the Two Perspectives

By Proposition G.1, the timeless criterion is, up to three small terms, the utilitarian criterion (28) with the weights $\hat{g}$ in place of $g_{ss}$, and by Proposition G.2 the decomposition (C.2) of Section 6.3 becomes efficiency, $\mathbb{E}_{ss}[\hat{\omega}]\,\mathbb{E}_{ss}[\mathrm{d}V^\lambda/\mathrm{d}\phi_r]$, plus redistribution, $\mathrm{Cov}_{ss}(\hat{\omega}, \mathrm{d}V^\lambda/\mathrm{d}\phi_r)$. Each household's willingness to pay is the same as in Figure 8; only the weights change. Measured in the timeless criterion's own units of perpetual consumption, as in Dávila and Schaab (2025), the efficiency term is identical to that figure's; in that figure's units, which Figure G.2 keeps, it is 0.81 times its value there, since $\mathbb{E}_{ss}[\hat{\omega}] = 0.81$ under every shock. It keeps its sign at every coefficient, and the coefficient it favors on its own is the same from both perspectives: $-0.18$ under the government-spending shock and $-0.04$ under the rental supply shock, while efficiency favors a higher coefficient at every

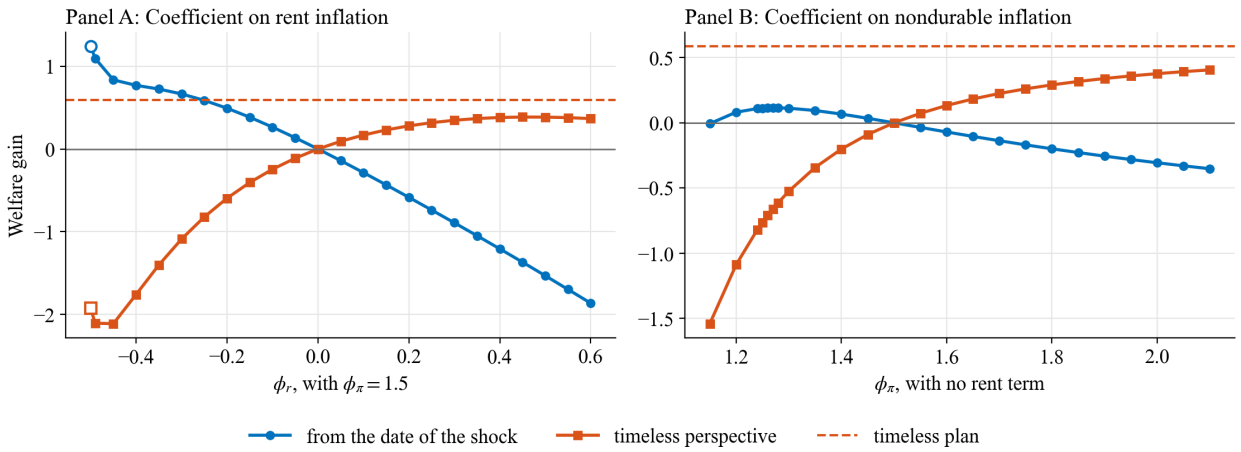



**Figure G.1.** Two Welfare Perspectives under the Patience Shock

**Note.** Welfare gain relative to the benchmark rule $(\phi_\pi, \phi_r) = (1.5, 0)$ after a one-standard-deviation patience shock, in basis points of permanent consumption. Panel A varies the coefficient on rent inflation with $\phi_\pi = 1.5$; Panel B varies the coefficient on nondurable inflation with no rent term. Blue lines evaluate welfare from the date of the shock, criterion (28); orange lines evaluate it from the timeless perspective of Appendix G.1. The dashed line is the timeless gain of the timeless Ramsey plan, computed to first order. Open markers lie beyond the determinacy frontier, $\phi_\pi + \phi_r \leq 1.0045$.

point of Figure G.2 under the patience shock and a lower one under the cost-push shock.

The timeless weights $\hat{g}$ do not depend on the shock, and the weights per unit of consumption are nearly identical under the four shocks. Together, the two promises leave renters 0.58 times their population weight and give landlords between 2.3 and 3.1 times theirs. The promise on values moves weight from renters to landlords, and on its own it slightly favors the optimized coefficient. The promise on marginal values moves weight within each tenure from poorer and indebted households toward richer ones, and it turns the optimized coefficient into a loss: it accounts for 108 percent of the difference between the two perspectives at $\phi_r = -0.49$. Under the patience shock, the average weight of renters falls from 1.95 to 1.08 and that of landlords rises from 0.53 to 1.39, so that renters count 3.7 times as much as landlords from the date of the shock and 0.8 times as much from the timeless perspective. Renters with less than one quarter's income in financial wealth still weigh more than the average household, but their weight falls from 3.38 to 1.36, while that of mortgagor-landlords rises from 0.58 to 1.83 and that of unlevered landlords from 0.51 to 1.17.

With these weights, redistribution is small and positive under all four shocks: between 0.7 and 3.1 under the patience shock, compared with $-53$ to $-27$ from the date of the shock, and between 0.1 and 0.8 under the other three (Figure G.2). The timeless total therefore favors a higher coefficient under the patience shock, a positive coefficient under the government-spending and rental supply shocks, and a lower one under the cost-push shock, where efficiency outweighs redistribution. Under every shock, however, each component of this split moves at first order in the shock while their sum moves only at second order, so each component mixes first-order terms that cancel in the total with the second-order terms that decide it; the second-order parts of the

split of Appendix G.5 show redistribution favoring a lower coefficient under the patience, rental supply, and government-spending shocks, and efficiency opposing it more strongly.

The terms that the reweighting leaves out are small under the patience shock (Appendix G.6). Where the timeless total is close to zero, as near $\phi_r = 0$ under the other three shocks, they are of the same order as the total but do not change its sign at any negative coefficient we evaluate. We therefore locate zeros with the timeless welfare values themselves, which place the zero under the rental supply shock just above $\phi_r = 0.05$.

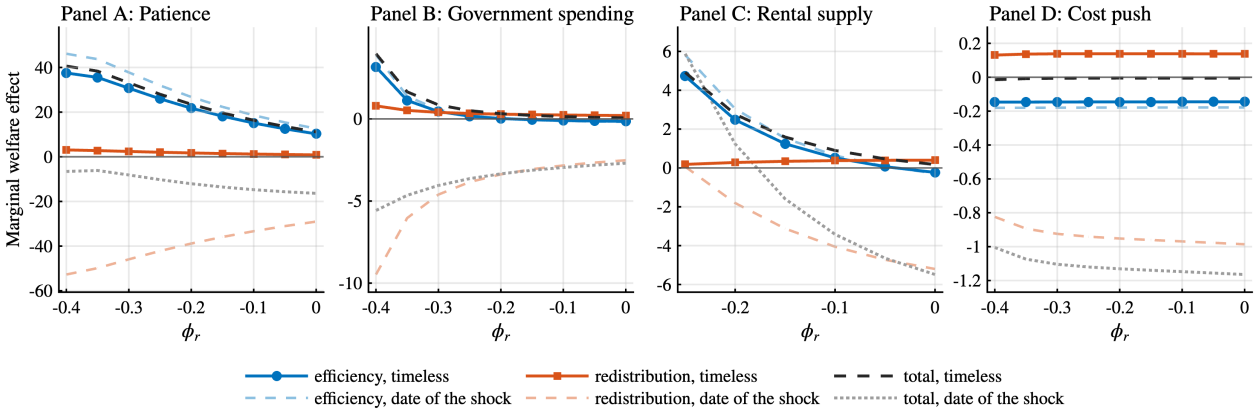



**Figure G.2.** Efficiency and Redistribution from the Two Perspectives

**Note.** Marginal welfare effect of the coefficient on rent inflation in rule (27), $d\mathcal{W}/d\phi_r$, holding $\phi_\pi = 1.5$, split into efficiency and redistribution. Solid lines use the timeless weights $\hat{g}$; light dashed and dotted lines repeat Figure 8, from the date of the shock. The black dashed line is the timeless total, the sum of the two solid lines. Positive values favor a higher coefficient. Units as in Figure 8.

## G.5 Shock Persistence and the Timeless Rent Coefficient

This subsection asks when the timeless criterion favors a negative rent coefficient. We optimize the rent coefficient of rule (27), with $\phi_\pi = 1.5$, on the timeless criterion (30), all its terms included, and vary the persistence $\rho$ of each shock and, under the rental supply shock, the probability $\varphi$ that a lease is not renegotiated in a quarter, holding every other parameter at its calibrated value. For computational reasons we use the version of the economy with leases pooled after eight quarters; under the rental supply shock at its calibrated persistence, its timeless optimum is 0.061, against 0.058 with twenty vintages. The stationary equilibrium does not depend on $\varphi$, but the stationary multipliers do, and we recompute them for each $\varphi$. At $\phi_\pi = 1.5$, determinacy requires $\phi_r > -0.4955$ (Appendix C.2); where welfare still rises at the frontier, we report the rule just above it, at $\phi_\pi + \phi_r = 1.01$, that is, $\phi_r = -0.49$, so an optimum there is a corner.

Under the rental supply and government-spending shocks, the timeless optimum is positive when the shock is persistent and negative when it is short-lived, crossing zero near $\rho = 0.77$ and 0.79 (Figure G.3, Panel A, and Table G.2). Under the cost-push shock it crosses between 0.67 and 0.85; its calibrated persistence, 0.67, is already below the crossing, which is why the best

rule evaluated under it in Table G.1 lies at the corner. Under the patience shock the optimum stays positive down to $\rho = 0.7$, although it is not monotone in the persistence; at $\rho = 0.5$ our computations do not determine its sign. More frequent renegotiation lowers the optimum at the calibrated persistence of the rental supply shock, from 0.061 with the calibrated $\varphi = 0.99$ to 0.039 with $\varphi = 0.75$, when leases are renegotiated about once a year, but when the shock is short-lived it pulls a negative optimum back toward zero (Panel B).

To see what decides the sign, we split the timeless welfare change from lowering the rent coefficient from 0 to $-0.10$ into efficiency and redistribution, with redistribution measured by the exact promise terms of Proposition G.1. Each component moves at first order in the size of the shock while their sum moves only at second order, so the split of Figure G.2 mixes first-order terms that cancel in the total with the second-order terms that decide it; we therefore compare the second-order parts. At the calibrated persistence, redistribution favors the negative coefficient and efficiency opposes it, with an efficiency cost four to five times the redistributive gain under the rental supply, government-spending, and patience shocks; the inflation penalty is negligible. As the shock becomes short-lived, the efficiency cost falls much faster than the redistributive gain, and under the rental supply and government-spending shocks redistribution decides the sign.

The stakes are small. At the negative optima, the timeless gain over $\phi_r = 0$ is between 0.001 and 0.04 percent of the timeless welfare cost of the shock under the benchmark rule, or a few thousandths of a basis point when the shock's cumulative size is held at its calibrated value; under the cost-push shock the relative gain is larger, up to 17 percent, only because the timeless cost of that shock is itself about 0.01 basis points.

| | Persistence of the shock, $\rho$ | | | | Calibrated |
|---|---|---|---|---|---|
| | 0.50 | 0.70 | 0.85 | 0.95 | ($\rho$) |
| Rental supply | $-0.105$ | $-0.021$ | 0.024 | 0.061 | 0.058 (0.95) |
| Government spending | $-0.285$ | $-0.134$ | 0.084 | – | 0.481 (0.96) |
| Cost push | $-0.49^{\mathrm{F}}$ | – | 0.399 | 0.317 | $-0.49^{\mathrm{F}}$ (0.67) |
| Patience | $(-0.49)^{\mathrm{F,\S}}$ | $0.57^{\ddagger}$ | 0.606 | 0.352 | 0.468 (0.917) |

**Table G.2.** The Timeless Rent Coefficient by Shock and Persistence

**Note.** Timeless-optimal $\phi_r$ in rule (27) with $\phi_\pi = 1.5$, in the economy with leases pooled after eight quarters, except the last column, computed with twenty vintages, and $\varphi = 0.99$. [F]Just above the determinacy frontier, at $\phi_\pi + \phi_r = 1.01$. A dash marks a value not computed. [‡]Only the sign is robust. [§]Sign not determined (see text).

## G.6 The Timeless Criterion as a Weighted Utilitarian Criterion

This subsection proves the representation of the timeless criterion as utilitarian welfare with the weights $\hat{g}$ and the timeless weights per unit of consumption, both used in Section 7.2. We work on the computational grid, where the statements are exact identities, and then give their continuum counterparts, which the text uses.

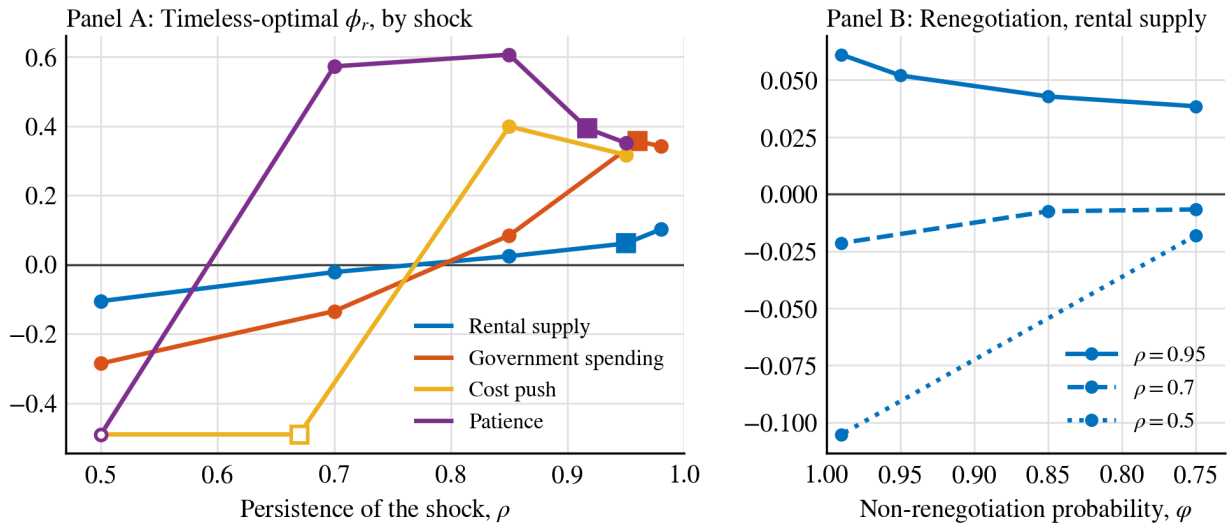



**Figure G.3.** The Timeless Rent Coefficient and the Persistence of Shocks

**Note.** Timeless-optimal coefficient on rent inflation in rule (27), with $\phi_\pi = 1.5$, in the economy with leases pooled after eight quarters. Panel A: against the persistence $\rho$ of each shock, with the non-renegotiation probability $\varphi$ at its calibrated value, 0.99; squares mark the calibrated persistence and open markers the rule just above the determinacy frontier, $\phi_r = -0.49$. Under the patience shock at $\rho = 0.5$ the sign is not determined (see text). Panel B: under the rental supply shock, against $\varphi$, for three persistence values.

**Setting.** A household state is $x = (a, h, y, s)$, with financial wealth $a$ on the grid $a_1 < a_2 < \cdots < a_N$, tenure $h$, productivity $y$, and lease state $s$. Sums $\sum_x$ run over all grid states. The stationary distribution assigns mass $g_{ss}(x) \geq 0$ to each state, with $\sum_x g_{ss}(x) = 1$; its support is $\mathcal{S} = \{x : g_{ss}(x) > 0\}$. For a path of the policy rate $\mathbf{i}$, three household objects matter. The first is $V(x; \mathbf{i})$, the welfare of a household that starts the transition in state $x$, as in Appendix C.1; it excludes the taste shocks, and criterion (28) is $\mathcal{W}(\mathbf{i}) = \sum_x g_{ss}(x) V(x; \mathbf{i})$. The second is $V_1(x; \mathbf{i})$, the value function (7) at the date of the shock, which includes the expected utility from the taste shocks of tenure choices: $\mathcal{E}_1(x; \mathbf{i}) \equiv V_1(x; \mathbf{i}) - V(x; \mathbf{i})$ is that utility. The third is $V_1^a(x; \mathbf{i})$, the marginal value of wealth at the date of the shock that enters the households' Euler equations at the date before; by the envelope condition, it is the derivative of $V_1$ in wealth wherever that derivative exists, and at $a = 0$, where the borrowing spread kinks the return on wealth, it is the right derivative. The timeless criterion adds to $\mathcal{W}$ the penalties on the date-0 promises, with stationary multipliers $\bar{\Psi}_V(x)$, $\bar{\Psi}_{V^a}(x)$, and $\bar{\Psi}_\pi$ that depend on neither the shock nor policy:

$$\mathcal{W}^{TL}(\mathbf{i}) = \mathcal{W}(\mathbf{i}) + \sum_x \left[ \bar{\Psi}_V(x)\, V_1(x; \mathbf{i}) + \bar{\Psi}_{V^a}(x)\, V_1^a(x; \mathbf{i}) \right] - \bar{\Psi}_\pi\, \pi_1^w \left(1 + \pi_1^w\right). \tag{G.2}$$

The household penalty is linear in the promised values, as DS's time-0 penalty is; the inflation penalty is the date-0 row of the wage Phillips curve, through which date-1 wage inflation enters,

and has the same first-order content as DS's linear inflation penalty.[54]

DS have one household promise, on the value: in continuous time the consumption-saving condition $u'(c) = \partial_a V$ holds within the instant, and its multiplier enters their distributional penalty through the births term $\partial_a \chi_{ss}$ of the promise-keeping Kolmogorov equation (their Proposition 8). In discrete time the Euler equation links the date before the shock to the date of the shock and carries a promise of its own. After summation by parts, $\bar{\Psi}_V + D'\bar{\Psi}_{V^a} = \hat{g} - g_{ss}$ is the counterpart of DS's distributional penalty $\phi_{ss}$, and $D'\bar{\Psi}_{V^a}$ that of its births term; the split between the two household promises is a feature of the discrete-time formulation.

A uniform shift of all values changes no decision and the transition preserves mass, so the stationary conditions determine $\bar{\Psi}_V$ only up to a multiple of $g_{ss}$. Along the path of the standard problem, the value multipliers sum to zero at every date: a common shift of all values moves no policy, and the criterion of the standard problem does not depend on $V_1$. The timeless multipliers are the limit of those of the standard problem, so $\sum_x \bar{\Psi}_V(x) = 0$; this is the standard timeless argument (Woodford, 2003; Marcet and Marimon, 2019), which DS also use for the same choice (their Section 4.2). This is not a free normalization. Every value of the multiple satisfies the stationary first-order conditions, since the condition for inflation holds at zero inflation for any multipliers and the remaining conditions form a square system in the other multipliers, so every value also passes the test of DS's Proposition 6; but another value changes every penalty, $\bar{\Psi}_{V^a}$ and $\bar{\Psi}_\pi$ included, and with them the criterion at second order, which is the order at which the rules we compare differ. Criterion (30) in the text is (G.2) written with integrals and with $\partial V_1/\partial a$ for $V_1^a$.

**The derivative along the wealth grid.** Let $D$ be the linear operator that differentiates a function on the grid with respect to wealth, holding $(h, y, s)$ fixed:

$$(DF)(a_j, h, y, s) = \sum_{k=1}^{N} D_{jk}\, F(a_k, h, y, s).$$

Our $D$ uses centered differences in the interior, a forward difference at $a = 0$, where the borrowing spread makes the value function kinked, and one-sided differences at the two ends of each state's feasible range of wealth, the borrowing limit and the top of the grid. Every row of $D$ sums to zero, $D\mathbf{1} = 0$: the derivative of a constant is zero. Its adjoint $D'$ is defined by

$$\sum_x \Psi(x)\,(DF)(x) = \sum_x (D'\Psi)(x)\,F(x) \qquad \text{for all functions } \Psi, F \text{ on the grid,} \tag{G.3}$$

[54] The household penalty follows DS's time-0 penalty, which is linear in the promised values; the inflation penalty is written on the date-0 row of the Phillips curve, as in the recursive-multiplier approach of Marcet and Marimon (2019), and the two conventions agree to first order. The date-0 Bellman and logit conditions are nonlinear in $V_1$, so a household penalty written on those conditions themselves would add a second-order term. In a check with leases pooled after eight quarters, under the patience shock, that term changes the timeless plan's gain over the benchmark rule by less than half a percent, and the loss of the coefficient of Section 6 by about two percent, making it larger.

that is, $(D'\Psi)(a_k,h,y,s) = \sum_j D_{jk}\Psi(a_j,h,y,s)$. Taking $F = \mathbf{1}$ in (G.3) gives $\sum_x (D'\Psi)(x) = \sum_x \Psi(x)(D\mathbf{1})(x) = 0$ for every $\Psi$.

Write $e(x;\mathbf{i}) \equiv V_1^a(x;\mathbf{i}) - (DV_1)(x;\mathbf{i})$ for the gap between the marginal value in the Euler equations and the grid derivative of the value. It is a discretization error, small in the welfare changes we compare.

**Proposition G.1** (The timeless criterion as a weighted utilitarian criterion)**.** *Define the timeless weights*

$$\hat{g}(x) = g_{ss}(x) + \bar{\Psi}_V(x) + (D'\bar{\Psi}_{V^a})(x). \tag{G.4}$$

*For every path* $\mathbf{i}$*,*

$$\mathcal{W}^{TL}(\mathbf{i}) = \sum_x \hat{g}(x)\, V(x;\mathbf{i}) + \underbrace{\sum_x \left[\hat{g}(x) - g_{ss}(x)\right]\mathcal{E}_1(x;\mathbf{i})}_{\text{taste shocks}} + \underbrace{\sum_x \bar{\Psi}_{V^a}(x)\, e(x;\mathbf{i})}_{\text{discretization}} \underbrace{- \bar{\Psi}_\pi\, \pi_1^w\left(1+\pi_1^w\right)}_{\text{inflation penalty}}, \tag{G.5}$$

*and* $\sum_x \hat{g}(x) = 1$.

*Proof.* Substitute $V_1^a = DV_1 + e$ into (G.2) and apply (G.3) with $\Psi = \bar{\Psi}_{V^a}$ and $F = V_1$:

$$\sum_x \bar{\Psi}_{V^a}(x) V_1^a(x;\mathbf{i}) = \sum_x (D'\bar{\Psi}_{V^a})(x)\, V_1(x;\mathbf{i}) + \sum_x \bar{\Psi}_{V^a}(x)\, e(x;\mathbf{i}).$$

The penalty in (G.2) is therefore $\sum_x [\hat{g}(x) - g_{ss}(x)] V_1(x;\mathbf{i}) + \sum_x \bar{\Psi}_{V^a}(x) e(x;\mathbf{i}) - \bar{\Psi}_\pi \pi_1^w(1+\pi_1^w)$. Write $V_1 = V + \mathcal{E}_1$ in its first sum and add $\mathcal{W}(\mathbf{i}) = \sum_x g_{ss}(x) V(x;\mathbf{i})$, which gives (G.5). Finally, $\sum_x \hat{g}(x) = \sum_x g_{ss}(x) + \sum_x \bar{\Psi}_V(x) + \sum_x (D'\bar{\Psi}_{V^a})(x) = 1 + 0 + 0$, by $\sum_x \bar{\Psi}_V(x) = 0$ and the property of $D'$ shown below (G.3). □

The representation in Section 7.2 keeps the first term of (G.5) and drops the other three: the taste shocks, the discretization error, and the inflation penalty. Under the patience shock, the three dropped terms account for between 4.5 and 8 percent of the timeless welfare changes between the rules we compare, and the taste-shock term is the largest of them.

**Weights per unit of consumption.** Section 6.3 measures each household's gain from a higher rent coefficient in units of perpetual consumption, $\mathrm{d}V^\lambda(x)/\mathrm{d}\phi_r = \lambda(x)^{-1}\mathrm{d}V(x;\phi_r)/\mathrm{d}\phi_r$, and weighs it by $\omega(x) = \lambda(x)/\bar{\lambda}$, with $\lambda$ and $\bar{\lambda}$ as in (C.1). On the support $\mathcal{S}$, define the timeless weight per unit of consumption

$$\hat{\omega}(x) = \omega(x)\,\frac{\hat{g}(x)}{g_{ss}(x)} = \omega(x) + \omega(x)\left[\frac{\bar{\Psi}_V(x)}{g_{ss}(x)} + \frac{(D'\bar{\Psi}_{V^a})(x)}{g_{ss}(x)}\right], \tag{G.6}$$

whose averages by cohort Figure G.4 reports.

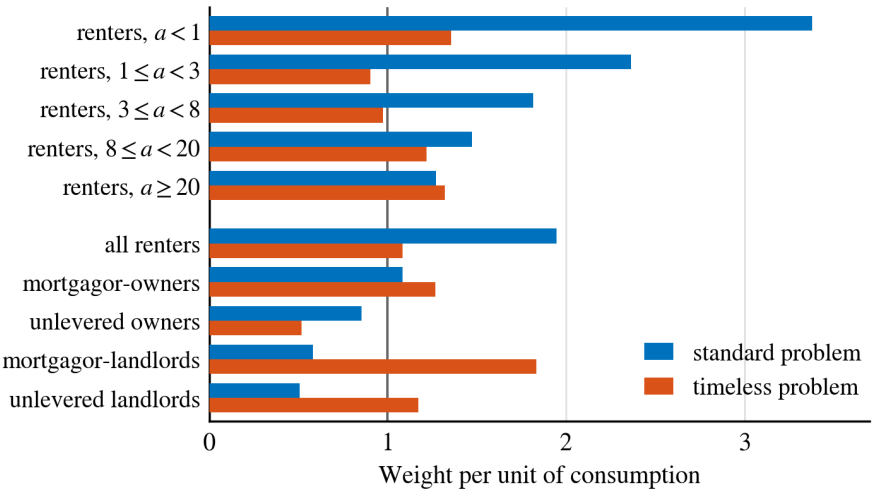



**Figure G.4.** Welfare Weights by Cohort

**Note.** Average welfare weight per unit of consumption by cohort at the date of the shock, in the standard problem and in the timeless problem, computed under the patience shock and the benchmark rule; under the other shocks and at the steady state they differ by less than 1.5 percent. The vertical line marks the average standard weight. Net financial assets $a$ in units of quarterly steady-state income.

**Proposition G.2** (Efficiency and redistribution from the timeless perspective)**.** *For any rule coefficient, here* $\phi_r$*,*

$$\frac{1}{\bar{\lambda}}\frac{\mathrm{d}\mathcal{W}^{TL}}{\mathrm{d}\phi_r} = \mathbb{E}_{ss}[\hat{\omega}]\,\mathbb{E}_{ss}\left[\frac{\mathrm{d}V^{\lambda}}{\mathrm{d}\phi_r}\right] + \mathbb{C}\mathrm{ov}_{ss}\left(\hat{\omega}, \frac{\mathrm{d}V^{\lambda}}{\mathrm{d}\phi_r}\right) + \frac{1}{\bar{\lambda}}\left[\sum_{x\notin\mathcal{S}} \hat{g}(x)\frac{\mathrm{d}V(x;\phi_r)}{\mathrm{d}\phi_r} + \frac{\mathrm{d}}{\mathrm{d}\phi_r}\Big(\textit{taste shocks} + \textit{discretization} + \textit{inflation penalty}\Big)\right], \quad \text{(G.7)}$$

*where* $\mathbb{E}_{ss}$ *and* $\mathbb{C}\mathrm{ov}_{ss}$ *are taken over* $g_{ss}$ *and the last three terms are those of* (G.5)*.*

*Proof.* Differentiate (G.5); the multipliers do not depend on $\phi_r$. On $\mathcal{S}$, $\hat{g}(x)\,\mathrm{d}V(x)/\mathrm{d}\phi_r = g_{ss}(x)\,\bar{\lambda}\,\hat{\omega}(x)\,\mathrm{d}V^{\lambda}(x)/\mathrm{d}\phi_r$, so $\bar{\lambda}^{-1}\sum_{x\in\mathcal{S}}\hat{g}(x)\,\mathrm{d}V(x)/\mathrm{d}\phi_r = \mathbb{E}_{ss}[\hat{\omega}\,\mathrm{d}V^{\lambda}/\mathrm{d}\phi_r]$, and $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] + \mathbb{C}\mathrm{ov}(X,Y)$. States outside $\mathcal{S}$ give the first term in brackets. □

Without the penalties, Proposition G.2 is the decomposition (C.2) of Section 6.3, since then $\hat{g} = g_{ss}$, $\hat{\omega} = \omega$, $\mathbb{E}_{ss}[\omega] = 1$, and the bracket vanishes. In general $\mathbb{E}_{ss}[\hat{\omega}] = 1 + \bar{\lambda}^{-1}\sum_{x\in\mathcal{S}}[\hat{g}(x) - g_{ss}(x)]\lambda(x)$, which is not one: the penalty moves weight between households whose marginal values of consumption differ. This is why, measured in the units of (C.2), efficiency (the sum of households' willingness to pay) enters the timeless criterion scaled by $\mathbb{E}_{ss}[\hat{\omega}] = 0.81$. Dividing instead by the timeless criterion's own value of perpetual consumption, $\sum_{x\in\mathcal{S}}\hat{g}(x)\lambda(x) = 0.81\,\bar{\lambda}$, as Dávila and Schaab (2025) do, makes the normalized weights $\hat{\omega}/\mathbb{E}_{ss}[\hat{\omega}]$ average one, leaves

efficiency exactly as in (C.2), and divides redistribution by 0.81. Nothing in either proposition requires $\hat{g} \geq 0$, and about 10 percent of the stationary mass, mostly unlevered owners, has $\hat{g} < 0$. The timeless weights are a way to read the timeless criterion, not weights a planner would choose.

**The continuum counterpart.** The text writes these objects with integrals and with $-\partial_a$ in place of $D'$. Let $\bar{\psi}_{V^a}(a)$ be the density of the multiplier $\bar{\Psi}_{V^a}$ along wealth on $[\underline{a}, \bar{a}]$, for given $(h, y, s)$, and let $F$ be absolutely continuous in $a$, as value functions are, kinks included. If $\bar{\psi}_{V^a}$ is absolutely continuous too, integration by parts gives

$$\int_{\underline{a}}^{\bar{a}} \bar{\psi}_{V^a}(a) \, \frac{\partial F(a)}{\partial a} \, da = \bar{\psi}_{V^a}(\bar{a}) F(\bar{a}) - \bar{\psi}_{V^a}(\underline{a}) F(\underline{a}) - \int_{\underline{a}}^{\bar{a}} \frac{\partial \bar{\psi}_{V^a}(a)}{\partial a} \, F(a) \, da. \tag{G.8}$$

The adjoint $D'$ is the discrete counterpart of the right-hand side as a whole: its interior rows approximate $-\partial_a \bar{\psi}_{V^a}$, and its rows at the two ends of each feasible range of wealth carry the two boundary terms, which in our solution are negligible. If $\bar{\psi}_{V^a}$ jumps at some wealth level, for instance where the borrowing limit starts to bind and the Euler equation holds as an inequality, $-\partial_a \bar{\psi}_{V^a}$ also carries a point mass equal to minus the jump, which $D'$ again contains. In the text, $-\partial_a \bar{\Psi}_{V^a}$ stands for $D' \bar{\Psi}_{V^a}$ in this sense, boundary terms and point masses included. Kinks of $V_1$, at $a = 0$ and where the saving rule switches branches, do not affect (G.8), which needs only absolute continuity of $V_1$; on the grid they show up in the discretization error $e$.

**The timeless problem resolves the time-0 problem.** Our calibrated steady state, with zero inflation, is a stationary Ramsey plan in the sense of DS (their Definition 4). As in their economy, monetary policy changes the stationary real allocation only through inflation: under the fiscal rule public debt returns to its steady-state level, and the real interest rate clears the asset market. Leases are indexed to consumer prices and debt is real, so no household decision depends on inflation, which enters only the wage Phillips curve and, through the adjustment cost, households' values. In the stationary first-order condition for inflation, the two Phillips-curve terms cancel for any multiplier, since the planner discounts as households do. What remains is the adjustment cost, counted once in the criterion and once in every household's Bellman equation weighted by its value multiplier, $-\theta_w \pi^w \big(1 + \sum_x \bar{\Psi}_V(x)\big)$; since the value multipliers sum to zero, it vanishes only at zero inflation. This is the counterpart of DS's Proposition 5.

At zero inflation the remaining stationary first-order conditions form a square linear system in the other multipliers, nonsingular at our steady state, so the stationary multipliers exist. The argument of DS's Proposition 6, in the dual form of their Proposition 21, which is the form of (G.2), then applies: with the penalties at their stationary values, the gradient of $\mathcal{W}^{TL}$ with respect to the policy rate is zero at the steady state, so a surprise brings no first-order gain. By Proposition G.1, this is a statement about the weights $\hat{g}$ together with the three terms that the representation in Section 7.2 drops, each of which enters the gradient to first order: with them, the weights $\hat{g}$ are those under which no surprise from the stationary equilibrium raises welfare to first order.